

Lecture Notes on Algebraic Geometry

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Remark 0.1. These are lecture notes based on the course Algebraic geometry I at Humboldt University Berlin SoSe2026 taught by Prof. Dr. Salvatore Floccari. I am not affiliated with him, and any typos and mistakes are strictly to be attributed to myself.

§1 Lecture 1

Remark 1.1. This lecture started with a long motivational part which highlighted some parallels to differential geometry and number theory. I have not included this in these notes, as that was mainly an informal discussion, which was best captured in person.

The main object of the study is to find solutions to polynomial equations.

Definition 1.2. let k be a field, we define **affine n -space** over k , denoted $\mathbb{A}_k^n$ as the set of n -tuples in k .

Notice that each polynomial $f \in k[x_1, \dots, x_n]$ may be viewed as a function $f: \mathbb{A}_k^n \rightarrow k$, $p \mapsto f(p)$. This allows us to define the zero locus of a collection of polynomials.

Definition 1.3. Let $S \subseteq k[x_1, \dots, x_n]$ be a collection of polynomials. Define $V(S)$ as the common vanishing of these polynomials:

$$V(S) = \{p \in \mathbb{A}_k^n \mid f(p) = 0 \forall f \in S\}.$$

Remark 1.4. One easily confirms $Z(S) = Z(\langle S \rangle)$.

Definition 1.5. Sets of the form $Z(S)$ are called **algebraic sets** over k , or algebraic subsets of $\mathbb{A}_k^n$.

Proposition 1.6 (1) Finite unions of algebraic sets are algebraic.

(2) Arbitrary intersections of algebraic sets are algebraic.

Proof. (1) By induction, it suffices to show that $Z(S_1) \cup Z(S_2)$ is algebraic for $S_i \subseteq k[x_1, \dots, x_n]$. Indeed, let us denote

$$S_1 \cdot S_2 = \{f \cdot g \mid f \in S_1, g \in S_2\}.$$

We claim $Z(S_1) \cup Z(S_2) = Z(S_1 \cdot S_2)$. Indeed, suppose first $p \in Z(S_i)$ and then clearly all polynomials in $S_1 \cdot S_2$ vanish on p as one of the factors does. Conversely suppose $p \notin Z(S_1) \cup Z(S_2)$, then there exists a polynomial $f \in S_1$ and $g \in S_2$ such that $f(p) \neq 0 \neq g(p)$. In particular $(f \cdot g)(p) \neq 0$.

(2) . Let $(S_i)_{i \in I}$ be a family of ideals. we claim

$$\bigcap_{i \in I} Z(S_i) = Z\left(\bigoplus_{i \in I} S_i\right).$$

Indeed, if $p \in \bigcap_{i \in I} Z(S_i)$, then every polynomial f which is contained in some S_i vanishes on p . In particular, as $\bigoplus_{i \in I} S_i$ is generated by such polynomials, all polynomials in $\bigoplus_{i \in I} S_i$ vanish for p . The reverse inclusion is even more trivial as $S_j \subseteq \bigoplus_{i \in I} S_i$ for all $j \in J$.

□

We may observe that $\emptyset = Z((1))$ and $\mathbb{A}_k^n = Z((0))$. Thus, the algebraic subsets of $\mathbb{A}_k^n$ satisfy the axioms for closed sets for a topology on $\mathbb{A}_k^n$. This topology is called **Zariski topology** of $\mathbb{A}_k^n$.

Example 1.7

Some examples of algebraic sets are:

1. $\{p\} = \{(a_1, \dots, a_n)\} = V(x_1 - a_1, \dots, x_n - a_n)$
2. Hypersurfaces: $V(f)$, $f \in k[x_1, \dots, x_n]$, nonzero.

Remark 1.8. Every algebraic set is an intersection of hypersurfaces as can be seen from proposition 1.4. In fact, using Noetherianity every algebraic set is a finite intersection of hypersurfaces, but more on this later.

§2 Lecture 2

Let us recall some basic definitions from commutative algebra.

Definition 2.1. A ring is called noetherian if it satisfies the a.c.c condition for ideals, or, equivalently if every ideal is finitely generated.

From Hilbert's basis theorem we know that if R is a Noetherian ring then $R[X]$ is also noetherian. In particular $k[x_1, \dots, x_n]$ is noetherian.

We have equipped our spaces $\mathbb{A}_k^n$ with a topology and hence a geometry. On the other hand $\mathbb{A}_k^n$ is also equipped with a ring $k[x_1, \dots, x_n]$. We will start building a bridge between the algebra and geometry. First note that we have a map

$$\begin{aligned} \{ \text{Ideals of } k[x_1, \dots, x_n] \} &\rightarrow \{ \text{algebraic sets of } \mathbb{A}_k^n \} \\ I &\mapsto Z(I). \end{aligned}$$

This map is by construction surjective, but it is not injective: $V((x)) = V((x^2))$.

Definition 2.2. For an ideal $a \subseteq R$, define its radical

$$\sqrt{a} := \{f \in R \mid \exists n \in \mathbb{N}, f^n \in a\}$$

. An ideal a is called **radical** if $\sqrt{a} = a$.

We want to construct a type of "inverse" to the above map. Clearly this doesn't make sense in the current setting as the map is not even injective. However, as we shall see, if we restrict ourselves to radical ideals, it will turn out to be injective.

Definition 2.3. Let $X \subseteq \mathbb{A}_k^n$ be any set, we define

$$I(X) = \{f \in k[x_1, \dots, x_n] \mid f|_X \equiv 0 \text{ (as functions)}\}.$$

since $0 \in I(X)$, this set is always nonempty, and in fact, it is an ideal as one easily verifies. We can already verify some properties of this map.

Proposition 2.4

Let $X, Y \subseteq \mathbb{A}_k^n$. Then we have:

1. If $X \subseteq Y$ then $I(X) \supseteq I(Y)$.
2. $I(X \cup Y) = I(X) \cap I(Y)$.
3. $I(X)$ is always a radical ideal.
4. $Z(I(X)) = \overline{X}$.

Proof. All of the statements except 4 follow immediately from the definition so we will only show 4. Suppose $X \subseteq Y$, where Y is close. applying $Z \circ I$ we have $Z(I(X)) \subseteq Z(I(Y))$, hence it suffice to show that for closed sets we have $Z(I(Y)) = Y$. Suppose first $Y = Z(\mathfrak{a})$. Then a point lies in $Z(I(Y))$ if and only if all polynomials in $I(Y)$ vanish at it. In particular the polynomials in $\mathfrak{a}$ must vanish in it, so $Z(I(Y)) \subseteq Y$. Conversely if $p \in Y$ then all polynomials in $I(Y)$ vanish at p , and so $p \in Z(I(Y))$. $\square$

Definition 2.5. Let $X \subseteq \mathbb{A}_k^n$ be an algebraic set. We define its **affine coordinate ring** as

$$A(X) = \frac{k[x_1, \dots, x_n]}{I(X)}$$

Remark 2.6. By the above proposition this is a finitely generated reduced k -algebra.

We have seen that $I(Z(\mathfrak{a}))$ is a radical ideal which contains $\mathfrak{a}$ but the following Theorem tells us more.

Theorem 2.7 (Hilberts Nullstellensatz)

Let k be an algebraically closed field and $\mathfrak{a} \subseteq k[x_1, \dots, x_n]$ be an ideal. Then we have

$$I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}.$$

We will not prove the full version here but rather show how it follows from the weak Nullstellensatz:

Theorem 2.8 (Weak Nullstellensatz)

Let k be an algebraically closed field. Then The maximal ideals of $k[x_1, \dots, x_n]$ are of the form $(x_1 - a_1, \dots, x_n - a_n)$ where $a_i \in k$.

Remark 2.9. One may prove the Weak nullstellensatz using Noethers Normalisation lemma.

Proposition 2.10

The weak Nullstellensatz implies Hilberts' nullstellensatz.

Proof. We employ the Rabinowitsch trick. let $\mathfrak{a} \subseteq k[x_1, \dots, x_n]$ be any ideal. We already know that $I(Z(\mathfrak{a})) \supseteq \sqrt{\mathfrak{a}}$ so we need to show the reverse inclusion. Thus let us choose

$f \in I(Z(\mathfrak{a}))$ nonzero. By Hilbert's basis theorem we may assume $\mathfrak{a} = (g_1, \dots, g_m)$ and so we wish to find an integer $r \in \mathbb{N}$ such that we may express f^r as

$$f^r = H_1 g_1 + \dots + H_m g_m, \quad H_i \in k[x_1, \dots, x_n].$$

for this consider the ideal

$$\mathfrak{b} = (g_1, \dots, g_m, (f \cdot x_{n+1} - 1)) \subseteq k[x_1, \dots, x_{n+1}].$$

Suppose now $p \in V(\mathfrak{b})$, then $g_1, \dots, g_m$ vanish on p and so f also does. but then $(f \cdot x_{n+1} - 1)(p) \neq 0$. Thus we have shown that $V(\mathfrak{b}) = \emptyset$. By the weak Nullstellensatz this yields $\mathfrak{b} = (1)$ and so there exists a polynomial equation

$$1 = (\tilde{H}_1 g_1 + \dots + \tilde{H}_m g_m) + \tilde{H}(X_{n+1} f - 1) \text{ with } \tilde{H}_i, \tilde{H} \in k[x_1, \dots, x_{n+1}].$$

Consider now the evaluation morphism

$$\Phi: k[x_1, \dots, x_{n+1}] \rightarrow k[x_1, \dots, x_n][\frac{1}{f}], \quad x_{n+1} \mapsto \frac{1}{f}.$$

Under this evaluation we receive

$$1 = H_1 g_1 + \dots + H_m g_m \text{ with } H_i \in k[x_1, \dots, x_n]_f.$$

Multiplying out the denominators we receive the result. □

Having proven Hilbert's Nullstellensatz, we have established a bijection

$$\begin{aligned} \{ \text{radical Ideals of } k[x_1, \dots, x_n] \} &\rightarrow \{ \text{algebraic sets of } \mathbb{A}_k^n \} \\ I &\mapsto Z(I) \end{aligned}$$

which is inverse inclusion and has the inverse $S \mapsto I(S)$. Under this bijection, points correspond to maximal ideals. Let us continue with this bridge between algebra and geometry. We will translate properties of ideals to properties of the algebraic sets and vice versa using the above established bijection.

Definition 2.11. A topological space X is called **irreducible** if whenever $X = C_1 \cup C_2$ for closed sets C_1, C_2 , we have $X = C_1$ or $X = C_2$.

Example 2.12

$\mathbb{A}_k^1$ is irreducible as the only closed subsets are the whole space, the empty set and finite collections of points. As k is algebraically closed, $\mathbb{A}_k^1$ is infinite and so $\mathbb{A}_k^1 = C_1 \cup C_2$ implies either $C_1 = \mathbb{A}_k^1$ or $C_2 = \mathbb{A}_k^1$.

Proposition 2.13

Let $X \subseteq \mathbb{A}_k^n$ be an algebraic set. Then X is irreducible if and only if $I(X)$ is a prime ideal.

Proof. Suppose first that $I(X) = \mathfrak{p}$ is a prime ideal. If $X \subseteq C_1 \cup C_2$ we obtain $\mathfrak{p} \supseteq I(C_1 \cup C_2)$. If $C_1 = Z(\mathfrak{a}_1)$ and $C_2 = Z(\mathfrak{a}_2)$ then $I(C_1 \cup C_2) \supseteq \mathfrak{a}_1 \cdot \mathfrak{a}_2$ and thus $\mathfrak{p} \supseteq \mathfrak{a}_1$ or $\mathfrak{p} \supseteq \mathfrak{a}_2$ since $\mathfrak{p}$ is prime. In particular, applying $Z(\cdot)$ we obtain the result. Conversely,

suppose that X is irreducible and $f \cdot g \in I(X)$. Then $V(f) \cup V(g) = V(f \cdot g) \supseteq X$. By irreducibility, we may assume $V(f) \supseteq X$. Applying I , we get

$$f \in \sqrt{(f)} \subseteq I(X),$$

which was to show. $\square$

Definition 2.14. An **affine variety** is defined as an irreducible algebraic subset of some $\mathbb{A}_k^n$.

Tasks for Week 1

Throughout, k denotes an algebraically closed field.

Exercise 2.1. (i) Let $Y \subset \mathbb{A}_k^2$ be the plane curve defined by $y - x^2$. Show that $A(Y)$ is a polynomial ring in 1 variable.

(ii) Let $Y \subset \mathbb{A}_k^2$ be the plane curve defined by $xy - 1$. Calculate $A(Y)$ and show that it is not isomorphic to a polynomial ring in 1 variable.

Solution. (i) By definition we have $Y = Z(y - x^2)$ and thus $I(Y) = \sqrt{(y - x^2)}$. we claim that $(y - x^2)$ is a prime ideal and thus $I(Y) = (y - x^2)$ and $A(Y) = \frac{k[x, y]}{(y - x^2)}$. Indeed consider the composition of k -algebra morphisms

$$k[x] \xleftarrow{i} k[x, y] \xrightarrow[\pi]{\pi \circ i} \frac{k[x, y]}{(y - x^2)}$$

on the one hand this map is surjective as every polynomial in the quotient is equivalent to one in only x by substituting $y = x^2$. This composition is also however injective as all nonzero polynomials in the ideal divisible by $y - x^2$ have a term in y which is nonzero. Thus this is a k -algebra isomorphism. In particular $(y - x^2)$ is a prime ideal and $A(X) \cong k[x]$

(ii) Consider the again the composition

$$k[x] \xleftarrow{i} k[x, y] \xrightarrow[\pi]{\pi \circ i} \frac{k[x, y]}{(xy - 1)}$$

This is again an injective map for the same reason as above but it is in general not a surjective map. However, as x gets mapped to a unit under this composition, by the universal property of localization there exists a unique map $\tilde{\varphi}: k[x]_x \rightarrow k[x, y]/(xy - 1)$ such that the following diagram commutes:

$$\begin{array}{ccccc} k[x] & \xleftarrow{i} & k[x, y] & \xrightarrow[\pi]{\pi \circ i} & k[x, y]/(xy - 1) \\ \downarrow \ell & & & \nearrow \tilde{\varphi} & \\ k[x]_x & & & & \end{array}$$

More specifically the map is given by $\tilde{\varphi}\left(\frac{p(x)}{x^n}\right) = p(x)y^n \in \frac{k[x, y]}{(xy - 1)}$. This map is indeed a surjective map. As the localization remains injective, it is also injective. Thus $A(Y) \cong K[x]_x$. The latter is not isomorphic to $k[x]$ as it contains units which are not in k . $\square$

- Exercise 2.2.** (i) Show that the Zariski topology on $\mathbb{A}_k^n$ is not Hausdorff, but that it satisfies the separation axiom T_1 (i.e., for any two distinct points there exists an open subset U of $\mathbb{A}_k^n$ containing one but not the other).
- (ii) There is a natural bijection $\mathbb{A}_k^2 \rightarrow \mathbb{A}_k^1 \times \mathbb{A}_k^1$. Show that the Zariski topology on $\mathbb{A}_k^2$ does not coincide with the product topology on $\mathbb{A}_k^1 \times \mathbb{A}_k^1$ induced by the Zariski topology on $\mathbb{A}_k^1$.
- (iii) Let $1 \leq m \leq n$. Let $\pi : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ be the projection onto the first m coordinates. Show that π is continuous for the Zariski topology. Show moreover that π is open, but not closed.

Solution. (i) We will show that any two nonempty open sets intersect which immediately implies that the topology is not Hausdorff. To show this it suffices to show that every open set is dense: indeed, if $U \cap V = \emptyset$, for two proper opens then $U \subseteq V^C$ which is not the whole space, which contradicts the density of V . Thus let $U = \mathbb{A}_k^n \setminus V(I)$. We find that $V(I(U) \cdot I) = \mathbb{A}_k^n$ as the generators vanish on the whole space. Thus, by Hilbert's nullstellensatz, we find that $\sqrt{I(U) \cdot I} = (0)$ and hence since we are working in an integral domain also $I(U) \cdot I = 0$. In particular, if U is nonempty then I is nonzero and $I(U)$ must be zero. Thus $\overline{U} = Z(I(U)) = Z(0) = \mathbb{A}_k^n$.

now let $p = (p_1, \dots, p_n)$ and $q = (q_1, \dots, q_n)$ be two distinct points in $\mathbb{A}_k^n$. In particular for some i , $p_i \neq q_i$ resulting in the open set $U = \mathbb{A}_k^n \setminus V(x_i - p_i)$ containing q but not p .

- (ii) Recall that the product topology on the product $X \times Y$ is the smallest topology which makes both projections continuous. This allows us to describe the closed sets of the product $\mathbb{A}_k^1 \times \mathbb{A}_k^1$ quite directly: they are the whole space, the empty set and finite unions of points, vertical lines and horizontal lines. Indeed, These sets do in fact verify the axioms for closed sets of a topology and it is also easy to see that any topology on the product which makes both projections continuous must contain these sets as they are achieved as preimages of points, the whole space, the empty set and the intersections of these. The Zariski topology on $\mathbb{A}_k^2$ contains of course many more closed sets such as $V(x - y)$ which is not in the product topology.
- (iii) Let $Z(I) \subseteq \mathbb{A}_k^m$ be any closed set. Then we have

$$\pi^{-1}(Z(I)) = \{p = (p_1, \dots, p_n) \in \mathbb{A}_k^n \mid f(p_1, \dots, p_m) = 0 \forall f \in I\} = Z(i(I)),$$

where i denotes the inclusion $i: k[x, \dots, x_m] \hookrightarrow k[x, \dots, x_n]$. Next let $U = \mathbb{A}_k^n \setminus I$ be any open set and compute

$$\begin{aligned} \pi(U) &= \{x \in \mathbb{A}_k^m \mid \exists q : (x, q) \in U\} \\ &= \mathbb{A}_k^m \setminus \{x \in \mathbb{A}_k^m \mid \forall q, (x, q) \notin U\} \\ &= \mathbb{A}_k^m \setminus \{x \in \mathbb{A}_k^m \mid \forall q, (x, q) \in V(I)\} \\ &= \mathbb{A}_k^m \setminus Z(S) \end{aligned}$$

where we define $S = \{f(X_1, \dots, X_m, q) \in k[X, \dots, X_m] \mid q \in \mathbb{A}_k^{n-m}, f \in I\}$.

Lastly we can see that π is not a closed map: consider the set $Z(\prod_{i=1}^n x_i - 1)$ This is a closed set which under the projection maps to $\mathbb{A}_k^m \setminus \{0\}$ which is open, and thus dense by 1. In particular as it is not the whole space it is not closed.

□

Exercise 2.3. Let $L \subset \mathbb{A}_k^n$ be a line, i.e., L is defined by $n - 1$ independent linear equations. Let $X \subset \mathbb{A}_k^n$ be an algebraic set. Show that $X \cap L$ is either empty, or finite, or equal to L .

Solution. By assumption L is given by $\ker A$ where A is $(n - 1 \times n)$ matrix of full rank. From basic linear algebra, we know that $L = \langle v \rangle_k$ for a $v = (v_1, \dots, v_n) \in \mathbb{A}_k^n \setminus \{0\}$. This allows us to define a polynomial map

$$\phi: \mathbb{A}_k^1 \rightarrow L, t \mapsto (tv_1, \dots, tv_n)$$

We claim this is an isomorphism. Indeed, since v is not zero let us assume $v_i \neq 0$ and thus we may write the inverse as $(p_1, \dots, p_n) \mapsto \frac{p_i}{v_i}$ which is polynomial (in fact linear). Thus, we have shown that L is isomorphic to $\mathbb{A}_k^1$. In particular any set of the form $X \cap L$ corresponds to a closed set of $\mathbb{A}_k^1$ for which we know what the closed subsets look like. $\square$

Exercise 2.4. (i) Consider the subset

$$C := \{(t, t^2, t^3) \mid t \in k\} \subset \mathbb{A}_k^3.$$

Show that C is an algebraic set and find generators for $I(C)$.

(ii) Let

$$Y := \{(t^3, t^4, t^5) \mid t \in k\} \subset \mathbb{A}_k^3.$$

Show that Y is an algebraic set and give generators for $I(Y)$. Prove that this ideal is not generated by less than 3 elements.

Solution. (i) To see that C is algebraic, we claim $C = V(y - x^2, z - x^3)$. Indeed points in C clearly vanish on this ideal and furthermore if $\lambda = (\lambda_1, \lambda_2, \lambda_3) \in V(y - x^2, z - x^3)$, then $\lambda_2 = \lambda_1^2$ and $\lambda_3 = \lambda_1^3$ and so $\lambda \in C$. In fact, we claim $I(C) = (y - x^2, z - x^3)$. For this simply note that $\frac{k[x, y, z]}{(y - x^2, z - x^3)} \cong k[x][x^2, x^3] = k[x]$ which is an integral domain.

(ii) $Y = V(y^3 - x^4, z^3 - x^5, z^4 - y^5)$. Indeed all points in Y are easily seen to verify these equations. Furthermore, if $p = (p_1, p_2, p_3)$ is nonzero and verifies these equations then $p = (t^3, t^4, t^5)$ for $t = p_2/p_1$. Now let us compute $I(C) = \sqrt{(y^3 - x^4, z^3 - x^5, z^4 - y^5)}$. we claim:

$$I(Y) = \mathfrak{a} := (y^2 - xz, x^3 - yz, z^2 - x^2y).$$

Indeed, one easily sees that a point in Y satisfies these equations as well. Furthermore a simple calculation shows that a point which satisfies the equations of $\mathfrak{a}$ satisfies the original equations $(y^3 - x^4, z^3 - x^5, z^4 - y^5)$ and so is in Y . Hence

$$I(Y) = \sqrt{\mathfrak{a}}.$$

Hence, to conclude we need to show that

$$R := \frac{k[x, y, z]}{\mathfrak{a}},$$

is a reduced ring. In fact it is an integral domain. For this, Consider the map

$$\tilde{\varphi}: k[x, y, z] \rightarrow k[t], \begin{cases} x \mapsto t^3 \\ y \mapsto t^4 \\ z \mapsto t^5. \end{cases}$$

This map is surjective onto its image $k[t^3, t^4, t^5] \subseteq k[t]$. Notice that this map vanishes on $\mathfrak{a}$ and thus by the homomorphism theorem we obtain a well defined map

$$\varphi : R \twoheadrightarrow k[t], \begin{cases} \bar{x} \mapsto t^3 \\ \bar{y} \mapsto t^4 \\ \bar{z} \mapsto t^5. \end{cases}$$

The goal is to show that this map is injective as well which would imply that $\mathfrak{a}$ is a prime ideal. For this we need to investigate the arithmetic of R a little closer. First we claim that every monomial $x^i y^j z^k \in k[x, y, z]$ is equivalent to a monomial of one of the forms: $x^n, x^n y, x^n z$ for some integer n . Indeed, since we have $(y^3 - x^4, z^3 - x^5, z^4 - y^5) \subseteq \mathfrak{a}$, it follows that every monomial is equivalent to one of the form

$$x^i y^j z^k, \text{ with } i, j, k \in \{0, 1, 2\}.$$

If both j and k are positive, then let $t := \min\{j, k\} > 0$. Then using the generators of $\mathfrak{a}$, we find

$$x^i y^j z^k \equiv x^i (yx)^t y^{j-t} z^{k-t} \equiv x^i x^{3t} y^{j-t} z^{k-t} \in \{x^n, x^n y, x^n z\},$$

where the last claim follows from the fact that at least one of $j - t, k - t$ is zero and the other is at most one. Hence, we only need to consider the monomials $x^n y^2$ and $x^n z^2$. For these we have,

$$\begin{aligned} x^n y^2 &\equiv x^{n+1} z \pmod{\mathfrak{a}} \\ x^n z^2 &\equiv x^{n+2} y \pmod{\mathfrak{a}} \end{aligned}$$

Hence, every element in $r \in R$ is represented by a term of the form.

$$r \equiv F(x) + G(x)y + H(x)z \pmod{\mathfrak{a}}$$

Suppose now that $\varphi(r) = 0$, and so

$$F(t^3) + G(t^3)t^4 + H(t^3)t^5 = 0 \in k[t].$$

Notice that the first summand only contributes to the terms which have degree 0 mod (3) the second term only contributes to terms with degree which are 1 mod 3 and the third to terms with degree 2 mod 3. Thus, the only way to have the above is if

$$F(t^3) = G(t^3) = H(t^3) = 0 \Leftrightarrow F = G = 0.$$

Hence $r \equiv 0 \pmod{\mathfrak{a}}$. Thus φ is injective as desired.

To complete this problem we need to show that $\mathfrak{a}$ is not generated by less than three elements. To see this note that

$$\mathfrak{a} = (y^2 - xz, x^3 - yz, z^2 - x^2y) \subseteq \mathfrak{m}^2,$$

where $\mathfrak{m}$ is the maximal ideal $(x, y, z) \subseteq k[x, y, z]$. Hence, we can consider the k vector space

$$V := \frac{\mathfrak{a}}{\mathfrak{m}^3 \cap \mathfrak{a}} \subseteq \frac{\mathfrak{m}^2}{\mathfrak{m}^3}$$

V has dimension at least three as the generators of $\mathfrak{a}$ reduce to: $\{xz, yz, z^2\} \in V$ and these are k linearly independent. We claim that any set of generators of $\mathfrak{a}$ must generate V . Indeed, since $\mathfrak{a} \subseteq \mathfrak{m}^2$, all of these generators do not have constant or linear terms, all there terms are at least quadratic. Then V must be contained in the k span of the degree two parts of these generators. Thus, there must be atleast three such parts.

□

Exercise 2.5. Let $k = \mathbb{C}$. Consider the subset

$$X := \{(z, \exp(z)) \mid z \in \mathbb{C}\} \subset \mathbb{A}_{\mathbb{C}}^2.$$

Is X an algebraic set? Describe the ideal $I(X)$ and the Zariski closure of X .

Solution. X is not algebraic. Indeed, let $k[x, y]$ denote the ring of regular functions on $\mathbb{A}_k^2$. If X were algebraic then the intersection

$$X \cap V(y - 1)$$

would either be empty, finite or the entire line $V(y - 1)$. However it is neither of these: □

$$X \cap V(y - 1) = \{(2\pi i k, 1) \mid k \in \mathbb{Z}\}.$$

Now, let us compute $I(X)$, we claim it is 0. In other words, suppose f is a function that vanishes on X , then, as we have seen before, $f(x, 1)$ contains infinity many points and thus is the zero polynomial. In particular, $V(f)$ must contain the entire $V(y - 1)$ and therefore $f \in (y - 1)$ and we can write $f = (y - 1) \cdot g$. If g is nonzero, then repeat the same argument with $V(y - 2)$ instead and find $f = (y - 1)(y - 2)g_2$ proceeding in this way we arrive at a contradiction because then f would have an infinite degree.

§3 Lecture 3

Affine varieties as introduced in the last lecture can be viewed as the building blocks of algebraic sets.

Theorem 3.1

Let $X \subseteq \mathbb{A}_k^n$ be an algebraic set. Then there exists a decomposition

$$X = X_1 \cup \dots \cup X_r$$

where $X_i \subseteq \mathbb{A}_k^n$ are varieties. If we require $X_j \not\subseteq X_i$ for all $i \neq j$ then this decomposition is unique up to reordering. For this case, X_i is called **irreducible components** of X

Before we can prove this theorem cleanly, we need some more terminology, which will be useful vocabulary for the future.

Definition 3.2. A topological space is called **Noetherian** if it satisfies the d.c.c for closed sets. Equivalently, a topological space is called Noetherian if every collection of closed sets contains a minimal element.

Lemma 3.3

Affine space is a noetherian space.

Proof. This follows simply due to the bijection of closed sets and radical ideals and the fact that $k[x_1, \dots, x_n]$ is a noetherian ring by Hilbert's basis theorem. □

Remark 3.4. In the exercises, we show that every subset of a noetherian space is also noetherian with respect to the subspace topology.

Theorem 4.1 is now the consequence of a more general phenomena in noetherian spaces:

Theorem 3.5

Let X be a topological space. Then any nonempty closed subset $Y \subseteq X$ can be written as a finite union

$$Y = Y_1 \cup \cdots \cup Y_r$$

where the Y_i are closed and irreducible. Assuming $Y_i \subsetneq Y_j$ for $j \neq i$, this decomposition is unique up to reordering.

Proof. First we show existence. consider the set

$$\Sigma = \{\emptyset \neq Y \subseteq X \mid Y \text{ is closed and } Y \text{ does not have decomposition}\}.$$

If Σ is empty we are done. Otherwise by noetherianity Σ admits a minimal element Y . Then Y itself is not irreducible and so $Y = Y_1 \cup Y_2$ for proper subsets Y_1, Y_2 . By the minimality of Y , these both have decompositions into irreducibles. This yields a contradiction however as it produces a decomposition of Y itself. Now let us show uniqueness. Assume

$$Y_1 \cup \cdots \cup Y_r = X_1 \cup \cdots \cup X_s$$

where the Y_i, X_i are closed irreducible and $Y_i \subsetneq Y_j$ for $i \neq j$ and similarly $X_i \subsetneq X_j$ for $i \neq j$. Then we have

$$Y_1 \subseteq X_1 \cup \cdots \cup X_s$$

, and so by the irreducibility of Y_1 there exists a j with $Y_1 \subseteq X_j$. Similarly, for X_j there exists i with $X_j \subseteq Y_i$. However, this implies $Y_1 \subseteq Y_j$ and thus $Y_1 = Y_i$ and therefore $Y_1 = X_j$. In this way we have produced a map

$$\{Y_i \mid i = 1, \dots, r\} \rightarrow \{X_j \mid j = 1, \dots, s\}.$$

This map is clearly injective, since we assume $Y_i \neq Y_j$ for $i \neq j$. It is also surjective as we may apply the same argument in reverse. In particular, $r = s$ and up to rearrangement, the factors agree. $\square$

In the category of finitely generated reduced k -algebras the morphisms are k -linear ring homomorphisms. In the current setting we do not have a notion of morphism for affine varieties. One option would be to only consider continuous maps between affine varieties as our morphisms. This is not very fruitful: If X is a topological space every map $X \rightarrow \mathbb{A}_k^1$ is continuous. Thus we need a more restrictive definition of morphism to ensure that these maps retain some algebraic information.

Definition 3.6 (Regular Functions on Affine Sets). Let $X \subseteq \mathbb{A}_k^n$ be an algebraic set. A function $f : X \rightarrow k$ is called a **regular function** on X if there exists a polynomial $F \in k[x_1, \dots, x_n]$ such that $f(P) = F(P)$ for all $P \in X$. we denote by $\mathcal{O}(X)$ the set of **regular functions** in X .

Remark 3.7. The following are easy to verify

- $\mathcal{O}(X)$ is a ring with respect to point wise operations in k .
- By construction we have a surjective map $k[x_1, \dots, x_n] \rightarrow \mathcal{O}(X)$ which in turn induces an isomorphism

$$A(X) = \frac{k[x_1, \dots, x_n]}{I(X)} \cong \mathcal{O}(X).$$

It is important to realize that $A(X)$ and $\mathcal{O}(X)$ are isomorphic rings, but they are not the same object. There is a natural notion of evaluating functions in $\mathcal{O}(X)$ but given just a finitely generated reduced k -algebra, $A(X)$, one needs to put more thought into what it means to evaluate its elements at points. That having been said, we will blur this distinction in the future appealing to the above isomorphism.

Definition 3.8. Let $X \subseteq \mathbb{A}_k^n$ be an algebraic set if $S \subseteq A(X)$ defines:

$$Z_X(S) := \{p \in X \mid f(p) = 0 \forall f \in S\},$$

and for a subset $Y \subseteq X$, define

$$I_X(Y) := \{f \in A(X) \mid f(p) = 0 \forall p \in Y\}.$$

If we equip X with the subspace topology of $\mathbb{A}_k^n$, then the closed sets of X are precisely of the form $Z_X(S)$ and, in fact, we have a relative version of the Nullstellensatz.

Theorem 3.9

There is a natural bijection

$$\{\text{closed subsets of } X\} \leftrightarrow \{\text{radical ideals of } A(X)\}.$$

Proof. The closed subsets of X are in bijection to the closed subsets of $\mathbb{A}_k^n$ which are contained in X . By the Nullstellensatz these are in bijection to radical ideals containing $I(X)$ which are in turn again in bijection to radical ideals of $A(X)$. The bijections are given by the operations $Z_X(\cdot)$ and $I_X(\cdot)$. $\square$

As promised, we will now discuss the right version of the morphism for our category. As hinted before, we want to consider those continuous maps that retain algebraic information. In other words, we will require them to be given by polynomials.

Definition 3.10. Let $X \subseteq \mathbb{A}_k^n$ and $Y \subseteq \mathbb{A}_k^m$ be affine algebraic sets. a map

$$\Phi : X \rightarrow Y,$$

is called a **regular map** or **morphism** if there exists polynomials $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ such that

$$\Phi(P) = (f_1(P), \dots, f_m(P)) \forall P \in X.$$

Φ is called an isomorphism if it has an inverse map which is regular.

Remark 3.11. We have a natural isomorphism $\text{Hom}(X, \mathbb{A}_k^1) \cong \mathcal{O}(X) \cong A(X)$

We can now use morphisms to pull back algebraic information:

Definition 3.12. If $\Phi: X \rightarrow Y$ is a morphism, define

$$\Phi^*: \mathcal{O}(Y) \rightarrow \mathcal{O}(X), f \mapsto \Phi^*(f) := f \circ \Phi$$

Using the natural isomorphism $\mathcal{O}(X) \cong A(X)$ we will usually regard the pullback as map

$$\Phi^*: A(Y) \rightarrow A(X).$$

It is essentially obvious to verify the following.

Lemma 3.13

The pullback is well defined, and a ring homomorphism

Let us compute a few examples

Example 3.14

if $\Phi: \mathbb{A}^1 \rightarrow \mathbb{A}^1, t \mapsto t^r$, the Pullback is given by

$$\Phi^* k[x] \rightarrow k[x], x \mapsto x^r.$$

Under this morphism, every point except zero has r preimages (Assume $\text{char } k = 0$), so this map describes a branched cover with a unique branch point 0. This can also be seen on the pullback map. The points mapped to a point t correspond to the maximal ideals lying above $\sqrt{(x^r - t)} \subseteq k[x]$.

Lemma 3.15

Morphisms between affine algebraic sets are continuous maps.

Proof. Let $\Phi: X \rightarrow Y$ be a morphism and $V = Z_Y(f_1, \dots, f_t)$ be a closed set in Y . then we compute :

$$P \in \Phi^{-1}(V) \Leftrightarrow \Phi(P) \in Z_Y(f_1, \dots, f_t) \Leftrightarrow P \in Z_X(\Phi^* f_1, \dots, \Phi^* f_t).$$

□

§4 Lecture 4

Last week we introduced regular maps and have seen that they induce pullback morphisms on the ring of regular functions. In fact, for affine algebraic sets we have a much stronger correspondence, which is the main theorem of today's lecture. Let us start with the first result towards this direction.

Theorem 4.1

Let X, Y be affine algebraic sets: the map $\Phi \mapsto \Phi^*$ induces a bijection:

$$\text{Hom}(X, Y)_{\text{algebraic sets over } k} \rightarrow \text{Hom}(A(Y), A(X))_{\text{reduced f.g } k\text{-algebras}}, \Phi \mapsto \Phi^*.$$

Proof. the i -th coordinate of the map Φ is uniquely determined by the value $\Phi^*(y_i)$ where $y_1, \dots, y_n$ are coordinates for Y . This shows that the pullbacks determine the morphism and the map described in the theorem is injective. Conversely if

$$F: A(Y) \rightarrow A(X)$$

is a morphism, the map $\Phi: X \rightarrow Y$, whose i -th coordinate is given by $F(y_i)$ is a morphism whose pullback is F . $\square$

Remark 4.2. The above bijection is natural in the sense that it behave well under compositions and sends the identity to the identity.

Under the above bijection we may translate algebraic properties of Φ^* to geometric properties of Φ and vice versa.

Proposition 4.3

Let $\Phi: X \rightarrow Y$ be a morphism of affine algebraic sets. Then

1. Φ^* is injective if and only if ϕ is **dominating** : $\overline{\Phi(X)} = Y$.
2. Φ^* is surjective if and only if Φ is a **closed immersion**, i.e there exists a closed subset $V \subseteq Y$ and an isomorphism $f: X \rightarrow V$, such that $\Phi = i \circ f$.

Proof. 1. Let $V \subseteq Y$ denote the closure of $\Phi(X)$. We then have a commutative diagram

$$\begin{array}{ccccc} & & \Phi & & \\ X & \xrightarrow{\quad} & V & \xrightarrow{\quad} & Y \end{array}$$

By the bijection to morphisms of coordinate rings we then have a comutative diagram

$$\begin{array}{ccccc} & & \Phi^* & & \\ A(Y) & \xrightarrow{\quad} & A(V) & \xrightarrow{\quad} & A(X) \end{array}$$

The first map is injective if and only if $V = Y$. We claim the second map: $A(V) \rightarrow A(X)$, is always injective. Indeed if g is in the kernel, then g must vanish on the image $\Phi(X)$. As $V(g)$ is closed, and $\Phi(X)$ is dense in V , g must vanish everywhere.

If Φ is a closed immersion, the claim is clear so we will only prove the converse. Suppose Φ^* is surjective and let $I = \ker \Phi^*$ be the kernel. Then we have $A(X) \cong A(Y)/I$ and this isomorphism is induced naturally by Φ . The claim then follows for the algebraic set $V := V_Y(I)$. $\square$

A formal way of of describing the correspondences we are discovering is:

Proposition 4.4

There exists an equivalence of categories

$\mathcal{F}: \{ \text{Algebraic sets over } k, \text{ regular maps} \} \rightarrow \{ \text{reduced, finitely generated } k\text{-algebras} \}$
which sends $X \mapsto A(X)$ and a morphism Φ to its pullback Φ^* .

and easily following from this is

Corollary 4.5

There exists an equivalence of categories

$$\{\text{affine varieties over } k, \text{ regular maps}\} \rightarrow \{\text{integral, finitely generated } k\text{-algebras}\}$$

Remark 4.6. Recall that a covariant functor $\mathcal{F}: \mathcal{C} \rightarrow \mathcal{C}'$ is called an equivalence of categories if it is:

- **Fully Faithful**, meaning that the induced maps $\text{Hom}(X, Y) \rightarrow \text{Hom}(\mathcal{F}(X), \mathcal{F}(Y))$ is a bijection for all $X, Y \in \mathcal{C}$.
- **Essentially surjective**, meaning that for any $X' \in \mathcal{C}'$ there is a $X \in \mathcal{C}$ with $X' \cong \mathcal{F}(X)$.

Proof of proposition 5.4. The functor $\mathcal{F}$ is fully faithful by Theorem 5.1. It is also essentially surjective: if R is a reduced, finitely generated k algebra choose generators $x_1, \dots, x_n$.

$$k[t_1, \dots, t_n] \rightarrow R, t_i \mapsto x_i.$$

is surjective with kernel I . Then $R \cong k[t_1, \dots, t_n]/I$. The latter is the coordinate ring of $V(I) \subseteq \mathbb{A}_k^n$. $\square$

Tasks for week 2**Exercise sheet 2**

Throughout, k denotes an algebraically closed field.

Exercise 4.1. Show that a topological space X is noetherian if and only if every non-empty collection of closed subsets has a minimal element. Show moreover:

- any subset of a noetherian space is also noetherian;
- a noetherian topological space is quasi-compact, i.e., every open cover has a finite subcover;
- a noetherian topological space which is Hausdorff must be a finite set equipped with the discrete topology.

Solution. Suppose first that X is a Noetherian space and $\{C_i\}_{i \in I}$ is any collection of closed sets. let C_1 be any closed set from this collection, if it is minimal we are done. If it is not minimal, choose a closed set C_2 from the collection which is strictly contained in C_1 . Iterate this process to obtain a strictly decreasing chain of closed sets. By noetherianity this process terminates after finitely many steps. The converse is easier, any chain

$$C_1 \supseteq C_2 \supseteq \dots$$

defines a collection $\{C_i\}_{i \in \mathbb{N}}$, which must have a minimal element C_N . Then $C_{N+j} = C_N$ for all $j \in \mathbb{N}$.

- Suppose $Y \subseteq X$ is a subset of a noetherian space and $\{Y_i\}_i$ is a collection of closed subsets of Y . By the definition of the subspace topology there exists closed sets $\{C_i\}_i$ of X with $Y_i = C_i \cap Y$. By the Noetherianity of X they admit a minimal element $C \in \{C_i\}_i$ with $C \subseteq C_i$ for all i . In particular $Y = C \cap Y$ is a minimal element of the family $\{Y_i\}_i$.

- (ii) Let X be a noetherian space and $X = \bigcup_{i \in I} U_i$ be an open cover of X . Consider the family of closed sets

$$\left\{ C_J := \left(\bigcup_J U_i \right)^c \mid J \subseteq I, |J| < \infty \right\}$$

By noetherianity this has a minimal elements C_J . If C_J is not empty, then we find $x \in C_J$ which must also lie in some U_i . In particular $C_J \cap U_i \subsetneq C_J$ which contradicts minimality. Thus $C_J = \emptyset$ and $X = \bigcup_J U_j$

- (iii) Let X be a noetherian Hausdorff space. we show that points are open. Indeed let $x \in X$. Then, for all $y \in Y$ by the Hausdorff property, we have open sets $U_{x,y}$ and U_y such that $x \in U_{x,y}$, $y \in U_y$ and $U_{x,y} \cap U_y = \emptyset$. Consider the open cover

$$X \setminus \{x\} = \bigcup_{y \in X \setminus \{x\}} U_y.$$

since $X \setminus \{x\}$ is quasi compact by (i) and (ii) it admits finite subcover

$$X \setminus \{x\} = \bigcup_{i=1}^n U_{y_i}$$

In particular the set $\bigcap_{i=1}^n U_{x,y_i}$ is a nonempty open set disjoint from $X \setminus \{x\}$, and thus

$$\{x\} = \bigcap_{i=1}^n U_{x,y_i}$$

Thus X is a compact discrete space hence finite.

□

Exercise 4.2. (i) Consider the ideal $I \subset k[x, y, z]$ generated by $x^2 - yz$ and $xz - x$. Show that $Z(I)$ is the union of 3 irreducible components Z_1, Z_2, Z_3 , and give generators for the ideals $I(Z_1), I(Z_2), I(Z_3)$.

- (ii) Find the irreducible components of $Z(xy, xz, yz) \subset \mathbb{A}_k^3$. Do the same for

$$Z(xy - z^2, x^2 + y^2 + z^2) \subset \mathbb{A}_k^3.$$

Solution. (i) Suppose $p = (x, y, z) \in Z(A^3)$, then p satisfies the relations

$$\begin{cases} x^2 - yz = 0 \\ xz - x = 0 \end{cases}$$

from the second equation we obtain $x(z - 1) = 0$. thus either $x = 0$ or $z = 1$. If $x = 0$, the second equation implies $y = 0$ or $z = 0$. Hence in the first case p is in $Z(x, y) \cup Z(x, z)$. Now consider the case $z = 1$. In this case we get $x^2 - y = 0$ and thus p is in $Z(z - 1, x^2 - y)$. In particular we have shown

$$Z(I) \subseteq Z(x, y) \cup Z(x, z) \cup Z(z - 1, y^2 - x).$$

The reverse inclusion clearly also holds. Finally all these components are irreducible as the ideals (x, y) , (x, z) , and $(z - 1, y^2 - x)$ are all prime ideals.

- (ii) consider first $p \in Z(xy, xz, yz)$. The equations dictate that atleast two of p 's coordinates must be zero

$$p \in I(x, y) \cup I(x, z) \cup I(y, z)$$

the reverse inclusion clearly also holds and thus we have a irreducible decomposition. Now let us consider $p \in Z(xy - z^2, x^2 + y^2 + z^2)$. Plugging the first equation into the second we obtain that p must satisfy

$$x^2 + y^2 + xy = 0$$

This if $\text{char } k \neq 3$ polynomial factors into linear factors:

$$(x - y\zeta)(x - y\zeta^2) = 0$$

where ζ is a primitive third root of unity. Let us consider first the case where $x = \zeta y$. then the first equation forces

$$z^2 = y^2\zeta \Leftrightarrow (z - y\sqrt{\zeta})(z + y\sqrt{\zeta}) = 0$$

in particular in this case we have

$$p \in Z(x - \zeta y, z - \sqrt{\zeta}y) \cup Z(x - \zeta y, z + \sqrt{\zeta}y)$$

Similarly if $x = y\zeta^2$ we obtain $(z - y\zeta)(z + y\zeta) = 0$ and in particular

$$p \in Z(x - \zeta^2 y, z - \zeta y) \cup Z(x - \zeta^2 y, z + \zeta y)$$

In total we have shown $Z(xy - z^2, x^2 + y^2 + z^2)$ is a subset of

$$Z(x - \zeta y, z - \sqrt{\zeta}y) \cup Z(x - \zeta y, z + \sqrt{\zeta}y) \cup Z(x - \zeta^2 y, z - \zeta y) \cup Z(x - \zeta^2 y, z + \zeta y)$$

Conversely one verifies that any point in the latter set also satisfies the original equation. These four sets are clearly all irreducible

If $\text{char } k = 3$ then there are only two components as in this case $\zeta = \zeta^2 = 1$. $\square$

Exercise 4.3. (i) Show that

$$f : \mathbb{A}_k^1 \rightarrow Z(y^2 - x^3) \subset \mathbb{A}_k^2, \quad t \mapsto (t^2, t^3)$$

is a morphism of affine varieties which is bijective (and even a homeomorphism) but not an isomorphism. Similarly,

$$t \mapsto (t^3, t^4, t^5)$$

gives a morphism

$$\mathbb{A}_k^1 \rightarrow \mathbb{A}_k^3$$

which is a bijection onto a closed subset, but not an isomorphism onto its image (which is the affine variety of Exercise 4.(ii), Sheet 1).

- (ii) Assume $\text{char}(k) = p > 0$. Define a map

$$F : \mathbb{A}_k^1 \rightarrow \mathbb{A}_k^1 \quad \text{via} \quad t \mapsto t^p.$$

Prove that F is a morphism which is bijective and a homeomorphism, but not an isomorphism. The morphism F is called the Frobenius morphism. What are its fixed points?

Solution. (i) This map is a morphism as it is given by polynomials. It is also surjective: if $p \in Z(y^2 - x^3) \setminus \{(0, 0)\}$ then $p = (t^2, t^3)$ for $t = y/x$ and $(0, 0) = f(0)$. It is also injective as an inverse is given by $(x, y) \mapsto y/x$ is an inverse where we interpret this expression as zero at $(0, 0)$. Let us investigate the topology of $Z(y^2 - x^3)$. The integral ring $k[x, y]/(y^2 - x^3) = k[x][\sqrt[3]{x^2}]$ has Krull dimension one and thus its nonzero prime ideals are all maximal. By Hilbert's Nullstellensatz, and irreducible decomposition, this means that the closed sets of $Z(y^2 - x^3)$ consist of finitely many points. This immediately causes f to be a homeomorphism as both spaces are equipped with the cofinite topology. f is not, however, an isomorphism as:

$$f^*: A(Z(y^2 - x^3)) \rightarrow k[t], \begin{cases} x \mapsto t^2 \\ y \mapsto t^3 \end{cases}$$

is not a surjective map.

The second one follows analogously.

- (ii) F is clearly a morphism as it is given by polynomial maps. F is also surjective as for any $a \in \mathbb{A}_k^1$, we have roots for the polynomial $x^p - a$, furthermore it is also injective since if $t^p = s^p$ we find $t^p - s^p = 0 \Leftrightarrow (t - s)^p = 0 \Leftrightarrow t = s$. As $\mathbb{A}_k^1$ carries cofinite topology, this makes f a homeomorphism. f is not however an isomorphism as the pullback

$$f^*: k[t] \rightarrow k[t], t \mapsto t^p$$

is not a surjective map. Lastly let us compute the fixed points: we have

$$t^p = t \Leftrightarrow t(t^{p-1} - 1) = 0$$

this polynomial has at most p roots, and using Fermat's little theorem the roots are precisely $\mathbb{F}_p \subseteq k$. □

Exercise 4.4 (Product of affine varieties). Let $X \subset \mathbb{A}_k^n$ and $Y \subset \mathbb{A}_k^m$ be affine varieties. Show that

$$X \times Y \subset \mathbb{A}_k^{n+m}$$

is a variety and that

$$A(X \times Y) = A(X) \otimes_k A(Y).$$

. Verify that $X \times Y$ with its natural projections to X and Y satisfies the universal property of the product in the category of varieties.

Solution. let $\pi_1: \mathbb{A}^{n+m} \rightarrow \mathbb{A}^n$ and $\pi_2: \mathbb{A}^{n+m} \rightarrow \mathbb{A}^m$ denote the canonical projection maps, with corresponding (injective) pullback maps π_1^*, π_2^* . Suppose furthermore, that $X = Z(I_X)$ and $Y = Z(I_Y)$ for prime ideals I_X, I_Y . then we claim

$$X \times Y = Z(\pi_1^*(I_Y) \cup \pi_2^*(I_X))$$

Indeed a point lies in $X \times Y$ if and only if the projections lie in X and Y respectively which is equivalent to saying that the original point is a zero for all polynomials in $\pi_1^*(I_Y) \cup \pi_2^*(I_X)$. Next, we show irreducibility. Suppose $X \times Y = Z_1 \cup Z_2$ were reducible. for a fixed $y \in Y$ consider the embedding

$$\varphi_y: X \rightarrow X \times Y, x \mapsto (x, y).$$

This map is clearly Zariski continuous, since the φ_y -pullback of a polynomial $f \in A(X \times Y)$ is given of $f(\cdot, y)$ which is a polynomial on X . In particular, the the set

$$\varphi_y^{-1}(Z_1) = \{x \in X \mid (x, y) \in Z_1\}$$

is a closed set. Taking the intersection of these sets over all $y \in Y$, we obtain that the set

$$\{x \in X \mid \{x\} \times Y \subseteq Z_1\} =: C_1$$

is a closed set in X . In the exact same way the set

$$\{x \in X \mid \{x\} \times Y \subseteq Z_2\} =: C_2$$

is also closed. We claim that $X = C_1 \cup C_2$. Indeed, suppose $x \in X$. This x gives rise to a union:

$$Y = \{y \in Y \mid (x, y) \in Z_1\} \cup \{y \in Y \mid (x, y) \in Z_2\}$$

where both sets are closed for the same reasons above. The irreducibility of Y implies that Y equal to one of the two sets and thus x is either in C_1 or in C_2 . Since now $X = C_1 \cup C_2$ we find that $X = C_1$ or $X = C_2$ implies $X \times Y = Z_1$ or $X \times Y = Z_2$.

Finally let us compute the coordinate ring. A small argument using the universal property of the tensor product ensures we have an isomorphism

$$A(X) \otimes_k A(Y) \cong \frac{k[x_1, \dots, x_n, y_1, \dots, y_m]}{(I_X + I_Y)}.$$

On the other hand $A(X \times Y)$ is given by

$$A(X \times Y) = \frac{k[x_1, \dots, x_n, y_1, \dots, y_m]}{\sqrt{(I_X + I_Y)}}.$$

Thus, it suffice to show that $A(X) \otimes_k A(Y)$ is reduced. In fact, we can show it is an integral domain. Suppose we have a zero divisor

$$\left(\sum_i a_i \otimes b_i\right) \left(\sum_j c_j \otimes d_j\right) = 0$$

By rearranging terms we may assume that the a_i are k -linearly independent, and similarly the c_j . Let us assume both factors are nonzero and so let us assume $b_1 \neq 0 \neq d_1$. By Hilberts Nullstellensatz there exist a maximal ideal $\mathfrak{m}$ in $A(Y)$ which does not contain b_1 and c_1 . Consider now the evaluation

$$A(X) \otimes_k A(Y) \rightarrow A(X) \otimes_k (A(Y)/\mathfrak{m}).$$

The above relation becomes

$$\left(\sum_i a_i \otimes \bar{b}_i\right) \left(\sum_j c_j \otimes \bar{d}_j\right) = 0$$

Now there is a canonical isomorphism

$$A(X) \otimes_k (A(Y)/\mathfrak{m}) \cong A(X) \otimes_k k \cong A(X),$$

and as the latter is an integral domain one of the two factors is zero and we may assume

$$\left(\sum_i a_i \otimes \bar{b}_i\right) = 0.$$

under the isomorphism $A(Y)/\mathfrak{m} \cong k$ let $b_i \equiv \lambda_i \in k$. then we have produced a relation

$$\sum_i a_i \lambda_i = 0, \lambda_1 \neq 0,$$

which contradicts that we have chosen the a_i to be k -linearly independent.

Lastly the fact that $X \times Y$ has the universal property of products is a purely formal consequence due to applying the equivalence of categories to finitely generated integral k -algebras and seeing the $A(X) \otimes_k A(Y)$ has the universal property of the tensor product. $\square$

Exercise 4.5. Find an irreducible polynomial $f \in \mathbb{R}[x, y]$ such that $Z(f) \subset \mathbb{A}_{\mathbb{R}}^2$ is not an irreducible topological space (with its Zariski topology). Find an irreducible polynomial $f \in \mathbb{R}[x, y]$ such that $Z(f) \subset \mathbb{A}_{\mathbb{R}}^2$ is irreducible for the Zariski topology but not for the classical topology.

Solution. For the first one consider $f = x^2(y - 1)^2 + y^2(x - 1)^2$ this polynomial is irreducible over $\mathbb{R}$ (a little annoying to check but doable). However. the zero locus is not even connected. For the second one let $y^2 - x(x - 1)(x + 1)$. this equation is clearly irreducible as $x(x - 1)(x + 1)$ is not a square in $k[x]$. Thus, as $k[x, y]$ is a UFD, (f) is a prime ideal and $Z(f)$ is Zariski irreducible. It is not real irreducible however.

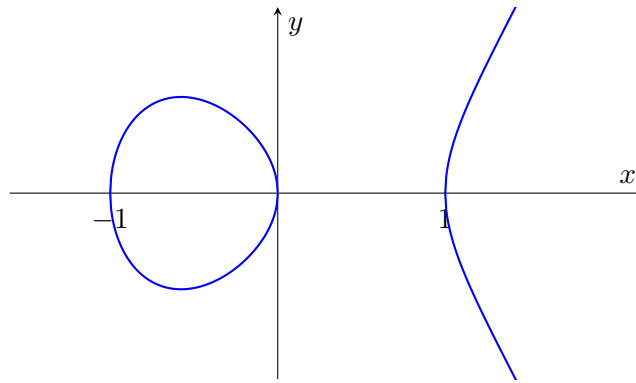


Figure 1: The real locus of the irreducible elliptic curve $y^2 = x^3 - x$, which has two connected components over $\mathbb{R}$.

$\square$

§5 Lecture 5

In the last lecture we introduced our first notion of morphism between algebraic sets, namely regular maps. Under this notion of morphism, we have the strongest form of bridge between the worlds of geometry and algebra: an equivalence of categories. we will now introduce a broader class of morphisms. The idea is to set up a theory for the studying of fractions of polynomials instead of just polynomials.

Suppose X is an affine variety over k , then $A(X)$ is an integral domain, and we can form its field of fractions $k(X) := \text{Frac}(A(X))$. An element in this field of fractions is represented by a term of the form $\frac{f}{g}$, $f, g \in A(X)$, $g \neq 0$. This fraction does not define a valid map to k for all points in X as the denominator may vanish. However, we do have a well defined map

$$\frac{f}{g}: X \supseteq D(g) \rightarrow k, P \mapsto \frac{f(P)}{g(P)}.$$

We call these functions **rational functions**. A rational function $h \in k(X)$ is called **regular** at a point $x \in X$, if there exists $f, g \in A(X)$, $g(x) \neq 0$, such that $h = \frac{f}{g}$.

Remark 5.1. Note again that there is a slight subtle difference between rational functions and elements of the function field $k(X)$. The first is the ring of functions taking in values in k represented by fractions. We will blur this distinction from now on.

Remark 5.2. One may also take a different point of view on rational functions: we may view them as equivalence classes $[U, \varphi]$, where U is a nonempty open, and φ is a rational function defined over all of U .

Definition 5.3. let X be an affine variety and $h \in k(X)$ be a rational function. Then we define the domain of h as

$$\text{dom}(h) := \{x \in X \mid h \text{ is regular at } x\}.$$

This set is always nonempty. It is also open as it is given by

$$X \setminus V(I), \quad I = \{f \in A(X) \mid fh \in A(X)\}.$$

Definition 5.4. The **local ring** of an affine variety X at a point $p \in X$, denoted $\mathcal{O}_{X,p}$, is defined as

$$\mathcal{O}_{X,p} = \{h \in k(X) \mid p \in \text{dom}(h)\}$$

We have an inclusion

$$A(X) \hookrightarrow \bigcap_{p \in A(X)} \mathcal{O}_{X,p} \hookrightarrow k(X).$$

The next lemma says that the first map is surjective as well.

Lemma 5.5

Suppose X is an affine variety and $h \in k(X)$ a rational function which is regular at every point in X . Then $h \in A(X)$.

Proof. h is regular at $p \in X$ if and only if there exists $f, g \in A(X)$ such that $g(p) \neq 0$ and $h = \frac{f}{g}$. In particular a point p is a regular point of h if and only if it is in

$$X \setminus V(I), \quad I = \{f \in A(X) \mid fh \in A(X)\}.$$

if h is regular everywhere, then $V(I) = \emptyset$. Now, the above I is clearly an ideal and thus the Nullstellensatz implies $1 \in I$. In other words $h = h \cdot 1 \in A(X)$. $\square$

The above lemma says that rational maps are a natural extension of our notion of morphism. Consider for example the map

$$\mathbb{A}^1 \setminus \{0\} \rightarrow \mathbb{A}^1, t \mapsto \frac{1}{t},$$

it is a rational function, which is regular everywhere on this domain. But it does not come from a polynomial map on $\mathbb{A}^1$.

The term local ring insinuates that the above ring is obtained by localizing $A(X)$. This is true as seen below

Lemma 5.6

There is a canonical isomorphism $\mathcal{O}_{X,p} \cong A(X)_{\mathfrak{m}_p}$, where $\mathfrak{m}_p$ is the maximal ideal corresponding to p .

Proof. By the universal property of localisation we have a commutative diagram

$$\begin{array}{ccccc} A(X) & \xrightarrow{i} & \mathcal{O}_{X,p} & \xrightarrow{i} & k(X) \\ \downarrow & \nearrow \tilde{i} & & \nearrow i & \\ A(X)_{\mathfrak{m}_p} & & & & \end{array}$$

which shows that $\tilde{i}$ is injective as the inclusion of $A(X)_{\mathfrak{m}_p}$ into $k(X)$ is injective. It is also surjective as if $f/g \in \mathcal{O}_{X,p}$ then $f/g \in A(X)_{\mathfrak{m}_p}$ as well. $\square$

We are now ready to broaden the objects in our category:

Definition 5.7. A quasi affine variety U is a open subset $U \subseteq X$, where $X \subseteq \mathbb{A}_k^n$ is an affine variety.

In this relative setting, we define

$$\mathcal{O}_X(U) = \{h \in k(X) \mid h \text{ is regular for all } p \in U\} = \bigcap_{p \in U} \mathcal{O}_{X,p}.$$

In general these rings can be hard to calculate but in some simple cases we can still make the computations:

Example 5.8

Let $f \in A(X)$ be nonzero and consider the open set $D(f)$, where $D(f) = X \setminus V(f)$. Then $\mathcal{O}_X(D(f)) \cong A(X)_f$.

Proof. As before, the universal property of localization yields a commutative diagram:

$$\begin{array}{ccccc} A(X) & \xrightarrow{i} & \mathcal{O}_X(D(f)) & \xrightarrow{i} & k(X) \\ \downarrow & \nearrow \tilde{i} & & \nearrow i & \\ A(X)_f & & & & \end{array}$$

which shows that $\tilde{i}$ is injective. We need to show it is surjective as well. Indeed, suppose $h \in k(X)$ with $\text{dom}(h) \supseteq D(f)$. Then we have for the ideal of denominators I_h of h (defined as in lemma 7.5)

$$X \setminus V(I_h) \supseteq X \setminus V(f) \Leftrightarrow V(I_h) \subseteq V(f).$$

Then applying the Nullstellensatz we have $f \in \sqrt{I_h}$. choosing n large enough f^n is a denominator of h and $h = \frac{g}{f^n} \in A(X)_f$. $\square$

Definition 5.9. A rational map $\varphi: X \dashrightarrow Y$ between affine varieties is a map

$$\varphi: U \rightarrow Y,$$

where $U \subseteq X$ is a nonempty open and where φ is of the form

$$\varphi(p) = (h_1(p), \dots, h_m(p)), \forall p \in U$$

where h_i are functions on U .

Note that shrinking U , does not change the h_i and so while for a rational function the U is not unique, the h_i are indeed unique. With this in mind, we will typically consider rational functions to be equivalence classes $[U, \varphi]$ of maps of the above form.

Remark 5.10. We can not a priori compose two rational maps since domains and codomains may not coincide. The right frame work to talk about compositions is that of dominating rational maps which we will introduce next lecture..

§6 Lecture 6

In the last lecture, we saw that rational maps cannot, in general, be composed. The appropriate way to address this problem is to introduce dominant rational maps.

Definition 6.1. A rational map is called **dominant** if it has dense image.

Remark 6.2. If $\varphi: X \dashrightarrow Y$ is a dominant rational map and $\psi: Y \dashrightarrow Z$ is any rational map, the domain of ψ is open and hence dense. Thus, it necessarily intersects $\varphi(X)$, and so the composition is well defined (although possibly only on a smaller open set).

As in the case of regular maps, rational maps induce morphisms on certain rings of functions:

Definition 6.3. Let $\varphi: X \dashrightarrow Y$ be a dominant rational map. We define the pullback of φ to be the map

$$\varphi^*: k(Y) \rightarrow k(X), f \mapsto \varphi^*(f) := f \circ \varphi.$$

We should take a moment to ensure that this makes sense. Indeed, φ is represented on an open subset of X by a regular map on U :

$$\varphi: U \rightarrow Y \subseteq \mathbb{A}_k^m, p \mapsto (h_1(p), \dots, h_m(p)).$$

the h_i are fractions of polynomials. Furthermore, a $f \in k(Y)$ is also a fraction of polynomials. Then, $f(h_1, \dots, h_m)$ is also a fraction of polynomials.

Since φ is dominant, this map is injective. and in this way, $k(X)$ becomes a field extension of $k(Y)$.

Definition 6.4. A dominant rational map is called **birational** if it has an inverse. That is to say there is a dominant rational map $\psi: Y \dashrightarrow X$, such that

$$\begin{aligned} (\psi \circ \varphi : X \dashrightarrow Y) &\sim Id_X \\ (\varphi \circ \psi : Y \dashrightarrow X) &\sim Id_Y \end{aligned}$$

Two affine varieties are called **birationally equivalent** if there exists a birational map between them.

Lemma 6.5

Let X and Y be affine varieties, and $\varphi: X \dashrightarrow Y$ a dominant rational map. then the following are equivalent:

- $\varphi: X \dashrightarrow Y$ is a birational.
- $\varphi^*: k(Y) \rightarrow k(X)$ is a field isomorphism.

Proof. to prove the claim we show that the pullback operator induces a bijection:

$$\{\text{dominant rational maps } X \dashrightarrow Y\} \leftrightarrow \{\text{field homomorphisms } k(Y) \rightarrow k(X)\}$$

Indeed, suppose $Y \subseteq \mathbb{A}_k^m$ and $A(Y) = k[x_1, \dots, x_m]/I$. Let $\pi: k[x_1, \dots, x_m] \rightarrow A(Y)$ denote the canonical projection and let $\beta: k(Y) \rightarrow k(X)$ be any field homomorphism. Then we have a commutative diagram

$$\begin{array}{ccc} k[x_1, \dots, x_m] & \xrightarrow{\pi} & A(Y) \\ & \searrow & \downarrow \\ & & k(X) \\ & \nearrow \beta & \uparrow \\ & & k(Y) \end{array}$$

In this way, by a slight abuse of notation, we will also denote the morphism from $k[x_1, \dots, x_m] \rightarrow k(X)$, by β . This allows us to define a rational map

$$\tilde{\beta}: X \dashrightarrow \mathbb{A}_k^m, p \mapsto (\beta(x_1)(p), \dots, \beta(x_m)(p)),$$

which is defined on the open set $\bigcap_i \text{dom}(\beta(x_m))$. Now suppose $f \in k[x_1, \dots, x_m]$ is a function that vanishes in the image of $\tilde{\beta}$. Then $0 = f(\beta(x_1), \dots, \beta(x_m)) = \beta(f)$. This occurs if and only if $f \in I$. Hence $\tilde{\beta}$ actually maps to Y and has dense image there. Finally

$$(\tilde{\beta})^*(x_i) = \beta(x_i) \Rightarrow (\tilde{\beta})^* = \beta.$$

□

Let us now return to the setting of quasi affines.

Definition 6.6. Let U , and V be quasi affine varieties. a **regular map** $\varphi: U \rightarrow V$ is a map whose coordinates are given by regular functions on U . We say φ is an isomorphism if it has an inverse which is also a regular map.

Definition 6.7. A quasi affine variety is called affine, if it is isomorphic to an affine variety in the above sense

Example 6.8

Consider $X = \mathbb{A}^1 \setminus \{0\} = D(x)$ and $Y = V(xy - 1) \subseteq \mathbb{A}^2$. A priori, X is only quasi affine but the regular map

$$\varphi: X \rightarrow Y, t \mapsto (t, \frac{1}{t})$$

defines an isomorphism, whose inverse is projection. In the exercises we will show however, that $\mathbb{A}^2 \setminus \{(0,0)\}$ is not affine.

For quasi affine varieties we do not have a result quite as nice as in Proposition 5.4. It turns out that the crucial assumption is that the target is affine:

Proposition 6.9

Let X be a quasi affine variety and Y an affine variety. Then there exists a bijection

$$\{\text{regular maps } \varphi: X \rightarrow Y\} \leftrightarrow \{\text{ring homomorphisms } A(Y) \rightarrow \mathcal{O}(X)\},$$

which is induced by the pullback map

$$\varphi \mapsto \varphi^*: A(Y) \rightarrow \mathcal{O}(X), g \mapsto g \circ \varphi$$

Proof. We may assume that $Y \subseteq \mathbb{A}^m$, and so we have a projection

$$\pi: k[x_1, \dots, x_m] \rightarrow A(Y).$$

Suppose now we have any ring homomorphism $\beta: A(Y) \rightarrow \mathcal{O}(X)$. We have a commutative diagram

$$\begin{array}{ccc} k[x_1, \dots, x_m] & \xrightarrow{\pi} & A(Y) \\ & \searrow & \downarrow \beta \\ & & \mathcal{O}(X) \end{array}$$

Let us also denote the morphism $\beta \circ \pi$, by β . In this way, we have defined a regular map

$$\tilde{\beta}: X \rightarrow \mathbb{A}^m, p \mapsto (\beta(x_1)(p), \dots, \beta(x_m)(p))$$

This map has image in Y , since $I(Y) \subseteq \ker \beta$ and so any function vanishing on Y must vanish on $\beta(X)$. In this way we have constructed a unique regular map which has pullback β . $\square$

Example 6.10

Let X be an affine variety, and $f \in A(X)$. Then the quasi affine variety $D(f)$, is affine.

Proof. Indeed, let $X = V(I_X) \subseteq \mathbb{A}^n$. Then consider the affine variety

$$Y = V(I_X, x_{n+1}f - 1).$$

We then have an isomorphism

$$\varphi: D(f) \rightarrow Y, p \mapsto (p, \frac{1}{f(p)})$$

$\square$

In fact, this example shows that for any quasi affine variety X and any $p \in U$, p has an open neighborhood which is affine. Indeed if $X \subseteq \bar{X} \subseteq \mathbb{A}^n$, then $X = \bar{X} \setminus V(I)$ for some ideal in $A(\bar{X})$ and we may take a function also not in $V(I)$ which does not vanish at P to receive an open neighborhood $D(f) \subseteq X$.

From this discussion, we have a rather obvious characterization of birational equivalences for affine varieties:

Proposition 6.11

Let X and Y be two affine varieties. Then the following are equivalent

- 1 X and Y are birationally equivalent
- 2 there exists open subsets $U \subseteq X$ and $V \subseteq Y$ such that U and V are isomorphic as quasi affine varieties.

Tasks for Week 3**Algebraic Geometry I**

Exercise sheet 3

Throughout, k denotes an algebraically closed field.

Exercise 6.1. Consider the quasi-affine variety $X := \mathbb{A}_k^2 \setminus \{(0, 0)\}$. Show that the inclusion

$$\iota: X \hookrightarrow \mathbb{A}_k^2$$

induces an isomorphism

$$\iota^*: A(\mathbb{A}_k^2) \rightarrow \mathcal{O}(X).$$

Deduce that X is not affine.

Proof. The pullback of the inclusion morphism is given by the map

$$i^*: k[x, y] \rightarrow \mathcal{O}(X), \quad g \mapsto g|_X.$$

First we see that this map is injective as if g is a polynomial which vanishes outside of 0, then it vanishes everywhere. For surjectivity suppose $h \in k(x, y)$ is regular on X . consider the ideal

$$\mathfrak{a} = \{f \in k[x, y] \mid fh \in k[x, y]\}.$$

By assumption $Z(\mathfrak{a}) \subseteq (0, 0)$. We claim $Z(\mathfrak{a}) = \emptyset$ which would, by The Nullstellensatz, imply $1 \in \mathfrak{a}$ and $f \in k[x, y]$. If $Z(\mathfrak{a}) = \{(0, 0)\}$ then again by The Nullstellensatz $\sqrt{\mathfrak{a}} = (x, y)$. In particular, on X , we have representations

$$h(x, y) = \frac{A}{x^n} = \frac{B}{y^m} \quad \text{with } A, B \in K[x, y]$$

if $n = 0$ we receive $y = 0$ we receive $h \cdot 1 \in k[x, y]$ which is a contradiction. Assume $n > 0$ By simplifying fractions we may assume that y does not divide B and x doesn't divide A . Now the set where the above equality holds is a closed one and it contains the open set U , thus this equality holds everywhere. multiplying out denominators and using the fact that $k[x, y]$ is a UFD we arrive at a contradiction. $\square$

Exercise 6.2. Consider the affine variety $C \subset \mathbb{A}_k^2$ which is the zero set of the polynomial

$$y^2 = x(x - 1)(x - 3).$$

Consider the point $P = (-1, 2)$. Write down a regular function on the quasi-affine variety $C \setminus \{P\}$ which is not a regular function on C .

Remark 6.3. The way the problem was posed P was not on the curve, so I have interpreted the problem for the curve $y^2 = x(x-1)(x+3)$

Solution. We claim that $h = \frac{y+2}{x+1}$ is such a function. To prove this, we need to show that h is regular in $C \setminus \{P\}$, but it is not regular in P . It is immediately apparent that h is regular on $C \setminus V_C(x+1)$. Furthermore, one easily verifies

$$V_C(x+1) = \{(-1, 2), (-1, -2)\},$$

Thus, we need only show that h is regular at $(-1, -2)$. Indeed using the arithmetic of the ring $A(C)$ we obtain

$$\begin{aligned} \frac{y+2}{x+1} &= \frac{y^2-4}{(x+1)(y-2)} \\ &= \frac{x(x-1)(x+3)-4}{(x+1)(y-2)} \\ &= \frac{x^2+2x-3x-4}{y-2} \end{aligned}$$

where we see that the last expression is clearly regular at $(-1, -2)$.

It remains to show that h is not regular on all of C . If not, we would have

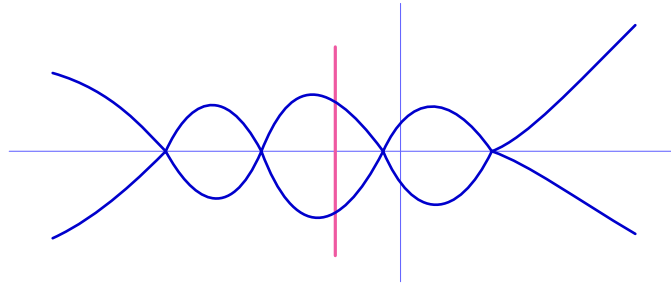
$$\frac{y+2}{x+1} = \bar{A} \in \mathcal{O}(C),$$

Thus there exists a polynomial $A \in k[x, y]$ such that

$$(y+2) \equiv A(x+1) \pmod{I(C)}.$$

Such an equivalence is, of course, impossible, as the right hand side vanishes for $(-1, 2) \in C$ but the left does not. $\square$

Remark 6.4. The way I came up with h is by taking the denominator to be a function which intersects C in two places and then making the numerator such that it also vanishes at the unwanted point. This is sketched below



Exercise 6.5. Let $P \in \mathbb{A}_k^2$ be the origin. Consider the varieties

$$C_1 = Z(y^2 - x^3 + x) \quad \text{and} \quad C_2 = Z(y^2 - x^3 + x^2).$$

Are the local rings $\mathcal{O}_{C_1, P}$ and $\mathcal{O}_{C_2, P}$ isomorphic?

Solution. We claim that the two local rings are not isomorphic. To show this let $\mathfrak{m}_1$ and $\mathfrak{m}_2$ denote the maximal ideals in the two respective local rings. The first thing to note, is that

$$\mathcal{O}_{C_1, P}/\mathfrak{m}_1 \cong k \cong \mathcal{O}_{C_2, P}/\mathfrak{m}_2,$$

via the map $\frac{F}{G} \mapsto \frac{F(P)}{G(P)} \in k$. In particular, we may regard $\mathfrak{m}_1/\mathfrak{m}_1^2$ and $\mathfrak{m}_2/\mathfrak{m}_2^2$ as k vector spaces. To prove the claim, it suffice to show that they have different dimensions. For $\mathfrak{m}_1/\mathfrak{m}_1^2$ we have

$$\mathfrak{m}_1/\mathfrak{m}_1^2 = \langle \bar{x}, \bar{y} \rangle_k.$$

Now The arithmetic of C ensures $0 \equiv y^2 \equiv x(1-x) \equiv x \pmod{\mathfrak{m}_1^2}$ and so $x \in \mathfrak{m}_1^2$. In particular $\mathfrak{m}_1/\mathfrak{m}_1^2 = \langle \bar{y} \rangle_k$. This ensures

$$\dim_k(\mathfrak{m}_1/\mathfrak{m}_1^2) \leq 1.$$

In fact $\dim_k(\mathfrak{m}_1/\mathfrak{m}_1^2) = 1$ to one by Nakayama's lemma for local rings: otherwise $\mathfrak{m}_1 = 0$ which is not the case. For C_2 we have a surjective morphism of local rings

$$k[x, y]_{\mathfrak{m}} \twoheadrightarrow \mathcal{O}_{C_2, P}.$$

The kernel of this map is precisely $I(C_2) \cdot k[x, y]_{\mathfrak{m}}$ and so we have an isomorphism of local rings

$$\frac{k[x, y]_{\mathfrak{m}}}{I(C_2) \cdot k[x, y]_{\mathfrak{m}}} \cong \mathcal{O}_{C_2, P}.$$

Hence we can conclude:

$$\mathfrak{m}_2/\mathfrak{m}_2^2 \cong \frac{(\mathfrak{m}/(I(C_2) \cdot k[x, y]_{\mathfrak{m}}))}{(\mathfrak{m}/(I(C_2) \cdot k[x, y]_{\mathfrak{m}}))^2} = \frac{(\mathfrak{m}/(I(C_2) \cdot k[x, y]_{\mathfrak{m}}))}{(\mathfrak{m}^2/(I(C_2) \cdot k[x, y]_{\mathfrak{m}}))} \cong \frac{\mathfrak{m}}{\mathfrak{m}^2 + I(C_2)} = \frac{\mathfrak{m}}{\mathfrak{m}^2}$$

where in the second last line we used the isomorphism theorem and in the last line we used $I(C) \subseteq \mathfrak{m}^2$. To conclude consider the maximal ideal

$$\mathfrak{m} \subseteq k[x, y]_P.$$

The dimension of $\mathfrak{m}/\mathfrak{m}^2$ is then clearly 2 as x, y are linearly independent in $k[x, y]_{\mathfrak{m}} \subseteq k(x, y)$ $\square$

Exercise 6.6. Let $\phi: U \rightarrow V$ be a regular map of quasi-affine varieties over k . Show that ϕ induces a homomorphism

$$\phi_P^*: \mathcal{O}_{V, \phi(P)} \rightarrow \mathcal{O}_{U, P}$$

of k -algebras such that

$$(\phi_P^*)^{-1}(\mathfrak{m}_P) = \mathfrak{m}_{\phi(P)}, \quad \text{for any } P \in U.$$

If ϕ is an isomorphism, then show that ϕ_P^* is an isomorphism for all $P \in U$.

Let X, Y be affine varieties over k . Prove that the following are equivalent:

- (1) X and Y are birational;
- (2) there exist points $P \in X$, $Q \in Y$, and an isomorphism of k -algebras between the local rings $\mathcal{O}_{X, P}$ and $\mathcal{O}_{Y, Q}$.

Solution. Suppose $h \in k(V)$ is a rational function which is regular at $\phi(P)$. Then there exists an open neighborhood $\tilde{U}$ of $\phi(P)$ such that $h = \frac{f}{g}$ for two regular functions f, g on U with g nonzero on $\tilde{U}$. Then the pullback $\phi^*h = \frac{\phi^*f}{\phi^*g}$ is a regular function on $\phi^{-1}(\tilde{U})$. Thus, the pullback maps rational functions which are regular at $\phi(P)$, to rational functions regular at P . Furthermore,

$$(\phi_P^*)^{-1}(\mathfrak{m}_P) = \{f \in \mathcal{O}_{V, \phi(P)} \mid (f \circ \phi)(P) = 0\} = \{f \in \mathcal{O}_{V, \phi(P)} \mid f(\phi(P)) = 0\} = \mathfrak{m}_{\phi(P)}$$

The statement about isomorphisms follows from the universal property of localisations. Lastly if X and Y are affine varieties then we have $k(Y) = \text{Frac}(\mathcal{O}_{Y, P})$ and $k(X) = \text{Frac}(\mathcal{O}_{X, Q})$ for any two points. Thus, if for two points these rings are the same, this implies, that the function fields are isomorphic and thus they are birationally equivalent. Conversely, if X and Y are birational, then there exists an open subset U of X and V of Y such that these quasi affine varieties are isomorphic. In particular if $\psi: U \rightarrow V$ is such an isomorphism, the statement then follows from the first part. $\square$

Exercise 6.7. Let X be an affine variety and let $Y \subset X$ be an irreducible closed subset. The local ring $\mathcal{O}_{X, Y}$ of Y in X is the set of equivalence classes of pairs $[U, \phi]$ where $U \subset X$ is open, $U \cap Y \neq \emptyset$, and ϕ is a regular function on U ; two pairs $[U, \phi], [V, \psi]$, are equivalent if $\phi = \psi$ on $U \cap V$. Show that $\mathcal{O}_{X, Y}$ is a local ring, isomorphic to the localization $A(X)_{\mathfrak{p}_Y}$ where we denote by $\mathfrak{p}_Y$ the prime ideal $I_X(Y)$. Deduce that the residue field of $\mathcal{O}_{X, Y}$ is isomorphic to the function field of Y (recall that the residue field of a local ring R is the quotient $R/\mathfrak{m}$ of R by its unique maximal ideal $\mathfrak{m}$).

Solution. Suppose $[U, \phi] \in \mathcal{O}_{X, Y}$ such that ϕ is not zero on $U \cap Y$. Then the set $D(\phi) \cap U$ is nonempty and $\frac{1}{\phi}$ is a regular function there. This shows that Every element in this ring which is not in the ideal of germs which vanish on Y is invertible. Now there is a natural map $f: A(X) \rightarrow \mathcal{O}_{X, Y}$. By the universal property of localization we receive a map $\tilde{f}$

$$\begin{array}{ccc} A(X) & \xrightarrow{f} & \mathcal{O}_{X, Y} \\ \downarrow & \nearrow \tilde{f} & \\ A(X)_{\mathfrak{p}_Y} & & \end{array}$$

since f is injective, so is $\tilde{f}$, we need to show it is also surjective. Indeed suppose $[U, \phi]$ is any germ in $\mathcal{O}_{X, Y}$. Then the open set $\text{dom}(\phi)$ meets Y . In particular, there is a point $p \in Y$ such that ϕ is regular at p . Near this point $\phi = \frac{f}{g}$ for two regular functions f, g where $g \neq 0$ on an open neighborhood. In particular, $g \notin \mathfrak{p}_Y$ since this would force g to vanish on all of Y . For the last statement we have:

$$\mathcal{O}_{X, Y}/\mathfrak{m} \cong (A(X)_{\mathfrak{p}_Y})/\mathfrak{p}_Y \cong \left(\frac{A(X)}{\mathfrak{p}_Y} \right)_{(0)} = \text{Frac}(A(Y)) = k(Y)$$

$\square$

§7 Lecture 7

The main goal of this lecture is to introduce projective spaces and to be able to talk about them in the language of algebraic geometry. Projective spaces turn out to be the correct replacement for compact spaces in algebraic geometry. Indeed, compactness is not a very useful notion in our setting, as we have seen that all our spaces were quasi

compact, and none of them were Hausdorff. The crucial application of compactness in topology is being able to control images. What we will find is that projective spaces have a similar property: the image of a projective variety will always be closed.

As a set we define **projective n space** to be the set of lines through the origin in $\mathbb{A}^{n+1}$. We would like to place a Zariski topology on this set in order to make sure that the topology captures algebraic information. To do this we review some basic notions from commutative algebra.

Definition 7.1. A Ring S is called a $\mathbb{N}$ graded ring if it admits a decomposition

$$S = \bigoplus_{d \geq 0} S_d,$$

such that $S_d \cdot S_{d'} \subseteq S_{d+d'}$. Notice that $A = S_0$ is itself a ring under these assumptions and so we will also call R a graded A -module.

Remark 7.2. Later we may also consider $\mathbb{Z}$ -gradings rather than $\mathbb{N}$ -gradings.

Example 7.3

The key example for us will be the graded k -module $k[x_0, \dots, x_n]$ which has degree d part:

$$k[x_0, \dots, x_n]_d = \{\text{homogeneous polynomials of degree } d\}.$$

Definition 7.4. Let S be a graded ring and $I \subseteq S$ an ideal. I is called a homogeneous ideal if it is a sum of its homogeneous parts:

$$I = \bigoplus_{d \geq 0} (I \cap S_d)$$

The key reason we want to consider homogeneous ideals is that if F is a homogeneous polynomial of degree d , we have

$$F(\lambda x_0, \dots, \lambda x_n) = \lambda^d F(x_0, \dots, x_n),$$

and therefore, if λ is nonzero we have

$$F(x) = 0 \Leftrightarrow F(\lambda x) = 0.$$

In particular, we may define

$$V_{\mathbb{P}^n}(F) := \{[x_0 : \dots : x_n] \mid F(x_0, \dots, x_n) = 0\}.$$

More generally, if S is any set of homogeneous polynomials, we can define $V_{\mathbb{P}^n}(S)$, as in the affine case. Similarly, for a homogeneous ideal I we define:

$$V_{\mathbb{P}^n}(I) = \{[x_0 : \dots : x_n] \mid F(x_0, \dots, x_n) = 0 \ \forall F \in I, \ F \text{ homogeneous}\}.$$

It is now again simple to verify that if S is a set of homogeneous polynomials, then $V_{\mathbb{P}^n}(S) = V_{\mathbb{P}^n}(I_S)$ where I_S is the (homogeneous) ideal generated by S .

Definition 7.5. An algebraic subset of $\mathbb{P}^n$ is a set of the form $V_{\mathbb{P}^n}(S)$.

Remark 7.6. As $k[x_0, \dots, x_n]$ is a Noetherian ring, algebraic subsets may always be written as $V_{\mathbb{P}^n}(F_1, \dots, F_k)$ for forms F_i .

Lemma 7.7

The algebraic subsets of $\mathbb{P}^n$ define the closed sets of a topology. We call it the Zariski topology on $\mathbb{P}^n$.

Proof. First note that the empty set and the whole space are algebraic subsets as:

$$\emptyset = V_{\mathbb{P}^n}(1) \quad \text{and} \quad \mathbb{P}^n = V_{\mathbb{P}^n}(0),$$

Now suppose $\{V(I_i)\}_{i \in I}$ is a collection of algebraic subsets, where I_i are homogeneous ideals. Then note, that the direct sum

$$\bigoplus_{i \in I} I_i \subseteq k[x_1, \dots, x_n]$$

is again a homogeneous ideal, as it has homogeneous generators (take for a generating set the union of the individual generators for example). Thus

$$V_{\mathbb{P}^n}\left(\bigoplus_{i \in I} I_i\right) \subseteq \mathbb{P}^n,$$

is a well defined closed set. We claim

$$\bigcap_{i \in I} V(I_i) = V\left(\bigoplus_{i \in I} I_i\right).$$

Indeed a point is in the LHS if and only if all homogeneous polynomial in all I_i vanish on it. But as these polynomials generate the ideal on the right hand side, this occurs if and only if, all polynomials in the ideal of the right hand side vanish on it. To show that finite unions of algebraic sets are algebraic works similar, one only needs to note that the products of two homogeneous ideals is again homogeneous. $\square$

We would like to establish a correspondence between homogeneous ideals and closed subsets of $\mathbb{P}^n$, as we had with the Nullstellensatz in the affine case. It turns out that this correspondence is slightly more subtle as certain non trivial homogeneous ideals may still have trivial vanishing locus. To investigate this, we define the annihilator of a subset of $\mathbb{P}^n$.

Definition 7.8. Let $S \subseteq \mathbb{P}^n$ be any subset, then we define

$$I(S) = \{f \in k[x_0, \dots, x_n] \mid f(p) = 0 \ \forall p \in S\}.$$

Note that this is a homogeneous ideal: Suppose

$$F = \sum_{d=0}^m F_d \in I(S),$$

where F_d is homogeneous of degree d . We need to show that $F_d \in I(S)$ as well. If $S = \emptyset$, then this is clear as $I(\emptyset) = (1)$. Now suppose $Q \in S$, say:

$$Q = [x_0 : \dots : x_n] \in S.$$

defining $P = (x_0, \dots, x_n)$ we have for all $\lambda \in k \setminus \{0\}$.

$$0 = F(\lambda P) = \sum_{i=0}^d \lambda^i F_i(P)$$

The right hand side is a polynomial in λ , with infinitely many zeros. thus it is the zero polynomial and all its coefficients are zero:

$$F_d(P) = 0 \quad \forall d = 0, \dots, m.$$

The operator $I(\cdot)$ has similar properties as the affine version:

Proposition 7.9

$I(\cdot)$ has the the following properties:

1. If $X \subseteq Y$, then $I(Y) \subseteq I(X)$
2. $I(X \cup Y) = I(X) \cap I(Y)$
3. $I(X)$ is a radical ideal
4. $V(I(X)) = \bar{X}$.

Proof. The proof works exactly as in Proposition 2.4. □

Using this we have set up maps between algebraic subsets of $\mathbb{P}^n$ and radical homogeneous ideals of $k[x_0, \dots, x_n]$. However the following observation shows that this is not a bijection. Consider the radical hompogeneous ideal $S_0 = (x_0, \dots, x_n)$. Its zero locust is the empty set, so

$$V(S_0) = \emptyset = V(1).$$

More generally, for a graded ring $S = \bigoplus_{d \geq 0} S_d$, we call the ideal $S_+ = \bigoplus_{d > 0} S_d$ the **irrelevant ideal**. As it turns out, other than this slight slip up, the Nullstellensatz from the affine case goes through in the projective case as well. To see this let us first prove the following lemma.

Lemma 7.10

Consider the graded ring $S = k[x_0, \dots, x_n]$ Let $\mathfrak{a} \subseteq S$ be a homogeneous ideal, then the following are equivalent:

- $V_{\mathbb{P}^n}(\mathfrak{a}) = \emptyset$
- $S_+ \subseteq \sqrt{\mathfrak{a}}$
- there exists a $d \geq 0$ with $S_d \subseteq \mathfrak{a}$.

Proof. Since $\mathfrak{a}$ is a homogeneous ideal we have:

$$P \in V_{\mathbb{A}^{n+1}}(\mathfrak{a}) \Rightarrow \lambda P \in V_{\mathbb{A}^{n+1}}(\mathfrak{a}) \quad \forall \lambda \in k.$$

Hence the affine Nullstellensatz implies:

$$V_{\mathbb{P}^n}(\mathfrak{a}) = \emptyset \Leftrightarrow V_{\mathbb{A}^{n+1}}(\mathfrak{a}) = \begin{cases} \emptyset \text{ or} \\ 0 \in \mathbb{A}^{n+1} \end{cases} \Leftrightarrow \sqrt{\mathfrak{a}} = \begin{cases} (1) \text{ or} \\ (x_0, \dots, x_n) \end{cases}$$

And so we have shown that one and two are equivalent. for the third, this follows since $S_d = (S_1)^d$ for $d > 0$. $\square$

We obtain the homogeneous Nullstellensatz, also sometimes called the projective Nullstellensatz.

Theorem 7.11 (The homogeneous Nullstellensatz)

If $\mathfrak{a} \subseteq k[x_0, \dots, x_n]$ is a homogeneous ideal with $V_{\mathbb{P}^n}(\mathfrak{a}) \neq \emptyset$, then

$$I(V_{\mathbb{P}^n}(\mathfrak{a})) = \sqrt{\mathfrak{a}}.$$

Proof. We can notice

$$V_{\mathbb{A}^n}(\mathfrak{a}) = \{(x_0, \dots, x_n) \neq 0 \mid [x_0, \dots, x_n] \in V_{\mathbb{P}^n}(\mathfrak{a})\} \cup \{0\}.$$

Denote the first set in the above union as S . Since $V_{\mathbb{P}^n}(\mathfrak{a})$ is not empty, S is dense in $V_{\mathbb{A}^n}(\mathfrak{a})$. Thus we obtain from the affine Nullstellensatz

$$\begin{aligned} \sqrt{\mathfrak{a}} &= I(V_{\mathbb{A}^n}(\mathfrak{a})) \\ &= \{F \in k[x_0, \dots, x_n] \mid F(P) = 0 \ \forall P \in V_{\mathbb{A}^n}(\mathfrak{a})\} \\ &= \{F \in k[x_0, \dots, x_n] \mid F(P) = 0 \ \forall P \in S\} \\ &= I(V_{\mathbb{P}^n}(\mathfrak{a})). \end{aligned}$$

$\square$

Combining Lemma 7.10 and Theorem 7.11 we obtain

Corollary 7.12

There is a one to one, bijection

$$\{\text{radical homogeneous ideals, not containing } S_+\} \leftrightarrow \{\text{nonempty closed subsets of } \mathbb{P}^n\},$$

which is given by the operators $I(\cdot)$ and $V_{\mathbb{P}^n}(\cdot)$.

Much like in differential geometry, where we investigate manifolds via charts, we would like to be able to view projective space as locally affine. There is a canonical choice of charts which we will reference often:

Example 7.13

Consider the open set

$$U_0 = \mathbb{P}^n \setminus V_{\mathbb{P}^n}(x_0).$$

This set is homeomorphic to $\mathbb{A}^n$ via the maps:

$$\begin{aligned}\varphi : U_0 &\rightarrow \mathbb{A}^n, [x_0 : \cdots : x_n] \mapsto \left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0} \right) \\ \psi : \mathbb{A}^n &\rightarrow U_0, (t_1, \dots, t_n) \mapsto [1 : t_1 : \cdots : t_n]\end{aligned}$$

These maps clearly define inverses to each other. To see that they are continuous, notice:

$$\psi^{-1}(V_{\mathbb{P}^n}(\mathfrak{a}) \cap U_0) = V(\mathfrak{a}_*),$$

where

$$\mathfrak{a}_* = \{F(1, x_1, \dots, x_n) \mid F \in \mathfrak{a}\}.$$

Similarly,

$$\psi(V(\mathfrak{a})) = V_{\mathbb{P}^n}(\mathfrak{a}^*) \cap U_0,$$

where $\mathfrak{a}^*$ is obtained from $\mathfrak{a}$ by homogenizing its elements.

Similarly, we can define $U_1, \dots, U_n$ in such a way that we can cover $\mathbb{P}^n$ with affine graphs $n + 1$. These are called the **standard charts** of $\mathbb{P}^n$.

§8 Lecture 8

In this lecture we continue to investigate the properties of projective space. It shares many properties of affine space, as we shall see.

Lemma 8.1

$\mathbb{P}^n$ is an irreducible space.

Proof. Suppose we have a decomposition $\mathbb{P}^n = Z_1 \cup Z_2$. and consider the canonical charts

$$\mathbb{A}^n \simeq U_0 \hookrightarrow \mathbb{P}^n$$

The decomposition of $\mathbb{P}^n$ yields a decomposition

$$\mathbb{A}^n = (Z_1 \cap \mathbb{A}^n) \cup (Z_2 \cap \mathbb{A}^n).$$

By the irreducibility of $\mathbb{A}^n$, we can assume wlog, $Z_1 \supseteq U_0$. To finish the proof, we only need to show that U_0 is dense in $\mathbb{P}^n$. Indeed, suppose F is a homogeneous polynomial such that

$$f_0 = F(1, x_1, \dots, x_n) = 0.$$

Then for each i , we have that the function

$$f_i = F(x_0, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_n)$$

vanishes on the open set $U_0 \cap U_i$. and thus everywhere. Hence, F vanishes everywhere as $\mathbb{P}^n$ is covered by these open sets. $\square$

Corollary 8.2

Every closed subset of $\mathbb{P}^n$ is a unique decomposition into irreducible spaces.

Definition 8.3. A **projective variety** is an irreducible closed subset of $\mathbb{P}^n$. A **quasi projective variety** is a nonempty open subset of a projective variety

Next we would like to introduce regular functions on projective space. Clearly it is not possible to just take homogeneous polynomials as in the affine case, as these do not define well defined functions on projective space. However, if $F, G \in k[x_0, \dots, x_n]$ are homogeneous forms of the same degree d , then and $P \in \mathbb{A}^{n+1}$ is a point such that $G(P) \neq 0$, then we have

$$\frac{F(P)}{G(P)} = \frac{F(\lambda P)}{G(\lambda P)},$$

for any nonzero $\lambda \in k$.

Definition 8.4. Let $U \subseteq \mathbb{P}^n$ be a quasi projective variety. A function $f: U \rightarrow k$ is called regular at a point $p \in U$, if there exists a neighborhood $W \subseteq U$ of p , and homogeneous forms $F, G \in k[x_0, \dots, x_n]$ of equal degree, such that G is nonzero on W and

$$f(q) = \frac{F(q)}{G(q)} \quad \forall q \in W.$$

A function is called regular on U if it is regular at every point in U . Denote

$$\mathcal{O}(U) = \{f: U \rightarrow k, f \text{ regular on } U\}$$

Remark 8.5. We can view $\mathbb{A}^n \cong U_0 \subseteq \mathbb{P}^n$ both as a quasi projective variety and also a affine variety. For the two different viewpoints, we have slightly different definitions of regular functions. However, they agree up to the isomorphism: If

$$F \in k[t_1, \dots, t_n]$$

is a regular function of the affine variety $\mathbb{A}^n$, then F agrees with the function

$$G = F\left(\frac{x_0}{x_0}, \frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

, which is a quotient of homogenization of F and x_0^d , where $d = \deg F$. Conversely, If $p \in U_0$ is a point and f is regular at p in the sense of quasi projective varieties, then we may shrink the neighborhood W to be contained in U_0 . Then, near p , we have:

$$f = \frac{F(x_0, \dots, x_n)}{G(x_0, \dots, x_n)} = \frac{F(1, \frac{x_1}{x_0}, \dots, \frac{x_n}{x_0})}{G(1, \frac{x_1}{x_0}, \dots, \frac{x_n}{x_0})} = \frac{F(t_0, \dots, t_n)}{G(t_0, \dots, t_n)} \in k(\mathbb{A}^n).$$

thus f is regular at p in the affine sense as well.

Note that being regular is a local condition. Thus $f: U \rightarrow k$ is regular on U if and only if $f|_{U \cap U_i}$ is regular for the canonical affine charts U_i . Having defined regular functions, we can also introduce rational functions on projective space.

Definition 8.6. let $U \subseteq \mathbb{P}^n$ be a quasi projective variety. A rational function on U is an equivalence class of functions $[V, \psi]$, where $V \subseteq U$ is a nonempty open subset and $\psi \in \mathcal{O}(U)$. The equivalence is determined by

$$[V, \psi] \sim [V', \psi'] :\Leftrightarrow \psi|_{V \cap V'} = \psi'|_{V \cap V'}.$$

The set of rational functions defines a field, where operations are defined in a pointwise direction. this field is called the function field of U and is denoted $k(U)$.

Remark 8.7. Once again, the way we defined rational functions here is easily seen to be the same as in the affine case, and the two definitions of the function field agree up to isomorphism.

Remark 8.8. Note that for any open subset U' of U , we have $k(U') = k(U)$. Indeed, every function in $k(U)$ is equivalent to one in $k(U')$. In particular, $k(\mathbb{P}^n) = k(U_0) \cong k(t_1, \dots, t_n)$

Let us now denote the right notion of morphisms in the setting of quasi projective varieties

Definition 8.9. Let U and V be two quasi projective varieties. A map

$$\psi: U \rightarrow V$$

is called a morphism if:

- ψ is continuous
- For every open subset $W \subseteq V$, the pullback of functions ψ^* maps regular functions to regular functions:

$$\psi^*(\mathcal{O}(W)) \subseteq \mathcal{O}(\psi^{-1}(W)).$$

A regular map is called an isomorphism if it has a regular inverse

Remark 8.10. Once again, one checks that this agrees with our definition from the affine case.

Proposition 8.11

Let $U \subseteq \mathbb{P}^n$ and $V \subseteq \mathbb{P}^m$ be quasi projective varieties, then a map

$$\varphi: U \rightarrow V,$$

is a morphism if and only if for every $p \in U$, there exist a neighborhood $W' \subseteq U$ of p , and homogeneous polynomials $F_0, \dots, F_m \in k[x_0, \dots, x_n]$ of equal degree, such that $V(F_0, \dots, F_m) \cap W = \emptyset$ and

$$\varphi = [F_0: \dots: F_m] \text{ on } W,$$

Proof. Let first φ be a regular map and let $p \in U$ be any point. The point $\varphi(p)$ is then contained in one of the canonical affine graphs U_i . Wlog let it be contained in U_0 so that $W = \varphi^{-1}(U_0)$ is an open neighborhood of p . By the regularity condition, the elements

$$\varphi^*\left(\frac{x_1}{x_0}\right), \dots, \varphi^*\left(\frac{x_m}{x_0}\right)$$

are regular on W . Thus, after shrinking W even more, we may assume them each to be given by quotients of homogeneous polynomials of equal degree. Let us write

$$\varphi^*\left(\frac{x_j}{x_0}\right) = \frac{F_j}{G_j}$$

if d_i is the the degree of F_i and d is the maximum degree of the F_i we may assume all the F_i to have degree d . Indeed, if $p \in U_k$ then on a small enough set:

$$\frac{F_i}{G_i} = \frac{x_k^{d-d_i} F_i}{x_k^{d-d_i} G_i}.$$

Thus, we have arrived at a representation

$$\varphi = [1 : \frac{F_1}{G_1} : \cdots : \frac{F_m}{G_m}]$$

Multiplying the denominators out, (which now have the same degree), we arrive at the desired representation.

Now let us assume φ is a map which is locally given by homogeneous polynomials of equal degree. First we show such a φ is continuous. Let $W \subseteq V$ be an open and let $p \in \varphi^{-1}(W)$. Then, we may express φ at an open neighborhood W' of p as:

$$\varphi = [H_0 : \dots : H_m].$$

If $W = V \setminus V(G_1, \dots, G_t)$, for homogeneous polynomials G_i . Since $p \in \varphi^{-1}$, we have that

$$p \in U \setminus V(G_1(H_1, \dots, H_m), \dots, G_t(H_1, \dots, H_m)), =: W''$$

where these polynomials are seen to be homogeneous again since the H_i ore of equal degree. In particular, $W' \cap W''$ is an open neighborhood of p , which is mapped to W .

Finally, let $U \subseteq V$ be an open subset and $f \in \mathcal{O}(U)$. Then locally, f is a fraction

$$f = \frac{H(Y_0, \dots, Y_m)}{G(Y_0, \dots, Y_m)},$$

, of homogeneous polynomials of equal degree. for a point $p \in U$, restrict to a small neighborhood such that $\varphi = [F_0 : \dots : F_m]$, where the F_i are homogeneous of equal degree. Then, near p , we have

$$\varphi^* f(q) = \frac{H(F_0, \dots, F_m)}{G(F_0, \dots, F_m)}$$

which is a regular expression of φ^* in p . □

Example 8.12

Let X be a variety, $U \subseteq X$ open, and $Z \subseteq X$ closed. Then both inclusions

$$i: U \hookrightarrow X,$$

and

$$i: Z \hookrightarrow X,$$

are regular as they have the form from the above proposition. The first inclusion is an example of an open immersion and the second a closed immersion. More generally, a regular map $\varphi: Y \rightarrow X$ is called an **open/closed immersion** if it factors through an isomorphism to a map such as above. A standard example is the open immersion

$$\mathbb{A}^n \cong U_0 \hookrightarrow \mathbb{P}^n.$$

Remark 8.13. Every quasiprojective variety admits an open cover by finitely many affine varieties.

To make these abstract definitions more understandable, let us consider one more example.

Example 8.14

Consider the map,

$$\varphi: \mathbb{P}^1 \rightarrow \mathbb{P}^2, [x_0 : x_1] \mapsto [x_0^2 : x_0x_1 : x_1^2].$$

It is a regular map according to Proposition 8.11. We claim it is an isomorphism to

$$X := \varphi(\mathbb{P}^1) = V(t_1^2 - t_0t_2).$$

Indeed consider the map

$$\psi: X \rightarrow \mathbb{P}^2, [t_0 : t_1 : t_2] \mapsto \begin{cases} [t_0t_1 : t_0t_2] & \text{if } t_0 \neq 0 \\ [t_2t_0 : t_2t_1] & \text{if } t_2 \neq 0. \end{cases}$$

Note first that the sets $\{t_0 \neq 0\}$ and $\{t_2 \neq 0\}$ cover V , since

$$V_{\mathbb{P}}^2(t_1^2 - t_0t_2, t_0, t_2) = \emptyset.$$

Furthermore, the governing equations of X ensure that these two maps agree on intersections. Then, one can easily verify that ψ is an inverse to φ . Thus, by the previous proposition this map is regular. and thus φ is a closed immersion.

Tasks for Week 4

Algebraic Geometry I

Exercise sheet 4

Throughout, k denotes an algebraically closed field.

Exercise 8.1. Show that $\mathcal{O}(\mathbb{P}_k^n) = k$. Deduce that any regular map

$$\mathbb{P}_k^n \rightarrow \mathbb{A}_k^m$$

is constant.

Solution. It is clear that $k \subseteq \mathcal{O} = (\mathbb{P}_k^n)$, so we need to show the reverse inclusion. Suppose $f \in \mathcal{O}(\mathbb{P}^n)$. Then in particular, $f \in \mathcal{O}(U_0)$. Since $U_0 \cong \mathbb{A}^n$, we know that in the coordinates of $\mathbb{A}^n$, we have that f is given by a polynomial in $t_1, \dots, t_n$. If $x_0, \dots, x_n$ are coordinates for $\mathbb{P}^n$, then we have that

$$f|_{U_0} = F\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right) = \frac{G(x_0, \dots, x_n)}{x_0^d},$$

Where G is a homogeneous polynomial of degree d . If $d = 0$, we are done. Otherwise, we may assume that x_0 does not divide G by removing common factors. Via the projection

$$\mathbb{A}^{n+1} \rightarrow \mathbb{P}^n, (x_0, \dots, x_n) \mapsto [x_0 : \dots : x_n],$$

This may be viewed as an equality in $\mathcal{O}(\mathbb{A}^{n+1} \setminus V(x_0))$. Now since f is regular, we have f extends to a regular function on $\mathbb{A}^{n+1} \setminus \{(0,0)\}$ and hence we get on a point in $p \in V(x_0) \setminus \{(0,0)\}$, locally an equality:

$$\frac{G(x_1, \dots, x_n)}{x_0^d} = \frac{A}{B},$$

where $A, B \in A(\mathbb{A}^{n+1})$, and $B(p) \neq 0$. But the set where

$$G(x_1, \dots, x_n)B = x_0^d A,$$

holds is a closed one and contains an open set, thus it is all of $\mathbb{A}^{n+1}$. This however is not possible, unless $d = 0$ since $A(\mathbb{A}^{n+1})$ is a unique factorization domain and x_0 does not divide the RHS. To conclude that

$$\text{Hom}(\mathbb{P}^n, \mathbb{A}^m) = \{\text{constant maps}\}$$

simply note that for any $f \in \text{Hom}(\mathbb{P}^n, \mathbb{A}^m)$, that all compositions

$$\pi \circ f: \mathbb{P}^n \rightarrow \mathbb{A}^1$$

are constant by the first part. □

Exercise 8.2. Let $Y \subset \mathbb{A}_k^n$ be an affine variety. We view Y as a subset of $\mathbb{P}_k^n$ by identifying $\mathbb{A}_k^n$ with the open subset

$$U_0 = \{[x_0 : \dots : x_n] \mid x_0 \neq 0\}.$$

The projective closure of Y is the Zariski closure $\bar{Y} \subset \mathbb{P}_k^n$ of Y .

- (i) Given a polynomial $f \in k[t_1, \dots, t_n]$, let $\beta(f) \in k[x_0, \dots, x_n]$ be the homogeneous polynomial

$$\beta(f) = x_0^{\deg(f)} f\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

Let $I \subset k[t_1, \dots, t_n]$ be the ideal of Y in $\mathbb{A}_k^n$. Show that the ideal of $\bar{Y}$ is the ideal generated by $\beta(f)$, for all $f \in I$.

If $f_1, \dots, f_n$ generate the ideal of $Y \subset \mathbb{A}_k^n$, then $\beta(f_1), \dots, \beta(f_n)$ do not necessarily generate the ideal of its projective closure, as the following example shows.

- (ii) Let $C \subset \mathbb{A}_k^3$ be the set

$$C = \{(t, t^2, t^3) \mid t \in k\},$$

and let $\bar{C} \subset \mathbb{P}_k^3$ be its projective closure. Give generators for the ideal I of C in $\mathbb{A}_k^3$ and for the homogeneous ideal $\bar{I}$ of $\bar{C}$ in $\mathbb{P}_k^3$. Show that $\bar{I}$ cannot be generated by less than 3 elements, while I can be generated by 2 elements.

Solution. (i) Let J be the ideal generated by the elements $\beta(f)$ where $f \in I(Y)$. We have

$$\beta(f)|_{Y \cap U_0} = 0 \Rightarrow \beta|_Y = 0$$

and thus $V(J) \supseteq V(I(\bar{Y}))$ which implies by the Nullstellensatz $J \subseteq I(\bar{Y})$. To show the reverse inclusion, we show that the homogeneous elements of $I(\bar{Y})$ are in J .

Since $I(\bar{Y})$ is a homogeneous ideal this is sufficient. Thus consider a homogeneous polynomial $F \in I(Y)$, and let us write it as

$$F = x_0^d \tilde{F}$$

where x_0^d does not divide $\tilde{F}$. Again, to show that J contains $I(\bar{Y})$ it suffices to show that it contains $\tilde{F}$. Hence we will assume, wlog that F is not divisible by x_0 . In particular, we may write it as

$$F = \sum_{i=0}^d \alpha_i x_0^i \cdot x_1^{s_{i,1}} \cdots x_n^{s_{i,n}},$$

where $\alpha_i \in k$, $\alpha_0 \neq 0$ and $s_{i,1} + \dots + s_{i,n} = d - i$ holds for each i .

Using the fact that F vanishes on $\bar{Y}$, it also vanishes on Y and thus using standard coordinates of U_0 , we have

$$F(t_1, \dots, t_n) := F\left(\frac{x_0}{x_0}, \dots, \frac{x_n}{x_0}\right) \in I(Y).$$

That is to say,

$$f := \sum_{i=0}^d \alpha_i t_1^{s_{i,1}} \cdots t_n^{s_{i,n}} \in I(Y)$$

The i -th term in this sum has degree $d - i$, and the zero'th term is nonzero by assumption. Thus this is a polynomial of degree d . In particular, we may calculate

$$\beta(f) = x_0^d f\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right) = F.$$

(ii) Consider now the curve $C \subseteq \mathbb{A}^3$, given by

$$C = \{(t, t^2, t^3) : t \in k\}$$

Then it is easy to see that the points in C satisfy the relations

$$x^2 - y = 0, \text{ and } xy - z = 0.$$

And thus $I(C) \supseteq (x^2 - y, xy - z)$. Conversely if $p \in \mathbb{A}^3$ is a point fulfilling the above described relations, we find $p = (t, t^2, t^3)$ for $t = p_1$. Thus $V(x^2 - y, xy - z) = C$. In particular, we need only show that $(x^2 - y, xy - z)$ is a radical ideal. In fact it is prime:

$$\frac{k[x, y, z]}{(x^2 - y, xy - z)} \cong \frac{k[x, z]}{(x^3 - z)} \cong k[x].$$

where the last ring is an integral domain. Thus, $I(C)$ is generated by two elements. Now Let $J \subseteq k[x, y, z, w]$ denote the ideal of the projective closure $\bar{C}$ of C . For sake of simplicity we will identify $\mathbb{A}^3$, with $\{\omega \neq 0\}$ instead of $\{x \neq 0\}$ as is usually done. we will first show that J can not be generated by two elements. Since I contains the elements:

$$\begin{cases} x^2 - y \\ xy - z \\ xz - y^2 \end{cases}$$

we see from (i) that J contains

$$\begin{cases} x^2 - yw \\ xy - zw \\ xz - y^2 \end{cases}$$

Note that these three elements are k -linearly independent. We claim that J does not contain constant or linear terms. If this were the case, then for any two $F, G \in I$ we can consider the ideal (F, G) , and see that its degree two part has dimension less than 3:

$$\dim_k((F, G)_2) = \dim_k(kF_2 + kG_2) \leq 2 < \dim_k(J_2).$$

Indeed, the elements of J are generated by elements of the form

$$\beta(f), f \in I.$$

The degree of $\beta(f)$, is the degree of f . We can easily see, that I does not contain polynomials of degree 0 or 1. Hence J is generated by homogeneous elements, all of which have degree atleast 2.

Now let us actually calculate J . First we claim that $\bar{C}$ is given as a set by

$$\bar{C} = \{[s^2t : st^2 : t^3 : s^3] \mid s, t \in k^2 \setminus \{(0, 0)\}\}.$$

Indeed this is a closed set with $J \cap \{w \neq 0\} = C$. Notice now that the points of $\bar{C}$ all satisfy:

$$xy - zw = 0 \tag{1}$$

$$x^2 - wy = 0 \tag{2}$$

$$y^2 - zx = 0 \tag{3}$$

$$x^3 - w^2z = 0 \tag{4}$$

$$y^3 - z^2w = 0. \tag{5}$$

Conversely if $P = [p_0 : p_1 : p_2 : p_3]$ satisfies these relations, then there are two options: either $p_3 \neq 0$ or $p_3 = 0$. In the first case we can rescale to have $p_3 = 1$, and see that P satisfies the equations of C . In the second case, the above equations dictate that p then must be of the form $[0 : 0 : p_2 : 0]$ which is $[s^2t : st^2 : t^3 : s^3]$ for $s = 0$ and $t = \sqrt[3]{p_2}$, hence in $\bar{C}$. Thus, we have proven

$$\bar{C} = V_{\mathbb{P}^n}(\mathfrak{a}), \text{ for } \mathfrak{a} := (xy - zw, x^2 - wy, y^2 - zx, x^3 - w^2z, y^3 - z^2w)$$

(note that the cubic generators are actually redundant, since they follow from the others, but we include them to simplify the upcoming arguments). We claim $I(\bar{C}) = \mathfrak{a}$. By the projective Nullstellensatz, it suffices to prove that $\mathfrak{a}$ is a radical ideal. In fact, we will prove that it is a prime ideal, or equivalently, that $R := \frac{k[x, y, z, w]}{\mathfrak{a}}$ is an integral domain. Consider the ring homomorphism

$$\Phi : k[x, y, z, w] \rightarrow k[s, t], \begin{cases} x \mapsto s^2t \\ y \mapsto st^2 \\ z \mapsto t^3 \\ w \mapsto s^3 \end{cases}$$

This map vanishes on the generators of $\mathfrak{a}$:

$$\begin{aligned}\Phi(xy - zw) &= s^2 t s t^2 - t^3 s^3 = 0 \\ \Phi(x^2 - wy) &= s^4 t^2 - s^3 s t^2 = 0 \\ \Phi(y^2 - zx) &= s^2 t^4 - t^3 s^2 t = 0 \\ \Phi(x^3 - w^2 z) &= s^6 t^3 - s^6 t^3 = 0 \\ \Phi(y^3 - z^2 w) &= s^3 t^6 - t^6 s^3 = 0.\end{aligned}$$

Hence we have a well defined morphism

$$R = \frac{k[x, y, z, w]}{\mathfrak{a}} \rightarrow k[s, t], \quad \begin{cases} x \mapsto s^2 t \\ y \mapsto s t^2 \\ z \mapsto t^3 \\ w \mapsto s^3 \end{cases}$$

To conclude the claim, we need to show that Φ is injective. For this we claim that every monomial $x^a y^b z^c w^d$ is equivalent $\pmod{\mathfrak{a}}$ to an element in the set

$$S := \{z^n w^m \mid n, m \in \mathbb{N}\} \cup \{z^n w^m x \mid n, m \in \mathbb{N}\} \cup \{z^n w^m y \mid n, m \in \mathbb{N}\}$$

Indeed let $r = x^a y^b z^c w^d$ be any such monomial. Using equation (1) from the generators (repeatedly) we see that r is equivalent to one of the two forms:

$$x^n z^m w^l \quad \text{or} \quad y^n z^m w^l$$

Let us consider the first case. Then write $n = 3k + j$ for $j \in \{0, 1, 2\}$ and use equation (4) to obtain that r is further equivalent to a term

$$x^j z^m w^l \quad \text{with } j \in \{0, 1, 2\} \quad (\text{for possibly new } m \text{ and } l)$$

If $j \in \{0, 1\}$ we are done. Otherwise, use equation (2) to reach the desired form. The other case, i.e $r \equiv y^n z^m w^l$ works exactly the same. Since we have proven that every monomial in $k[x, y, z, w]$ is equivalent to an element of $S \pmod{\mathfrak{a}}$, every element in R may be represented as

$$A(z, w) + B(z, w)x + C(x, w)y.$$

Suppose now that $f \in R$ is in the kernel of Φ . Then using the above representation, we find that

$$\Phi(f) = A(s^3, t^3) + B(s^3, t^3)s^2 t + C(s^3, t^3)st^2 = 0 \in k[s, t]$$

But notice now that the first term contributes only to terms where the degree of both s and t is $0 \pmod{3}$. The second term only to terms where the degree of s is $2 \pmod{3}$ and that of t is $1 \pmod{3}$. Similarly the last term only contributes to terms where the degree of s is $1 \pmod{3}$ and that of t is $2 \pmod{3}$. In particular, we must have

$$A(s^3, t^3) = B(s^3, t^3) = C(s^3, t^3) = 0.$$

But this occurs if and only if A, B and C were already zero. Thus

$$\Phi: R \rightarrow k[s, t]$$

is injective and $\mathfrak{a}$ is prime. □

Remark 8.3. The argument above shows that $\bar{C}$ is nothing more than the image of the Veronese embedding ϕ_3^1 .

Exercise 8.4 (Veronese embedding). (i) Given $n, d > 0$, let $\{M_0, M_1, \dots, M_N\}$ be the set of all monomials of degree d in $n + 1$ variables $x_0, \dots, x_n$. Let

$$\phi_d^n : \mathbb{P}^n \rightarrow \mathbb{P}^N$$

be the map

$$[a_0 : \dots : a_n] \mapsto [M_0(a_0 : \dots : a_n) : M_1(a_0 : \dots : a_n) : \dots : M_N(a_0 : \dots : a_n)].$$

Show that ϕ_d^n is a closed embedding, i.e. that ϕ_d^n gives an isomorphism onto a closed subset of $\mathbb{P}^N$.

(ii) Let $X \subset \mathbb{P}^n$ be a hypersurface of degree d , that is, $H = Z(F)$ for a homogeneous polynomial F of degree d . Show that $\mathbb{P}^n \setminus X$ is an affine variety. What is its affine coordinate ring?

Hint: consider the map $\phi_d^n : \mathbb{P}^n \hookrightarrow \mathbb{P}^N$ and observe that there exists a hyperplane $H \subset \mathbb{P}^N$ such that

$$X = (\phi_d^n)^{-1}(H).$$

Solution. (i) First, let us introduce some notation. The set of monomials $\{M_0, \dots, M_N\}$ is in bijection to tuples:

$$S = \{I = (a_0, \dots, a_n) \in \mathbb{N}^{n+1} \mid a_0 + \dots + a_n = d\}.$$

Let us therefore index the monomials as M_I where $I \in S$. Furthermore, let us define

$$I_j = (0, \dots, d, \dots, 0),$$

where the d is in j 'th position. With this in mind, we will denote X_I coordinates on $\mathbb{P}^N$, and consider the ring

$$k[X_I \mid I \in S].$$

With this notation in place let us prove the result. The Veronese embedding

$$\phi_d^n : \mathbb{P}^n \rightarrow \mathbb{P}^N,$$

is well defined as

$$V(M_I \mid I \in S) = V(\sqrt{(M_I \mid I \in S)}) = V(S_+) = \emptyset,$$

where S_+ denotes the irrelevant ideal in $k[x_0, \dots, x_n]$. Thus, by proposition 8.11, the Veronese embedding is a morphism. We claim it is an isomorphism to

$$X = V_{\mathbb{P}^N}(X_I X_J - X_{I'} X_{J'} \mid I, I', J, J' \in S, I + J = I' + J')$$

Indeed, the Veronese clearly maps to this set. We will now provide an inverse from this set to the $\mathbb{P}^n$. First we claim that for any point $x \in X$, not all the coordinates x_{I_j} for $j = 0, \dots, n$, can be zero. Assume for sake of contradiction this were not the case i.e there exists a point which violates this. We know that not all coordinates

$$x_I : I \in S,$$

can be zero. Hence we can choose index

$$I^* = (a_0, \dots, a_n),$$

Such that $x_{I^*} \neq 0$ and such that the supremum norm $\|I^*\|_{\text{sup}}$ is maximal among indices where x has nonzero coordinate. By assumption, I^* is not one of the "pure" indices I_j for $j = 0, \dots, n$ since we assume these all to be zero. Hence, at least two entries in the tuple I^* are nonzero. wlog let us assume

$$\begin{aligned} a_0 &= \|I^*\|_{\text{sup}} \\ a_n &> 0. \end{aligned}$$

In particular the tuples

$$J_1 = (a_0 + 1, \dots, a_n - 1), \text{ and } J_2 = (a_0 - 1, \dots, a_n + 1)$$

are both members of S . The governing equations of X now ensure:

$$0 = x_{I^*}^2 - x_{J_1} x_{J_2} \Rightarrow x_{J_1} \neq 0$$

But this is a contradiction since J_1 has larger supremum norm than I^* . This observation allows us to construct the inverse of φ . Before we do this, let us introduce one more piece of notation. Let us denote by $I_{i,j}$ the multi-index in S which consists of the tuple which has entry 1 in position i and entry $d-1$ in position j . With this notation define

$$\psi: X \rightarrow \mathbb{P}^n, x \mapsto \left[\frac{X_{I_{0,j}}}{X_{I_j}} : \dots : \frac{X_{I_{n,j}}}{X_{I_j}} \right] \text{ on } X_{I_j} \neq 0.$$

We need to show that this is well defined. The fact that this is a morphism then follows immediately from proposition 8.11 again. Consider now the map ψ on the intersection $\{X_{I_t} \neq 0\} \cap \{X_{I_s} \neq 0\}$. we have

$$\left[\frac{X_{I_{0,s}}}{X_{I_s}} : \dots : \frac{X_{I_{n,s}}}{X_{I_s}} \right] = \left[\frac{X_{I_{0,t}}}{X_{I_t}} : \dots : \frac{X_{I_{n,t}}}{X_{I_t}} \right] \Leftrightarrow [X_{I_{0,s}} X_{I_t} : \dots : X_{I_{n,s}} X_{I_t}] = [X_{I_{0,t}} X_{I_s} : \dots : X_{I_{n,t}} X_{I_s}]$$

Thus, let us try to prove the second holds. We want to Provide a nonzero scalar $\lambda \in k$, such that

$$\begin{pmatrix} X_{I_{0,s}} X_{I_t} \\ \vdots \\ X_{I_{n,s}} X_{I_t} \end{pmatrix} \cdot \lambda = \begin{pmatrix} X_{I_{0,t}} X_{I_s} \\ \vdots \\ X_{I_{n,t}} X_{I_s} \end{pmatrix}$$

Such a solution exists, if and only if the matrix

$$A := \begin{pmatrix} X_{I_{0,s}} X_{I_t} & X_{I_{0,t}} X_{I_s} \\ \vdots & \vdots \\ X_{I_{n,s}} X_{I_t} & X_{I_{n,t}} X_{I_s} \end{pmatrix}$$

has rank 1. This can be computed by computing all its (2×2) minors. But these all vanish as these are in the governing equations of X . Lastly, a small but tedious calculation shows $\psi \circ \varphi = Id$ and also $\varphi \circ \psi = Id$.

- (ii) Let $V_d^n \subseteq \mathbb{P}^N$ denote the Veronese of degree d of $\mathbb{P}^n$. If $F = \sum_{I \in S} \lambda_I F_I$ is a homogeneous polynomial of degree d . For this F , consider the hyperplane

$$H = V\left(\sum_{I \in S} X_I\right) \subseteq \mathbb{P}^N.$$

Clearly $\phi_d^n(X) \subseteq H$. In fact it is not hard to see,

$$\phi_d^n(X) = H \cap V_d^n.$$

In particular,

$$\mathbb{P}^n \setminus X \cong V_{n,d} \cap U,$$

where $U = \mathbb{P}^n \setminus H$. Now U is isomorphic to the standard chart U_0 , via a change of variables and $U_0 \cong \mathbb{A}^N$. Hence we conclude that $V_{n,d} \cap U$ may be identified with a closed subset of $\mathbb{A}^N$, and is thus affine. □

Exercise 8.5 (Segre embedding). Let $n, m > 0$ and consider

$$\phi : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^N$$

given by

$$([x_0 : \cdots : x_n], [y_0 : \cdots : y_m]) \mapsto [x_0 y_0 : x_0 y_1 : \cdots : x_i y_j : \cdots : x_n y_m].$$

Show that the image of ϕ is a projective subvariety of $\mathbb{P}^N$. In this way we regard $\mathbb{P}^n \times \mathbb{P}^m$ as a projective variety. Given projective varieties $X \subset \mathbb{P}^n$ and $Y \subset \mathbb{P}^m$, show that $X \times Y$ is naturally a projective subvariety of $\mathbb{P}^n \times \mathbb{P}^m$.

Solution. Let us again introduce some notation. We will denote by

$$\{X_{i,j} \mid 0 \leq i \leq n, 0 \leq j \leq m\},$$

the variables on $\mathbb{P}^N$: It is not hard to see that the image of ϕ is contained in the projective subvariety:

$$X := V(X_{i,j} X_{i',j'} - X_{i,j'} X_{i',j} \mid 0 \leq i, i' \leq n, 0 \leq j, j' \leq m).$$

In fact, it maps bijectively to this set as we will now show by constructing an inverse. Indeed, consider the map:

$$X \rightarrow \mathbb{P}^n \times \mathbb{P}^m, [x_{i,j}]_{i,j} \mapsto \left[\frac{x_{0,s}}{x_{t,s}} : \cdots : \frac{x_{n,s}}{x_{t,s}} \right] \times \left[\frac{x_{t,0}}{x_{t,s}} : \cdots : \frac{x_{t,m}}{x_{t,s}} \right] \text{ on } X_{t,s} \neq 0$$

Checking that this is well defined and is an inverse to ϕ comes down to computing (2×2) minors of a certain matrix again, similar to 4.3. □

§9 Lecture 9

In the last map we have introduced regular maps between affine/quasi-affine/projective/quasi-projective varieties. One should take a moment to realize that regular maps $f: X \rightarrow \mathbb{A}^1$ are the same as regular functions $f \in \mathcal{O}_X(X)$. Indeed, if $f \in \mathcal{O}_X(X)$, then f is continuous when viewed as a map to $\mathbb{A}^1$ and locally, f is given by quotients of homogeneous polynomials, which allows one to verify that f also pulls back regular functions to regular functions. In fact, as we have seen before, for affine varieties X, Y we have

$$\text{Hom}_k(X, Y) \cong \text{Hom}_{k\text{-algebras}}(A(Y), A(X)).$$

Crucially, this fails for general varieties. However, it turns out that the crucial point is that the target is affine.

Proposition 9.1

Let X and Y be varieties and suppose that Y is affine. Then the pullback $\varphi \mapsto \varphi^*$ defines a bijection

$$\mathrm{Hom}_k(X, Y) \cong \mathrm{Hom}_{k\text{-algebras}}(A(Y), \mathcal{O}_X(X)).$$

Proof. Suppose $Y \subseteq \mathbb{A}^m$ and let

$$\pi: k[x_1, \dots, x_m] \twoheadrightarrow A(Y),$$

be the canonical projection. Assume now $\beta \in \mathrm{Hom}_{k\text{-algebras}}(A(Y), \mathcal{O}_X(X))$ is any morphism. Then we have a commutative diagram

$$\begin{array}{ccc} A(Y) & \xrightarrow{\beta} & \mathcal{O}_X(X) \\ \pi \uparrow & \nearrow \tilde{\beta} & \\ k[x_1, \dots, x_m] & & \end{array}$$

This allows us to define a map

$$\beta_*: X \rightarrow \mathbb{A}^m, p \mapsto (\tilde{\beta}(x_1)(p), \dots, \tilde{\beta}(x_m)(p))$$

For any $F \in I(Y)$, we have

$$F((\tilde{\beta}(x_1)(p), \dots, \tilde{\beta}(x_m)(p))) = F((\beta(x_1)(p), \dots, \beta(x_m)(p))) = \beta(F)(P) = 0,$$

where we used the fact that β is a morphism of k -algebras. This shows that β_* is in fact a map to $Y \subseteq \mathbb{A}^m$. The map β^* is a regular map since its coordinates are regular functions and $\mathbb{A}^m$ carries the universal property of products. Note also, that $(\beta_*)^*(x_i) = \beta(x_i)$. Hence, for each morphism of k -algebras there is a regular map which pulls back to this morphism. This morphism is unique, since $\beta_1 = \beta_2$ occurs if and only if they agree on all coordinates, which is equivalent to saying that the pullbacks of x_i all agree which, finally, is equivalent to $\beta_1^* = \beta_2^*$. $\square$

Remark 9.2. In the Exercises we have shown that $\mathcal{O}(\mathbb{P}^n) \cong k$ and hence any regular map to an affine variety is constant. In fact, we will see, for any projective variety we have $\mathcal{O}(X) \cong k$. This is the first hint that projective varieties are the right replacement for compactness in algebraic geometry; they carry some finiteness properties. The next theorem, which is the main goal of this lecture, will make this heuristic more clear.

Theorem 9.3 (The Closed Mapping Theorem)

Let $\varphi: X \rightarrow Y$ be a regular map of k -varieties. If X is projective, then $\varphi(X) \subseteq Y$ is a closed set.

Before we are ready to prove this theorem we will need quite a few preparations. However, For the moment let us explore some of its consequences.

Example 9.4

Consider the map variety $\{xy - 1 = 0\} \subseteq \mathbb{A}^2$. It is not projective as the projection to $\mathbb{A}^1$ has image $\mathbb{A}^1 \setminus \{0\}$ which is not closed.

Corollary 9.5

For any projective variety X we have $\mathcal{O}(X) \cong k$.

Proof. Indeed, let $f \in \mathcal{O}(X)$ and interpret f as a regular map to $\mathbb{A}^1$. Since X is irreducible, $f(X)$ is a single point or all $\mathbb{A}^1$. In the first case we are done, but in the second we see that the composition:

$$X \xrightarrow{f} \mathbb{A}^1 \xrightarrow{i} \mathbb{P}^1$$

has an open image, which contradicts the closed mapping theorem. $\square$

Corollary 9.6

For any projective variety X and any regular map $\varphi: X \rightarrow Y$, $\varphi(X) \subseteq Y$ is projective.

Proof. If Y is quasi projective, we have an embedding $Y \hookrightarrow \mathbb{P}^n$ for some large n . The composition

$$X \xrightarrow{\varphi} Y \hookrightarrow \mathbb{P}^n$$

has an image isomorphic to $\varphi(X)$, and is closed. $\square$

With a few consequences of the closed mapping theorem highlighted, let us now work towards a proof. This will require a small digression into the theory of products, which will be useful in its own right later.

Recall from the tasks for week 4, the Segre embedding:

$$\sigma: \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^{nm+n+m}, [x_0 : \dots x_n] \times [y_0 : \dots y_m] \mapsto [x_i y_j]_{i,j}.$$

We have shown that it has closed image onto a projective variety. Using this, we are able to define the product of any quasi projective varieties: for $X \subseteq \mathbb{P}^n$, and $Y \subseteq \mathbb{P}^m$, we have an inclusion of sets $\Phi: X \times Y \hookrightarrow \mathbb{P}^n \times \mathbb{P}^m$. This allows us to define an inclusion

$$\Phi: X \times Y \hookrightarrow \mathbb{P}^n \times \mathbb{P}^m \xrightarrow{\sigma} \mathbb{P}^{nm+n+m}$$

Then we define the topology and structure sheaf of $X \times Y$, via its image in $\mathbb{P}^{nm+n+m}$. Now to show all these details work, and define a categorical product, we will need to understand the Segre embedding more.

Lemma 9.7

$V \subseteq \mathbb{P}^n \times \mathbb{P}^m$ is closed (w.r.t the topology pulled back from the segre embedding) if and only if it is the zero set of a collection of bihomogeneous polynomials

$$f_i \in k[x_0, \dots, x_n, y_0, \dots, y_m]$$

Remark 9.8. Here bihomogeneous means that it is homogeneous when regarded as a polynomial in either the x_i or the y_i . An example is $x_0^3 y_0 + x_0 x_1 x_2 y_1$. Such a polynomial is called bihomogeneous of bi degree (s, t) if it is homogeneous in the x_i of degree s and in the y_i of degree t . In particular a bihomogeneous polynomial of bidegree (s, t) is homogeneous of

degree $s + t$.

Proof. The set V is closed, if and only if

$$\sigma(V) = V(G_1, \dots, G_r),$$

for homogeneous polynomials in the variables $\{W_{i,j} \mid 0 \leq i \leq n, 0 \leq j \leq m\}$. Restricting these G_i to the image of σ , they can be viewed as homogeneous polynomials

$$f_i := G_i|_{\text{Im}(\sigma)} = G_i(x_0y_0, \dots, x_ny_m)$$

which is homogeneous in the variables $W_{i,j} = x_iy_j$. This means that for each G_i there is an integer d_i such that every monomial appearing in G_i is of degree d_i in the variables x_iy_j , and so in particular it is bihomogeneous of bidegree (d_i, d_i) . Thus $V = V(f_1, \dots, f_r)$. Conversely suppose now that

$$S = V(G_1, \dots, G_r),$$

where the G_i are bihomogeneous of bidegree (d_i, d'_i) . Then if $d_i = d'_i$, we may view G_i as a homogeneous polynomial in $W_{i,j}$ by pairing $x'_i s$ with $y'_j s$. If we have $d_i > d'_i$ then consider

$$G_{i,j} = y_j^{d_i - d'_i} G_i,$$

and notice that

$$\text{Im}(\sigma) \cap V(G_i) = \text{Im}(\sigma) \cap V(G_{i,0}, \dots, G_{i,m}).$$

Thus, we can replace G_i with homogeneous polynomials in $W_{i,j}$ with the same vanishing. Analogously, we may proceed if $d_i < d'_i$. Thus, replacing all G_i with the corresponding $G_{i,j}$ we arrive at the conclusion. $\square$

Let us continue our preparations towards the closed mapping theorem.

Theorem 9.9

Let X be a projective variety, and Y any variety. Then, the projection

$$\pi: X \times Y \rightarrow Y,$$

is a closed map.

Remark 9.10. Note that the projection is a morphism as it is given by homogeneous polynomials of equal degree and hence, when interpreted as a map on $\mathbb{P}^N$ via the segre-embedding, it is given by homogeneous polynomials of equal degree.

Proof. By assumption we have that $X \subseteq \mathbb{P}^n$ is a closed subvariety. If we can show that the projection $\mathbb{P}^n \times Y \rightarrow Y$, is closed, then the statement follows for Y as well. Hence, we will assume, wlog $X = \mathbb{P}^n$. Moreover, if $(U_i)_i$ is a finite open cover of Y , the map

$$\pi: \mathbb{P}^n \times Y \rightarrow Y$$

is closed if and only if, the maps

$$\pi|_{U_i}: \mathbb{P}^n \times U_i \rightarrow U_i$$

are all closed. Indeed, if π is closed this immediately implies that $\pi|_{U_i}$ is closed for all U_i . Conversely, for any closed set $C \subseteq Y$, we have

$$C = \bigcup_i (C \cap U_i),$$

and so

$$\pi(C) = \bigcup_i \pi(\mathbb{P}^n \times (C \cap U_i)) = \bigcup_i \pi|_{U_i}(\mathbb{P}^n \times (C \cap U_i)),$$

Since the cover is finite, the last set is closed as a finite union of closed sets. Since Y is covered by finitely many affine charts, we may assume that $Y \subseteq \mathbb{A}^m$ is closed. Then we have a commutative diagram

$$\begin{array}{ccc} \mathbb{P}^n \times Y & \hookrightarrow & \mathbb{P}^n \times \mathbb{A}^m \\ \downarrow \pi & & \downarrow \pi \\ Y & \hookrightarrow & \mathbb{A}^m \end{array}$$

Since the inclusion $\mathbb{P}^n \times Y \hookrightarrow \mathbb{P}^n \times \mathbb{A}^m$ is closed, it suffices to show that the projection

$$\pi: \mathbb{P}^n \times \mathbb{A}^m \rightarrow \mathbb{A}^m,$$

is closed. Suppose $V \subseteq \mathbb{P}^n \times \mathbb{A}^m$ is a closed set. Then V is given by

$$V = V(f_1, \dots, f_r),$$

where $f_i \in k[x_0, \dots, x_n, y_1, \dots, y_m]$, are polynomials which are homogeneous in x_i . A point $a \in \mathbb{A}^m$ lies in the projection $\pi(V)$ if and only if

$$V_{\mathbb{P}^m}(f_1(\cdot, a), \dots, f_r(\cdot, a)) \neq \emptyset.$$

In other words if we define $J_a = (f_1(\cdot, a), \dots, f_r(\cdot, a))$, then we obtain

$$\begin{aligned} a \notin \pi(V) &\Leftrightarrow V_{\mathbb{P}^m}(f_1(\cdot, a), \dots, f_r(\cdot, a)) = \emptyset \\ &\Leftrightarrow S_+ \subseteq \sqrt{J_a}, \end{aligned}$$

where $S_+ \subseteq k[x_0, \dots, x_n]$ denotes the irrelevant ideal. The last condition means that there exists a d such that $S_d \subseteq J_a$, and so

$$\mathbb{A}^m \setminus \pi(V) = \bigcup_{d \geq 0} \{a \mid J_a \supseteq S_d\}.$$

To finish the proof, we will show that the sets $\{a \mid J_a \supseteq S_d\}$ are all open sets. let $d_i = \deg(f_i(\cdot, a))$ be the respective degrees of the generators of J_a . Then we have

$$J_a \cap S_d = \text{Im}(L_a : \bigoplus_{i=1}^r S_{d-d_i} \rightarrow S_d),$$

where this map is given by multiplying the i -th component by $f_i(\cdot, a)$. Hence $J_a \cap S_d = S_d$ occurs if and only if this k -linear map is surjective. Choosing a basis, this can be determined by computing whether all $n \times n$ minors of a matrix $[L_a]$, whose coefficients are dependent on a , vanish. This is an algebraic condition in a . $\square$

Now we can prove the closed mapping theorem.

Proof of the closed mapping theorem. Let $\varphi: X \rightarrow Y$, be the regular map in question, and X projective. Consider the graph

$$\Gamma_\varphi = \{(x, \varphi(x)) \mid x \in X\} \subseteq X \times Y.$$

We claim it is closed. If this were the case, then we would be done, as the previous theorem would imply that $\pi_Y(\Gamma_\varphi) = \varphi(X)$ is a closed set. First note, that

$$\Delta_Y = \{(y, y) \mid y \in Y\} \subseteq Y \times Y,$$

is closed. To see this, it suffice to show that

$$\Delta_{\mathbb{P}^n} \subseteq \mathbb{P}^n \times \mathbb{P}^n$$

is closed. By the characterization of closed sets in $\mathbb{P}^n \times \mathbb{P}^n$, it suffices to find bi homogeneous equations in $k[x_0, \dots, x_n, y_0, \dots, y_n]$ which produce this set. We have

$$[x_0 : \dots : x_n] = [y_0 : \dots : y_n] \Leftrightarrow \exists \lambda \in k, (x_0, \dots, x_n) = \lambda(y_0, \dots, y_n).$$

This occurs if and only if

$$\text{Rank} \begin{pmatrix} x_0 & y_0 \\ \vdots & \vdots \\ x_n & y_n \end{pmatrix} = 1$$

In otherwords, all 2×2 minors vanish

$$x_i y_j - x_j y_i = 0 \quad \forall i, j.$$

Hence Δ_Y is carved out by bihomogeneous equations and is therefore closed. consider the two maps $X \times Y \rightarrow Y$, defined via:

$$\begin{array}{ccc} & X \times Y & \\ \swarrow \pi & & \searrow \pi \\ X & \xrightarrow{\varphi} & Y \end{array}$$

By the universal property of products this produces a morphism

$$X \times Y \xrightarrow{\varphi \times Id} Y \times Y,$$

and under this map

$$\Gamma_\varphi = (\varphi \times Id)^{-1}(\Delta_Y).$$

Thus Γ_φ is the preimage of a closed set under a continuous map and therefore also closed. $\square$

To finish this lecture, we briefly introduced rational maps for varieties.

Definition 9.11. Let X and Y be k -varieties. A rational map

$$\varphi : X \dashrightarrow Y$$

is an equivalence class of pairs $[U, \psi]$, where $U \subseteq X$ is a nonempty open and $\psi : U \rightarrow Y$ is a regular map, and where the equivalence is given by

$$[U, \psi] \sim [U', \psi'] : \Leftrightarrow \psi|_{U \cap U'} = \psi'|_{U \cap U'}.$$

A rational map is called **dominant** if it has dense image. A rational map is called **birational**, if it has an rational inverse.

Remark 9.12. Strictly speaking it doesn't make sense to ask whether a rational map $\varphi : X \dashrightarrow Y$ has dense image, but since X is irreducible, this quality is independent of choice of representative $[U, \psi]$ of the rational map φ .

As rational maps are only defined up to open subsets and any variety contains affine open subsets, the entire theory of rational maps for affine varieties also applies to general varieties. In particular, there is a bijection of dominant rational maps $\varphi : X \dashrightarrow Y$ and field morphisms $k(Y) \rightarrow k(X)$. Moreover, the following are equivalent:

- $k(Y) \cong k(X)$
- X is birational to Y
- there are open affine subvarieties $U \subseteq X$ and $V \subseteq Y$, such that $U \cong V$ (here the isomorphism is meant as a regular map).

§10 Lecture 10

In the last lecture we introduced rational maps for general varieties. Before we continue with general theory building, we will explore some examples.

Example 10.1 (The standard Cremona Transformation)

Consider the rational map

$$\phi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2, [x : y : z] \mapsto \left[\frac{1}{x} : \frac{1}{y} : \frac{1}{z} \right] = [yz : xz : xy]$$

. It is regular on $U = \mathbb{P}^2 \setminus \{[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]\}$. It is a birational equivalence, as applying it twice gives the identity.

Example 10.2 (The blowup)

Consider the quasi projective variety $\mathbb{A}^n \setminus \{0\}$ and the morphism

$$f : \mathbb{A}^n \setminus \{0\} \rightarrow \mathbb{P}^{n-1}, (x_1, \dots, x_n) \mapsto [x_1 : \dots : x_n].$$

which sends a point to the projective line it lays on. By the universal property of products this induces a morphism

$$\phi : \mathbb{A}^n \setminus \{0\} \rightarrow \mathbb{A}^n \times \mathbb{P}^{n-1}, x \mapsto (x, f(x)).$$

If W is the closure of the graph Γ_f , then ϕ defines a rational map between $\mathbb{A}^n$ and W . In fact, we may give equations of W : notice that the points in Γ_ϕ are of the form

$$(x_1, \dots, x_n) \times [x_1 : \dots : x_n],$$

thus, if $(x, y) \in \Gamma_f$, the matrix

$$A_{x,y} = \begin{pmatrix} x_1 & y_0 \\ \vdots & \vdots \\ x_n & y_{n-1} \end{pmatrix}$$

has rank 1 (independent of choice of representative of y). In other words, all the (2×2) minors vanish. using $u_1, \dots, u_n$ to be coordinates on $\mathbb{A}^n$ and $x_1, \dots, x_n$ to be homogeneous coordinates of $\mathbb{P}^{n-1}$, we see the points in Γ_ϕ , all satisfy the equations

$$u_i x_j - u_j x_i = 0 \text{ for all } i, j.$$

Thus, also all all points in the closure W of Γ_ϕ satisfy these equations. Now consider a point $(x, y) \in \mathbb{A}^n \times \mathbb{P}^{n-1}$ that satisfies these equations. If x is nonzero, then there is clearly only one projective point y such that $\text{Rank}(A_{x,y}) = 1$. Hence, $x, y \in \Gamma_\phi$. On the other hand, if $x = 0$, then any $y \in \mathbb{P}^{n-1}$ makes $\text{Rank}(A_{x,y}) = 1$, and so we have found:

$$V(u_i x_j - u_j x_i = 0 \mid \text{ for all } i, j) = \Gamma_\phi \cup (\{0\} \times \mathbb{P}^{n-1})$$

Thus, we see that $W = V(u_i x_j - u_j x_i = 0 \mid \text{ for all } i, j)$. In particular, ϕ is a birational map between $\mathbb{A}^n$ and W , whose inverse is given by projection to the first coordinate.

The blow up is useful in a process called resolving singularities. The idea is as follows: given any variety, can we find one which is birationally equivalent to it but has no singular points. This is a delicate problem, in fact for $\text{char}(k) = p$ and $n > 3$ it remains open. Let us see an example of a resolving a singularity for a curve.

Example 10.3

Consider the case $n = 2$, and let u, v be coordinates of $\mathbb{A}^2$, along with X, Y be homogeneous coordinates of $\mathbb{P}^1$. Then, the blow up of $\mathbb{A}^2$ at the origin, is given by

$$W = V(uY - vX)$$

Covering $\mathbb{A}^2 \times \mathbb{P}^1$ by the affines $\mathbb{A}^3 \cong \{X \neq 0\}$ and $\mathbb{A}^3 \cong \{Y \neq 0\}$, we can investigate what W looks like on these charts. In the first chart, dehomogenizing at X yields

$$W \cap \{X \neq 0\} = V(u - v\frac{x}{y}) \subseteq \mathbb{A}^3,$$

and similarly

$$W \cap \{Y \neq 0\} = V(u\frac{Y}{X} - v) \subseteq \mathbb{A}^3.$$

Consider now the curve

$$C = V(v^2 - u^2(u + 1)).$$

The **total transform** of C is defined to be the set

$$\pi^{-1}(C) \subseteq W,$$

which is given by

$$V(v^2 - u^2(u + 1), uY - vX) \subseteq \mathbb{A}^2 \times \mathbb{P}^1.$$

On the affine $X \neq 0$, we see that the total transform is given by

$$V(v^2 - u^2(u + 1), u\frac{Y}{X} - v).$$

Subbing the second equation for the first, this yields

$$0 = \left(u\frac{Y}{X}\right)^2 - u^2(u + 1) = u^2\left(\left(\frac{Y}{X}\right)^2 - (u + 1)\right)$$

Which has two irreducible components. In particular, $\pi^{-1}(C)$ has two components. Now consider, the **strict transform** of C which we define as

$$\tilde{C} = \pi^{-1}(C \setminus \{0\}).$$

This corresponds in our above calculation, exactly to the the component

$$W \cap V\left(\left(\frac{Y}{X}\right)^2 - (u + 1)\right) \subseteq \mathbb{A}^3,$$

since $u \neq 0$. This is ofcourse a smooth curve.

Tasks for Week 5

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Algebraic Geometry I

Exercise sheet 5

Throughout, k denotes an algebraically closed field.

Exercise 10.1. Let $P_1, \dots, P_r$ be points in $\mathbb{P}_k^n$. The linear span of the P_i is the smallest linear subvariety of $\mathbb{P}_k^n$ containing $P_1, \dots, P_r$. Notice that the dimension of the linear span of r points is at most $r - 1$.

- (i) Let $d \geq 1$ be an integer and consider the Veronese embedding

$$\phi_d : \mathbb{P}^1 \rightarrow \mathbb{P}^d$$

given by

$$[x_0 : x_1] \mapsto [x_0^d : x_0^{d-1}x_1 : \dots : x_1^d].$$

Given $d + 1$ distinct points $Q_0, \dots, Q_d$ in $\mathbb{P}^1$, show that the linear span of

$$\phi_d(Q_0), \dots, \phi_d(Q_d)$$

is the entire $\mathbb{P}^d$.

- (ii) Consider the Veronese embedding

$$\psi : \mathbb{P}^2 \rightarrow \mathbb{P}^5$$

given by

$$[x_0 : x_1 : x_2] \mapsto [x_0^2 : x_0x_1 : x_0x_2 : x_1^2 : x_1x_2 : x_2^2].$$

Given six distinct points $P_0, \dots, P_5$ in $\mathbb{P}^2$, show that the linear span of

$$\psi(P_0), \dots, \psi(P_5)$$

is the entire $\mathbb{P}^5$ unless the points P_i lie on a conic, i.e., there exists a homogeneous polynomial

$$F \in k[x_0, x_1, x_2]$$

of degree 2 vanishing at $P_0, \dots, P_5$. Show moreover that it is possible to choose $P_0, \dots, P_5$ so that they do not lie on any conic; this is the case for “most” choices.

Solution. (i) Let us denote

$$\phi(Q_i) = [q_{i,0}^d : q_{i,0}^{d-1}q_{i,1} : \dots : q_{i,1}^d].$$

Then, it suffice to show that the points

$$Q'_i = (q_{i,0}^d, q_{i,0}^{d-1}q_{i,1}, \dots, q_{i,1}^d) \in \mathbb{A}^{d+1}$$

are k -linearly independent. This occurs if, and only if, the matrix

$$A = \begin{pmatrix} q_{1,0}^d & q_{1,0}^{d-1}q_{1,1} & \dots & q_{1,1}^d \\ \vdots & \vdots & \ddots & \vdots \\ q_{d,0}^d & q_{d,0}^{d-1}q_{d,1} & \dots & q_{d,1}^d \end{pmatrix},$$

has nonzero determinant. Now for all the points Q_i , we have $q_{i,0} \neq 0$, then we may rescale such that $q_{i,0} = 1$, which turns the above matrix into a Vandermonde matrix and hence has nonzero determinant. Otherwise, since the Q_i are distinct, at most one of the points has $q_{j,0} = 0$, and we may scale the others to have this coordinate equal to one and also assume $q_{j,1} = 1$. This makes this matrix differ from a vandermonde matrix by a permutation matrix and hence it is invertible once more

- (ii) Suppose the linear span of the $\psi(Q_i)$ is not the whole space. Then, there exists a hyperplane H , containing all the points. the preimage of $Im(\psi) \cap H$ under ψ , is a conic by Exercise sheet 4. It suffice to show that the image of the veronese is not contained in any hyperplane. let $Y_0, \dots, Y_5$ be homogeneous coordinates for $\mathbb{P}^5$. Via a change of coordinates, it suffices to prove that the image is not contained in $Y_0 = 0$. But this is obvious. Now consider the set

$$S = \{(P_0, \dots, P_5) \in \underbrace{\mathbb{P}^5 \times \dots \times \mathbb{P}^5}_{6 \text{ times}} \mid \text{The } P_i \text{ lie on a conic}\}$$

Then S is a closed set since it is given by

$$S = V \left(\det \begin{pmatrix} \psi(P_0) \\ \vdots \\ \psi(P_5) \end{pmatrix} \right).$$

and it is not the whole space by the previous argument. Hence the complement of S is a nonempty open set consisting of points which fail to lie on a conic. $\square$

Exercise 10.2. (i) Let X be an affine variety of dimension n and take the intersection of r hypersurfaces

$$Z = \mathcal{Z}_X(f_1, \dots, f_r)$$

in X . If $Z \neq \emptyset$, show that any irreducible component of Z has dimension at least $n - r$.

- (ii) Let F_1 and F_2 be homogeneous polynomials in $k[x_0, x_1, x_2]$, and let

$$X_i = \mathcal{Z}_{\mathbb{P}_k^2}(F_i), \quad i = 1, 2.$$

Show that the intersection $X_1 \cap X_2$ is not empty.

Hint: consider the affine cones

$$C(X_i) := \mathcal{Z}_{\mathbb{A}_k^3}(F_i)$$

in $\mathbb{A}_k^3$.

- (iii) Let $X \subset \mathbb{P}_k^n$ be a projective variety of dimension r , and let

$$F \in k[x_0, \dots, x_n]$$

be a homogeneous polynomial of degree ≥ 1 . If

$$X \not\subset \mathcal{Z}_{\mathbb{P}^n}(F),$$

show that any irreducible component of

$$X \cap \mathcal{Z}_{\mathbb{P}^n}(F)$$

has dimension $r - 1$.

- (iv) Let $X \subset \mathbb{P}_k^n$ be a projective variety of dimension $n - 1$. Show that there exists a homogeneous polynomial F such that

$$X = \mathcal{Z}(F).$$

Solution. 1. We prove the claim by induction over r . Suppose $r = 1$ and so $x = \mathcal{Z}_X(f)$. Since $Z \neq \emptyset$, we know that f is not a unit. If $f = 0$, then $Z = X$ and the claim is also true. Otherwise we may apply Krull's principle ideal theorem to see that the minimal primes of (f) all have height one in $A(X)$. But the minimal primes of (f) are precisely the irreducible components of Z and the height of a prime is precisely the co dimension of its corresponding algebraic set. For $r > 0$ let $Z = \mathcal{Z}(f_1, \dots, f_r)$ and suppose Z_i is any irreducible component of Z . Then Z corresponds to a minimal prime $\mathfrak{p}$ of the ideal $(f_1, \dots, f_r)$. Since Z contains $(f_1, \dots, f_{r-1})$ it contains a minimal prime $\mathfrak{q}$ of $(f_1, \dots, f_{r-1})$. By induction this prime has height $\leq r - 1$. If Y denotes the vanishing of $\mathfrak{q}$ Then we have

$$A(X)/\mathfrak{p} \cong \frac{A(X)/\mathfrak{q}}{\mathfrak{q} + \mathfrak{p}} \cong \frac{A(Y)}{(\bar{f}_r)}.$$

By Krull's principle ideal theorem the last ring has dimension at least $\dim A(Y) - 1 \geq n - \text{ht } \mathfrak{q} - 1 = r$

(ii) It suffice to show that the cones have nonzero intersection in $\mathbb{A}^3$. But this follows from (i) as each irreducible component of the intersection has dimension at least one.

(iii) We have

$$X \cap \mathcal{Z}_{\mathbb{P}^n}(F) = \mathcal{Z}_X(F).$$

By (1) all the irreducible components of this have dimension at least $r - 1$. However, they don't have dimension equal to r , as this would imply $X \subseteq \mathcal{Z}_{\mathbb{P}^n}^n(F)$. thus they have dimension $r - 1$.

(iv) Suppose

$$X = \mathcal{Z}_{\mathbb{P}^n}(f_1, \dots, f_r).$$

Suppose $f_i \neq 0$. If X is not contained in, say $\mathcal{Z}_{\mathbb{P}^n}(f_1)$, then this implies that the dimension of X is at most $n - 2$ unless X is an irreducible component of $\mathcal{Z}_{\mathbb{P}^n}(f_1)$. But an irreducible component of f_1 corresponds simply to a factor of f_1 and so $X = \mathcal{Z}_{\mathbb{P}^n}(F)$ for this factor F .s

□

Exercise 10.3. Let $Q \subset \mathbb{P}^3$ be the quadric surface defined by

$$xy - zw = 0.$$

(i) Show that Q contains two families L_t and L'_t of lines, each parametrized by

$$t \in \mathbb{P}_k^1.$$

Show that $L_t \cap L'_{t'}$ consists of a single point for any t, t' , and that L_t and L_u , respectively L'_t and L'_u , are disjoint for $u \neq t$.

Hint: recognize Q as the image of the Segre embedding

$$\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^3.$$

(ii) Show that Q is birational to $\mathbb{P}^2$ but it is not isomorphic to it.

Hint: use Exercise 2.

Solution. (i) . Consider the Segre embedding

$$\phi: \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^3, [s_0 : s_1] \times [t_0, t_1] \mapsto [s_0 t_0 : s_1 t_1, s_0 t_1 : s_1 t_0].$$

We know from the previous sheet that ϕ defines an isomorphism to Q . Consider now the two families of lines

$$\begin{aligned} F_1 &= \{S_t = \{t\} \times \mathbb{P} \mid t \in \mathbb{P}^k\} \\ F_1 &= \{S'_t = \mathbb{P} \times \{t\} \mid t \in \mathbb{P}^k\} \end{aligned}$$

These two families of lines satisfy the intersection properties required from the task. Furthermore, for a line $S_t = \{[p_0, p_1]\} \times \mathbb{P}$ we have that

$$\phi(S_t) = \{[p_0 s_0 : p_1 s_1 : p_0 s_1 : p_1 s_0] \mid [s, t] \in \mathbb{P}^1\}$$

which is clearly a projective line (in fact we have parametrized it already).

- (ii) Q is not birational to $\mathbb{P}^2$, because in $\mathbb{P}^2$ any two hypersurfaces intersect while this is not true for Q . Let us show that Q is birational to $\mathbb{P}^2$. Dehomogenize Q at w to see that Q is birational to the affine variety

$$X = V_{\mathbb{A}^3}(xy - z).$$

For this variety we have

$$A(X) = \frac{k[x, y, z]}{xy - z} \cong k[x, y].$$

Thus X is isomorphic to $\mathbb{A}^2$. In particular Q and $\mathbb{P}^2$ contain open sets which are isomorphic which completes the claim. $\square$

Exercise 10.4. Let X be a noetherian topological space, and let

$$X = \bigcup_i U_i$$

be an open cover of X . Show that $\dim(X) = n$ if and only if $\dim(U_i) \leq n$ for all i , and $\dim(U_i) = n$ for at least one i .

One way to show the last assertion is to use induction on n to prove that, for any noetherian topological space X and any open cover

$$X = \bigcup_i U_i,$$

given any chain

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \subset X$$

of irreducible closed subsets, there exists i such that $\dim(U_i) \geq n$.

Solution. Let us first prove the claim for when X is irreducible and let $X = \bigcup_i U_i$. Let U be any open set of this member and suppose we have an increasing chain of irreducible closed sets in U

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \subsetneq U.$$

Then for each of the Z_i there are closed set $C_i \subseteq X$ such that $Z_i = U \cap C_i$. We claim that we may choose them to be irreducible: if not, say $C_i = A \cup B$, then $Z_i = (A \cap U) \cup (B \cap U)$, and since Z_i is irreducible, one of these sets already equals Z_i . Thus, we arrive at a chain

$$C_0 \subsetneq C_1 \subsetneq \cdots \subsetneq C_n \subsetneq X.$$

Thus, $n \leq \dim X$. Hence all U_i have dimension at most $\dim X$. Suppose now $n = \dim X$ and

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \subsetneq X.$$

is a chain of increasing irreducible closed subsets of X . Then since the U_i cover X , one of the U_i intersects Z_0 non trivially. Denoting $C_i = Z_i \cap U_i$, we have arrived at a chain

$$C_0 \subseteq C_1 \subseteq \cdots \subseteq C_n \subseteq U.$$

The first thing to note, is that this chain is increasing: C_i is dense in Z_i and thus $C_i = C_{i+1}$ would imply that $Z_i = Z_{i+1}$. The second thing to note is that C_i is irreducible: If $C_i = A \cup B$, then taking closures, $Z_i = \overline{(A \cup B)} = \overline{A} \cup \overline{B}$. Hence, since Z_i is irreducible, Z_i is one of the two factors and hence C_i is as well. Thus $\dim U = \dim X$ $\square$

§11 Lecture 11

In this lecture, we want to motivate the notion of dimension of a variety. For this let us consider vector spaces. for a k -vector space V , we have

$$\dim V = \sup\{n \in \mathbb{N} \mid \exists V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n \subseteq V, V_i \in k\text{-Vec}\}$$

That is to say, the largest increasing chain of subspaces contained in V . This motivates the following definition.

Definition 11.1. For a noetherian topological space X , we define its dimension $\dim X$, as

$$\dim X = \sup\{n \in \mathbb{N} \mid \exists X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_n \subseteq X, X_i \text{ closed irreducible}\} \in \mathbb{N} \cup \{\infty\}$$

Remark 11.2. Requiring our spaces to be Noetherian ensures that the dimension can also be viewed as the supremum over all irreducible closed chains in X , since they are all finite.

Recall that for a commutative ring R , its Krull dimension is defined as

$$\dim R = \sup\{n \in \mathbb{N} \mid \exists \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n \subseteq R, \mathfrak{p}_i \in \text{Spec } R\}$$

Hence, by Hilberts Nullstellensatz, we have

Corollary 11.3

For an affine algebraic set X we have

$$\dim X = \dim(A(X)).$$

The main tool in computing dimensions will be Krulls principle ideal theorem, which we will not prove and treat as a black box.

Theorem 11.4 (Krull's Principle ideal theorem)

Let R be a finitely generated integral algebra k . Then $\dim R < \infty$. moreover if $f \neq 0$ is not a unit, we have

$$\dim (R/(f)) = \dim R - 1.$$

Sometimes, we will need the stringer version:

Theorem 11.5

Let A be a noetherian ring and $x \in A$ be neither a unit nor a zero divisor. Then the minimal primes associated to (x) are all height one,

Lemma 11.6

$$\dim \mathbb{A}^n = n$$

Proof. First, $\dim \mathbb{A}^0 = \dim k = 0$. Furthermore, using Krulls PID theorem and Corollary 11.3 we have

$$\dim \mathbb{A}^{n+1} = \dim k[x_1, \dots, x_{n+1}] = \dim (k[x_1, \dots, x_n]) + 1 = \dim \mathbb{A}^n + 1,$$

The claim then follows from induction. $\square$

Corollary 11.7

Let X be an affine variety and suppose $f \in A(X) \setminus \{0\}$ is not a unit. Then, $V_X(f)$ has dimension $\dim X - 1$.

Proof. We have

$$\dim V_X(f) = \dim \left(A(X)/\sqrt{(f)} \right).$$

Furthermore, By Krulls principle ideal theorem, we have

$$\dim (A(X)/(f)) = \dim A(X) - 1 = \dim X - 1.$$

Thus, it suffice to show that

$$\dim (A(X)/(f)) = \dim \left(A(X)/\sqrt{(f)} \right)$$

In general, we claim we have $\dim R = \dim \text{red}(R)$, where $\text{red}(R)$ is the reduced ring. Indeed, we have a surjection $R \twoheadrightarrow \text{red}(R)$ which yields $\dim R \geq \dim \text{red}(R)$. On the otherhand any prime ideal in R , contains the nilradical, and so any chain of prime ideals in R , contains the nilradical, and hence may be viewed as a chain in the reduced ring. $\square$

Let us now see how the dimension of basic open sets relates to the dimension of the whole space.

Lemma 11.8

Let X be an affine variety and $f \in A(X) \setminus \{0\}$ a non unit. Then

$$\dim D(f) = \dim(X \setminus V_X(f)) = \dim X$$

Proof. From the lemma about localisation, we have that $D(f)$ is again an affine variety and its coordinate ring is given by

$$A(D(f)) = A(X)_f = \frac{A(X)[t]}{(ft - 1)}$$

Now $\frac{A(X)[t]}{(t)} = A(X)$ and thus, $\dim A(X)[t] = \dim A(X) + 1$ by Krull's principle ideal theorem. Applying Krull's PID theorem once more, we find

$$\dim \left(\frac{A(X)[t]}{(ft - 1)} \right) = \dim A(X) + 1 - 1 = \dim A(X).$$

□

In fact, this allows us to show that the dimension is invariant when passing to open sets:

Theorem 11.9

Let X be a projective variety. Then for all nonempty opens $\emptyset \neq U \subseteq X$, we have

$$\dim U = \dim X.$$

Proof. First, we clearly have

$$\dim U \leq \dim X,$$

Moreover, from sheet 5, we see that if $\bigcup_i U_i = X$ is an open cover, that $\dim X = n$ if and only if $\dim U_i \leq n$ for all i and for one of the U_i we have $\dim U_i = n$. Suppose now X is a projective variety. We may cover X by affine open sets X_i and therefore for one of these X_i , we have $\dim X_i = \dim X$. Now, for any open subset U of X , $U \cap X_i$ is a nonempty open subset of the affine set X_i , and therefore contains a smaller open subset $D(f)$ where $f \in A(X_i)$. By the previous theorem, we have $\dim D(f) = \dim X_i = \dim X$. In conclusion, we have

$$\dim X \geq \dim U_i \geq \dim D(f) = \dim X.$$

□

Corollary 11.10

If X and Y are birational, then $\dim X = \dim Y$.

Now the above corollary suggests, that the dimension of a variety is hidden in some invariant in the function field since two varieties are birational if and only if they have isomorphic function fields. Recall that if E/F is a field extension, we define a

transcendence basis $\{\alpha_i | i \in I\}$ of E over F , to be a collection of elements in E , which are algebraically independent over F , such that

$$E/F(\alpha_i | i \in I),$$

is an algebraic extension. We define **transcendence degree** $\text{tr deg}_F(E)$ as the cardinality of any transcendence basis, which we can check to be independent of choices. Another tool from commutative algebra we will need is:

Theorem 11.11 (Noether's Normalization)

Let E/F be a finitely generated field extension and F be a perfect field. Then there exists a transcendence basis $\{\alpha_1, \dots, \alpha_r\}$ of E over k , such that $E/F(\alpha_1, \dots, \alpha_r)$ is a finite separable extension. In particular,

$$E = F(\alpha_1, \dots, \alpha_r)(\gamma) = \frac{F(\alpha_1, \dots, \alpha_r)[t]}{(f)},$$

where f is the minimal polynomial of γ over $F[\alpha_1, \dots, \alpha_r]$.

Translating this theorem to a geometric interpretation, we find.

Corollary 11.12

Let X be a quasi projective variety over k such that $\text{tr deg}_k(X) = n$. Then X is birationally equivalent to an irreducible hypersurface in $\mathbb{A}^{n+1}$.

Proof. k is algebraically closed and thus perfect. Hence, we may apply Noether's Normalization to $k(X)/k$ to find

$$k(X) = \frac{k(\alpha_1, \dots, \alpha_n)[t]}{(f)},$$

A priori, f has coefficients in $k(\alpha_1, \dots, \alpha_n)$. However, by multiplying denominators out (i.e multiplying by a unit) we can assume f has coefficients in $k[\alpha_1, \dots, \alpha_n]$ without changing the ideal (f) . But since f is irreducible, by dividing out the gcd of the coefficients in $k[\alpha_1, \dots, \alpha_n][t]$, Gauß's lemma implies that it is irreducible in $k[\alpha_1, \dots, \alpha_n][t]$. This induces an inclusion

$$\frac{k[\alpha_1, \dots, \alpha_r, t]}{(f)} \hookrightarrow \frac{k(\alpha_1, \dots, \alpha_n)[t]}{(f)}.$$

passing to the field of fractions,

$$\text{Frac}\left(\frac{k[\alpha_1, \dots, \alpha_r, t]}{(f)}\right) = k(X).$$

Since f is irreducible, the left hand side is the function field of $Y = V(f) \subseteq \mathbb{A}^{n+1}$. $\square$

Remark 11.13. Another fact from commutative algebra is that the transcendence degree is additive: if

$$E/F/K,$$

is a tower of fields, then

$$\text{tr deg}_K(E) = \text{tr deg}_F(E) + \text{tr deg}_K(F).$$

In particular, $\text{tr deg}_K(E) = \text{tr deg}_K(F)$ if E/F is a finite extension-

Corollary 11.14

if $\varphi: X \dashrightarrow Y$ is a dominant rational map, then

$$\dim Y \leq \dim X$$

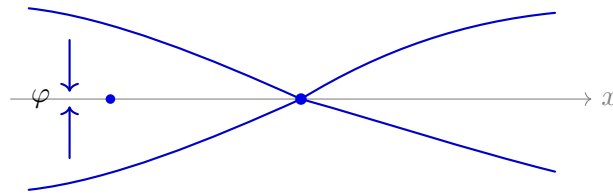
Proof. The pullback of φ induces an inclusion $k(Y) \hookrightarrow k(X)$. □

This yields us to the notion of the degree of a rational map.

Definition 11.15. Let $\varphi: X \rightarrow Y$ be a dominant rational map of varieties of equal dimension. Then we define the **degree** of φ to be the integer

$$\deg \varphi = [k(Y) : \varphi^*k(X)] < \infty.$$

The degree of a map of such a φ will be something similar to the degree of a covering map in topology. Indeed, we can view φ as a type of branched covering map and the degree will give the number of preimages of a generic point in Y . We will prove this here and also in the exercises, but the below figure should yield us some intuition for now.



We continue our investigation of fibers $\varphi^{-1}(y)$, for $y \in Y$. Note that if

$$\varphi: X \rightarrow Y,$$

is dominant rational map, then

$$\dim X = \dim Y + \text{tr deg}(k(Y)/k(X))$$

Thus $\dim X - \dim Y = \text{tr deg}(k(Y)/k(X)) \geq 0$

Theorem 11.16 (Theorem of the dimension of fibers)

If $\varphi: X \rightarrow Y$ is a regular dominant map and we denote by $n = \dim X$ and $m = \dim Y$, then we have

- (1) For all $y \in \varphi(X)$, we have $\dim \varphi^{-1}(y) \geq n - m$.
- (2) There exists a nonempty open subset $U \subseteq Y$, such that $\dim \varphi^{-1}(y) = n - m$ for all $y \in U \cap \varphi(X)$.

Corollary 11.17

$$\dim(X \times Y) = \dim X + \dim Y$$

Proof. The projections π_X and π_Y are both regular maps. Furthermore, for any $x \in X$, we have $\pi_X^{-1}(x) \cong Y$. Thus the theorem of the dimension of fibers ensures

$$\dim Y = \dim \pi_X^{-1}(x) = \dim(X \times Y) - \dim X.$$

□

Let us now prove the first part of the fibers theorem.

Part (1) of the theorem of the dimension of fibers. Suppose $y \in Y$ is any point. Then $\varphi^{-1}(y)$ depends only on an open neighborhood of y , and so we may replace Y by an affine $Y' \subseteq Y$ containing y , and X by $\varphi^{-1}(Y')$. Hence, we will assume that Y is an affine variety. Now if $Z \subseteq X$ is any irreducible component of $\varphi^{-1}(y)$, then passing to an affine open subset $U \subseteq X$, with $U \cap Z \neq \emptyset$ we have that $\dim U \cap Z = \dim Z$ and $\varphi|_U$ is still a regular dominant map. Thus, we may also assume X to be affine.

We now proceed by induction over $m = \dim Y$. If $m = 0$, then Y is a point and $\varphi^{-1}(Y) = X$, so the statement is clear. Assume now $m > 0$ and choose a nonzero regular function $f \in A(Y)$ such $f(y) = 0$. Then we have

$$V_X(\varphi^*f) = \varphi^{-1}(V_Y(f)).$$

Let Z be any irreducible component of $\varphi^{-1}(y)$, then since $Z \subseteq V_X(\varphi^*f)$, we have Z is contained in an irreducible component S of $V_X(\varphi^*f)$. By Krull's PID theorem

$$\dim S = \dim X - 1$$

. Furthermore, S maps entirely to $V_Y(f)$, and so if T is the closure of $\varphi(S)$ in Y , then $\dim T = \dim Y - 1$. Hence, we may apply the induction hypothesis on the map

$$\Phi := \varphi|_S: S \rightarrow T,$$

to obtain that any component of $\Phi^{-1}(y)$ has dimension $\geq \dim S - \dim T = \dim X - \dim Y$. Since Z is entirely contained in S , the theorem follows. □

For the second part of the theorem we will need to build some more theory.

Definition 11.18. A regular map $\varphi: X \rightarrow Y$ is called **quasi finite** if $\varphi^{-1}(y)$ consists of finitely many points for all $y \in Y$.

Lemma 11.19

If $\varphi: X \rightarrow Y$ is a quasi finite dominant map, then $\dim X = \dim Y$.

Proof. Applying the first part of the dimension theorem, we find that for a generic point in Y satisfies

$$\dim \varphi^{-1}(y) \geq \dim X - \dim Y \geq 0.$$

By quasi finiteness we have $\dim \varphi^{-1}(y) = 0$ and the claim follows. □

Related to quasi finiteness is the following algebraic notion, which seems unrelated at first glance.

Definition 11.20. Let X and Y be affine varieties and $\varphi: X \rightarrow Y$ be a regular map. We say φ is **finite** if $A(X)$ is a finitely generated $\varphi^*A(Y)$ module.

Let us see some examples of finite morphisms

Example 11.21 1. Let $i: X \hookrightarrow Y$ be a closed immersion. Then the pullback of i is given up to isomorphism by

$$i^*: A(Y) \rightarrow \frac{A(Y)}{I(i(X))} \cong A(X).$$

In particular, $A(X)$ is a finitely generated $A(Y)$ module since $\bar{1}$ is a generator.

2. The projection $\pi: \mathbb{A}^{n+1} \rightarrow \mathbb{A}^n$ is not a finite morphism as the pullback induces an inclusion

$$k[x_1, \dots, x_n] \hookrightarrow k[x_1, \dots, x_{n+1}],$$

which is not a module finite extension, since x_{n+1} is not integral over $k[x_1, \dots, x_n]$.

3. Consider the inclusion $\mathbb{A}^1 \setminus \{0\} \hookrightarrow \mathbb{A}^1$, is not finite. Indeed, its pullback corresponds to the map

$$i^*: k[t] \hookrightarrow k[t, \frac{1}{t}].$$

This is not a module-finite extension: indeed, $\frac{1}{t}$ has minimal polynomial $f(x) = xt - 1 \in k[t][x]$ and so $\frac{1}{t}$ is not integral over $k[t]$.

Remark 11.22. Suppose $f: X \rightarrow Y$ is a finite morphism and $V \subseteq Y$ is an affine subvariety (either closed or open). Then, let $W := f^{-1}(V)$ and consider the morphism

$$f|_W: W \rightarrow V.$$

We have a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{f|_W} & V \\ \downarrow i & & \downarrow i \\ X & \xrightarrow{f} & Y \end{array}$$

Applying the pullback functor, we receive

$$\begin{array}{ccc} A(W) & \xleftarrow{(f|_W)^*} & A(V) \\ i^* \uparrow & & \uparrow i^* \\ A(X) & \xleftarrow{f^*} & A(Y) \end{array}$$

Now, if V is closed, both i^* are surjective. Choosing an $A(Y)$ module-generating set of $A(Y)$, this set projects to an $A(V)$ -generating set of $A(W)$. For an open subset this is also true but requires more work...

Lemma 11.23

Let $\varphi: X \rightarrow Y$ be a dominant finite morphism, then φ is also quasi finite.

Remark 11.24. The converse is not true, as we have seen in example 11.20.3

Proof. Let $y \in Y$ be such that the preimage $\varphi^{-1}(y)$ is nonempty. we need to show that it is finite. let $\mathfrak{m}_y$ be the maximal ideal in $A(Y)$ corresponding to the point y . A point $x \in X$ maps to y if and only if all regular functions in $A(Y)$ which vanish on y , pullback

to functions which vanish on x . In otherwords,

$$\varphi^*(\mathfrak{m}_y) \subseteq \mathfrak{m}_x.$$

Hence, we have

$$\varphi^{-1}(y) = Z_X(\varphi^*\mathfrak{m}_y) \subseteq X$$

which is in particular an affine algebraic set and we have

$$A(\varphi^{-1}(y)) = \frac{A(X)}{\sqrt{\varphi^*\mathfrak{m}_y A(X)}}.$$

In any case, $A(\varphi^{-1}(y))$ is a quotient of $A(\varphi^{-1}(y)) = \frac{A(X)}{\varphi^*\mathfrak{m}_y}$. Thus it suffice to prove that the ring $\frac{A(X)}{\varphi^*\mathfrak{m}_y}$ only has finitely many maximal ideals. Now we know, that $A(X)$ is a finitely generated $A(Y)$ -module. Thus, as $A(Y)$ -modules we have

$$\frac{A(X)}{\varphi^*\mathfrak{m}_y A(X)} \cong A(X) \otimes_{A(Y)} \frac{A(Y)}{\mathfrak{m}_y} \cong A(X) \otimes_{\varphi^* A(Y)} k$$

Now notice that if $A(X)$ is generated, as an $A(Y)$ module by $x_1, \dots, x_r$, then these generate $A(X) \otimes_{\varphi^* A(Y)} k$ as a k vector space, and thus $A(\varphi^{-1}(y))$ is a finite dimensional k vector space and therefore has dimension 0. in particular, $\varphi^{-1}(y)$ consists of finitely many points. $\square$

Recall commutative algebra,

Theorem 11.25 (Noether's Normalization)

Let E be a field and F a finitely generated E -algebra with transcendence degree $n = \text{tr deg}_E(F)$ over E . Then there exists $\alpha_1, \dots, \alpha_n \in F$ which are algebraically independent over E , such that F is a finite $E[\alpha_1, \dots, \alpha_n]$ module.

Corollary 11.26

Let X be any affine variety of dimension n . Then there exists a finite dominant morphism

$$X \rightarrow \mathbb{A}^n.$$

Proof. Apply Noether's normalization to obtain an inclusion $k[x_1, \dots, x_n] \hookrightarrow A(Y)$ which is an injective finite module extension. The corresponding morphism corresponds to the desired map $\square$

§12 Lecture 12

In the last lecture, we proved part (1) of Theorem 11.15, and we begin this lecture by proving the second statement of this theorem

Part (2) of Theorem 11.15. We start the proof by reducing to a more simple case, namely that Y and X are both affine. Let us show how these reductions can be made. First, by passing to an open affine subset of Y , and replacing X by the φ preimage of this set, we can assume that Y is affine. Indeed passing to these sets does not change any of the assumptions from the theorem. Furthermore, let us cover X by open affine sets

$X = \bigcup_{i=1}^r U_i$. For any $y \in Y$, and any irreducible component Z of $\varphi^{-1}(y)$, we have that $Z \cap U_j$ is nonempty for at least one of the j . And thus, $\dim Z = \dim Z \cap U_j$ and $Z \cap U_j$ is an irreducible component of $(\varphi|_{U_j})^{-1}(y)$. Let us denote by φ_i for $i = 1, \dots, r$, the corresponding restrictions of φ to the U_i . If we knew the claim to be true for all the φ_i , then we would obtain open sets $Y_i, \dots, Y_r$ such that $\varphi_i|_{\varphi^{-1}(Y_i)}$ satisfies the conditions from (2) in the theorem. Then consider $V = \bigcap_{i=1}^r Y_i \subseteq Y$, which is open and nonempty. Then, for any $y \in V$, an irreducible component Z of $\varphi^{-1}(y)$ meets one of the U_i , and $\dim Z = \dim Z \cap U_i = n - m$. Hence we have reduced to the case where both X and Y are affine, and we may study the morphism via its pullback

$$\varphi^* : A(Y) \rightarrow A(X).$$

Since φ is dominant, the pullback φ^* is injective. Furthermore, consider the ring

$$R := A(X) \otimes_{A(Y)} k(Y).$$

Which may be viewed as a subring of $k(X)$ consisting of fractions with denominators in $A(Y)$. In diagrams we have:

$$\begin{array}{ccccc} A(Y) & \xrightarrow{\varphi^*} & A(X), & & \\ \downarrow i & & \downarrow & \searrow i & \\ k(Y), & \xrightarrow{i} & R & \xrightarrow{i} & k(X) \\ & \searrow \varphi^* & & & \end{array}$$

R is a finitely generated $k(Y)$ -algebra since $A(X)$ is a finitely generated k -algebra. Furthermore,

$$\text{tr deg}(\text{Frac}(R)/k(Y)) = \text{tr deg}(k(X)/k(Y)) = n - m$$

Thus we can apply Noethers normalization to obtain elements $t_1, \dots, t_{n-m}$, such that R is a finite extension over the polynomial ring $k(Y)[t_1, \dots, t_{n-m}]$.

Hence, R is generated by finitely many elements $\gamma_1, \dots, \gamma_r$ which are all algebraic over $k(Y)[t_1, \dots, t_{n-m}]$. Let $F_i \in k(Y)[t_1, \dots, t_{n-m}][X]$ be the corresponding minimal polynomials of the γ_i . For the t_i , the γ_i and the F_i , they all have denominators in $A(Y)$. Thus, by inverting finitely many elements in Y , and passing to this smaller open subset, we may assume that $t_i, \gamma_i \in A(X)$, and also $F_i \in A(Y)[t_1, \dots, t_{n-m}][X]$. The elements in $A(X)$ correspond to the elements in R which do not have a non unit from $A(Y)$ as a denominator. In fact, with our assumptions above, every element in $r \in R$, can be written as

$$r = \frac{P}{y}, P \in A(Y)[t_1, \dots, t_{n-m}][\gamma_1, \dots, \gamma_r], y \in A(X) \setminus \{0\}$$

If this is in $A(X)$, y can be taken to be a unit. In particular, $A(X)$ is module finite over $A(Y)[t_1, \dots, t_{n-m}]$. The ring homomorphism,

$$A(Y)[t_1, \dots, t_{n-m}] \hookrightarrow A(X),$$

corresponds to a map

$$\psi : A(X) \rightarrow Y \times \mathbb{A}^{n-m},$$

which is dominant and finite. We have a commutative diagram

$$\begin{array}{ccc}
 X & \xrightarrow{\psi} & Y \times \mathbb{A}^{n-m} \\
 & \searrow \varphi & \downarrow \pi \\
 & & Y
 \end{array}$$

Suppose now that $y \in Y$ is such that $\varphi^{-1}(y)$ is nonempty and let $Z \subseteq \varphi^{-1}(y)$ be an irreducible component. Since ψ is dominant and finite, it is also quasi finite. Thus, $\varphi|_Z$ is still a quasi finite map and it may be viewed as a map $Z \rightarrow \mathbb{A}^{n-m}$. Thus, $\dim Z \leq n - m$. But by part (1) the reverse inequality holds as well. $\square$

Let us view a few more consequences of the theorem on fibers

Corollary 12.1

Let $\varphi: X \dashrightarrow Y$ be a dominant rational map between varieties of equal dimension. The, φ is generically finite. In otherwords, there exists a nonempty open subset $U \subseteq Y$, which is open and affine, such that

- $\varphi^{-1}(U)$ is affine and φ is regular here,
- $\varphi: \varphi^{-1}(U) \dashrightarrow U$ is a finite morphism.

Proof. Not entirely sure.... $\square$

We continue with a discussion of codimension.

Definition 12.2. Let X be an irreducible noetherian topological space and $Y \subseteq X$ an irreducible closed subset. Then the codimension of Y in X , denoted $\text{codim}_X(Y)$, is defined as

$$\text{codim}_X(Y) = \sup\{n \in \mathbb{N} \mid \exists \text{ chain } : Y \subseteq Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \subseteq X, Z_i \text{ closed irreducible} \}$$

As to be expected by now, this has an algebraic counterpart:

Definition 12.3. Let R be a ring and $\mathfrak{p} \in \text{Spec } R$. Then the height of $\mathfrak{p}$, denoted $\text{ht } \mathfrak{p}$, is defined as

$$\text{ht } \mathfrak{p} = \sup\{n \in \mathbb{N} \mid \exists \text{ chain } : \mathfrak{p} \supseteq \mathfrak{p}_0 \supsetneq \mathfrak{p}_1 \supsetneq \cdots \supsetneq \mathfrak{p}_n \supseteq (0), \mathfrak{p}_i \in \text{Spec } R\}$$

Remark 12.4. We can also calculate the height as

$$\text{ht } \mathfrak{p} = \dim R_{\mathfrak{p}}.$$

Lemma 12.5

Let X be an affine variety and $Y \subseteq X$ an affine subvariety. then

$$\text{codim}_X(Y) = \text{ht}(I(Y)) = \dim A(X)_{I(Y)}$$

From commutative algebra we have Krull's theorem:

Theorem 12.6 (Krull)

Let R be a ring and $\mathfrak{p} \in \operatorname{Spec} R$, then

$$\dim R = \operatorname{ht} \mathfrak{p} + \dim(R/\mathfrak{p})$$

Translating this to geometry, we obtain:

Corollary 12.7

$$\operatorname{codim}_X(Y) = \dim X - \dim Y.$$

This concludes our discussion of dimension for now. We continue with smoothness

Smoothness

Let us start with a motivating example.

Example 12.8

Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a smooth function and consider the zero set

$$C = \{(x, y) \in \mathbb{R}^2 \mid F(x, y) = 0\}.$$

Consider furthermore, the derivative of F

$$DF_P = \begin{bmatrix} \frac{\partial F}{\partial x}(P) & \frac{\partial F}{\partial y}(P) \end{bmatrix}$$

Let $P = (p_1, p_2)$ be a point in C , and suppose $\frac{\partial F}{\partial y}(P) \neq 0$. Then the implicit function theorem asserts the existence of an open neighbor U of p_1 and W of p_2 along with a smooth function

$$f: U \rightarrow W$$

which maps p_1 to p_2 such that for all $x \in U$ there is a unique point $f(x) \in W$, which satisfies

$$F(x, f(x)) = 0.$$

In particular, the curve C , is locally parametrized near P by $g: U \rightarrow C$, $x \mapsto (x, f(x))$. In particular, C is smooth there.

Now if the derivative of f is zero at P , then C can not be smooth there. The key fact was that the tangent space $T_P C$ is given as the zero set of

$$\frac{\partial F}{\partial x}(P)(x - p_x) + \frac{\partial F}{\partial y}(P)(y - p_y) = \begin{bmatrix} \frac{\partial F}{\partial x}(P) & \frac{\partial F}{\partial y}(P) \end{bmatrix} \begin{pmatrix} x - p_x \\ y - p_y \end{pmatrix}$$

has dimension 1.

Let us take inspiration from analysis to generalize this to any field.

Definition 12.9. let k be a field, and $f \in k[x_1, \dots, x_n]$. Representing for each i , f as

$$f = \sum_{i=0}^{n_i} \alpha_i x^i \quad \alpha_i \in k[x_1, \dots, \hat{x}_i, \dots, x_n],$$

we define its partial derivative with respect to x_i as

$$\frac{\partial f}{\partial x_i} = \sum_{i=0}^{n_i} i \alpha_i x^{i-1}$$

Definition 12.10. Let $X \subseteq \mathbb{A}^n$ be an affine algebraic set, and suppose

$$I(X) = (f_1, \dots, f_r).$$

For a point $p \in X$, we define the **tangent space** of X in p as

$$T_p X = Z(d_p f_1, \dots, d_p f_r),$$

where

$$d_p f_i = \sum_{i=1}^n \frac{\partial f_i}{\partial x_i}(p)(x_i - p_i)$$

Remark 12.11. We should check that this is independent of choice of generators f_i . Indeed suppose,

$$(f_1, \dots, f_r) = (g_1, \dots, g_s).$$

Then we may write

$$f_i = \sum_{j=1}^s A_{i,j} g_j.$$

If $p \in X$, we can take the derivatives to obtain

$$\begin{aligned} d_p f_i &= d_p \left(\sum_{j=1}^s A_{i,j} g_j \right) \\ &= \sum_{j=1}^s d_p (A_{i,j} g_j) \\ &= \sum_{j=1}^s \left(\sum_{k=1}^n \frac{\partial (A_{i,j} g_j)}{\partial x_k}(p)(x_i - p_i) \right) \\ &= \sum_{j=1}^s \left(\sum_{k=1}^n \left(g_j \frac{\partial A_{i,j}}{\partial x_k} + A_{i,j} \frac{\partial g_j}{\partial x_k} \right)(p)(x_i - p_i) \right) \\ &= \sum_{j=1}^s \left(\sum_{k=1}^n A_{i,j}(p) \frac{\partial g_j}{\partial x_k}(p)(x_i - p_i) \right) \\ &= \sum_{j=1}^s A_{i,j}(p) \left(\sum_{k=1}^n \frac{\partial g_j}{\partial x_k}(p)(x_i - p_i) \right) \\ &= \sum_{j=1}^s A_{i,j}(p) d_p g_j \end{aligned}$$

This shows that

$$Z(d_p g_1, \dots, d_p g_s) \subseteq Z(d_p f_1, \dots, d_p f_r).$$

The reverse inclusion follows symmetrically.

Definition 12.12. Let X be an affine algebraic set and $p \in X$. We say that X is **smooth** at p , or that p is a smooth point of X , if the tangent space of X at p has dimension equal to $\dim X$. A point $p \in X$ is called a singular point if it is not smooth. X is called smooth if all its points are smooth points.

In other words, If $I(X) = (f_1, \dots, f_r)$ we want the matrix

$$J_P = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(P) & \dots & \frac{\partial f_1}{\partial x_n}(P) \\ \vdots & & \vdots \\ \frac{\partial f_r}{\partial x_1}(P) & \dots & \frac{\partial f_r}{\partial x_n}(P) \end{bmatrix}$$

To have kernel of dimension $\dim X$.

Proposition 12.13

Let $f \in k[x_1, \dots, x_n] \setminus \{k\}$ be a polynomial and $X = Z(f)$. Then

$$\dim T_p X \geq n - 1 \quad \forall p \in X,$$

and equality holds on a nonempty open subset. In particular, since $\dim X = n - 1$ by Krull's PID, we have that X is generically smooth.

Proof. Let us assume that f is square free, and so $I(X) = (f)$. Then the tangent space of X at a point $p \in X$ is given by

$$T_p X = Z\left(\frac{\partial f}{\partial x_1}(p)(x_1 - p_1) + \dots + \frac{\partial f}{\partial x_n}(p)(x_n - p_n)\right)$$

This has dimension $\geq n - 1$, and dimension equal to n if and only if

$$\frac{\partial f}{\partial x_1}(p) = \dots = \frac{\partial f}{\partial x_n}(p) = 0.$$

Thus, the smooth points of X are those where these equations don't vanish. It suffice to show that there exists a point where not all partials vanish. If $\text{char } k = 0$, this is obvious since f is nonconstant. If $\text{char } k = p$, we find that $f \in k[x_1^p, \dots, x_n^p]$, and is therefore a power of p , which contradicts that we chose f square free. $\square$

In general we define

Definition 12.14. Let X be an affine algebraic set. The **Singular locus of X** , denoted $\text{Sing}(X)$ consists of the set of points where X is not smooth.

Remark 12.15. Let $p = (0, 0)$ and suppose $f \in k[x_1, \dots, x_n]$. Then we may write

$$f = f_0 + f_1 + \dots + f_n,$$

where f_i is a homogeneous form of degree i . If $f(p) = 0$, then $f_0 = 0$. In fact,

$$d_p f = f_1.$$

Indeed, note that if $F, G \in k[x_1, \dots, x_n]$, then

$$d_p(FG) = F(p)d_p(G) + G(p)d_p(F).$$

Thus, $d_p(f_i) = 0$ for $i > 1$. In general, if p is any point, and $f(p) = 0$, then we may represent $f = f_0 + \dots + f_n$, but where the f_i are homogeneous forms in the variables $t_i = x_i - p_i$.

Tasks For Week 6

Algebraic Geometry I

Exercise sheet 6

Throughout, k denotes an algebraically closed field.

Exercise 12.1. Let X, Y be projective varieties in $\mathbb{P}^n$. Assume that

$$\dim X + \dim Y \geq n.$$

Prove that $X \cap Y \neq \emptyset$ and that any irreducible component of $X \cap Y$ has dimension at least

$$\dim X + \dim Y - n,$$

via the following argument.

- (1) Let $\Delta \subset \mathbb{A}^r \times \mathbb{A}^r$ be the diagonal, i.e., the subset

$$\{(x, x) \mid x \in \mathbb{A}^r\}.$$

Show that Δ is closed and that

$$\Delta = Z_{\mathbb{A}^r \times \mathbb{A}^r}(f_1, \dots, f_r)$$

for r polynomials.

- (2) Assume V and W are subvarieties of $\mathbb{A}^r$, of dimension n and m respectively. Show that if $V \cap W$ is not empty, then any of its irreducible components has dimension at least

$$n + m - r.$$

Hint: consider the intersection

$$(V \times W) \cap \Delta$$

in $\mathbb{A}^r \times \mathbb{A}^r$ and use Exercise 2 from sheet 5.

- (3) Conclude the proof by considering the affine cones

$$C(X), C(Y) \subset \mathbb{A}^{n+1}$$

of the projective varieties X and Y .

Solution. (1) Let $x_1, \dots, x_r, y_1, \dots, y_r$ be coordinates for $\mathbb{A}^r \times \mathbb{A}^r$, then the diagonal is given by

$$\Delta = V(x_i - y_i \mid i = 1, \dots, r).$$

- (2) We claim that

$$V \cap W \cong (V \times W) \cap \Delta.$$

Indeed the maps

$$V \cap W \rightarrow (V \times W) \cap \Delta, x \mapsto (x, x) \tag{1}$$

$$(V \times W) \cap \Delta \rightarrow V \cap W, (x, x) \mapsto x \tag{2}$$

are morphisms and define inverses to each other. Furthermore, we have

$$\Delta \cap (W \times V) = Z_{V \times W}(f_1, \dots, f_r),$$

where the f_i are given from (1). Applying Exercise 2 from sheet 5, the result follows.

(3) Consider the affine cones

$$\begin{aligned} C(X) &= \overline{\pi^{-1}(X)} \\ C(Y) &= \overline{\pi^{-1}(Y)} \end{aligned}$$

where π is the projection from $\mathbb{A}^{n+1} \setminus \{0\}$ to $\mathbb{A}^n$. Consider first the cone over X . for it we have a dominant rational map

$$\pi: C(X) \dashrightarrow X, (x_1, \dots, x_n) \mapsto [x_0 : \dots : x_n].$$

All the fibres are one dimensional, and thus, the theorem about the dimension of fibres ensures:

$$\dim C(X) - \dim X = 1.$$

In otherwords, $\dim C(X) = \dim X + 1$ and similarly for Y . Consider now the intersection of cones

$$C(X) \cap C(Y).$$

by (2), any of the irreducible components of this intersection have dimension a tleast

$$\dim C(X) + \dim C(Y) - (n + 1) \geq 1.$$

In particular, $C(X)$ and $C(Y)$ intersect in a nonzero point. This nonzero point corresponds to an intersection point of X and Y .

□

Exercise 12.2. Let $\varphi: X \rightarrow Y$ be a finite surjective morphism of degree d of affine k -varieties. Assuming that k is of characteristic 0, show that there exists a non-empty open subset U of Y such that $\varphi^{-1}(y)$ consists of exactly d points. What could happen if $\text{char}(k) > 0$?

Prove the above statement in the following steps.

Since $\text{char}(k) = 0$, the field extension

$$k(X)/\varphi^*(k(Y))$$

is separable. By the theorem of the primitive element,

$$k(X) = k(Y)[\beta] = k(Y)[t]/(f)$$

for some irreducible polynomial $f \in k(Y)[t]$, the minimal polynomial of β . We then have

$$\deg(f) = d.$$

Moreover, f has d distinct roots in the algebraic closure $\overline{k(Y)}$ of $k(Y)$.

Remark 12.3. I was not able to prove the task without the following fact: finite morphisms are closed maps. This is a result due to the going up theorem.

(i) Show that up to shrinking X and Y ; that is, replacing X and Y with some non-empty open subsets, you may assume that

$$\beta \in A(X) \quad \text{and} \quad f \in A(Y)[t],$$

and that

$$A(X) = A(Y)[t]/(f).$$

Shrinking X and Y further, you may also assume that f is irreducible and monic.

(ii) Let $y \in Y$. Consider the polynomial

$$f_y \in k[t]$$

obtained evaluating the coefficients of f , which are regular functions on Y , at the point y . Show that the cardinality of $\varphi^{-1}(y)$ equals the number of distinct roots of f_y .

(iii) Deduce that $\varphi^{-1}(y)$ consists of at most d points, and that there exists a non-empty open subset of Y such that $\varphi^{-1}(y)$ consists of exactly d points.

Hint: to prove the last point use that for any field L and a polynomial $p \in L[t]$, there exists a polynomial $\Delta(p)$, the discriminant of L , such that p has some multiple root in the algebraic closure $\bar{L}$ if and only if $\Delta(p)$ is the zero polynomial. Apply this to $f \in k(Y)[\beta]$: since f has no repeated roots, $\Delta(f)$ is not the zero polynomial. Upon shrinking further X and Y if necessary, you may assume

$$\Delta(f) \in A(Y)[t];$$

evaluating the coefficients of $\Delta(f)$ at $y \in Y$ gives the discriminant of $f_y \in k[t]$.

Proof. (i) Let the minimal polynomial f , of β be

$$f = a_d t^d + \cdots + a_1 t + a_0 \in k(Y)[t]$$

Each a_i is represented as a fraction $a_i = \frac{A_i}{B_i}$ with $A_i, B_i \in A(Y)$. On the open affine set

$$U = \bigcap_i D(B_i) = D\left(\prod_i B_i\right),$$

the a_i define regular functions and hence, if we replace Y by U we may assume $a_i \in A(Y)$. To ensure φ still fulfills the properties, we replace X by

$$\varphi^{-1}(U) = D(\varphi^*\left(\prod_i B_i\right)),$$

which is evidently still affine. Similarly, we replace Y again with $D(a_d)$ and X by $\varphi^{-1}(D(a_d)) = D(\varphi^*(a_d))$ to assume that the leading coefficient of f is a unit and hence, up to rescaling by a unit we can assume that f is monic and hence, β is actually integral over $A(Y)$. We can write it as $\beta = \frac{g}{h}$ where $g \in A(X)$ and $h \in A(Y)$. Replacing Y by the affine $D(g)$ and X by the affine $\varphi^{-1}(D(g)) = D(\varphi^*(g))$, we may assume $\beta \in A(X)$. As such, we have

$$A(Y) \subseteq A(Y)[\beta] \subseteq A(X).$$

On the other hand $\text{Frac}(A(Y)[\beta]) = \text{Frac}(A(X)) = k(X)$. Since $k(X)/k(Y)$ is a finite algebraic extension, we then have:

$$k(Y) \otimes_{A(Y)} A(Y)[\beta] \cong \text{Frac}(A(Y)[\beta]) = \text{Frac}(A(X)) \cong k(Y) \otimes_{A(Y)} A(X)$$

since localization is exact, this implies

$$0 = k(Y) \otimes_{A(Y)} \left(\frac{A(X)}{A(Y)[\beta]} \right) = S^{-1} \left(\frac{A(X)}{A(Y)[\beta]} \right) \quad \text{for } S = A(Y) \setminus \{0\} \subseteq A(Y).$$

We also know that the quotient $\left(\frac{A(X)}{A(Y)[\beta]}\right)$ is a finitely generated $A(Y)$ module. If $\bar{b}_1, \dots, \bar{b}_n$ are generators, they are all zero when localizing at S . Thus, there exists $s_i \in A(Y) \setminus \{0\}$ such that $s_i \bar{b}_i = 0$. Consider the affine open subset $U = D(\prod_i s_i) \subseteq Y$. On this subset the s_i are all units and thus for this affine subset we have $\bar{b}_i = 0$. If we replace Y by U and X by $\varphi^{-1}(U)$, we can assume $A(X) = A(Y)[\beta]$. One should take a moment to verify that in the last step we didn't change the assumptions. The fact that f is irreducible is automatic.

- (ii) We will use this simplifying reduction proven in (i). Let $y \in Y$ be any point, and $\mathfrak{m}_y$ be its corresponding maximal ideal in $A(Y)$. Then a point $x \in X$ maps to y if and only if its maximal ideal $\mathfrak{m}_x \subseteq A(X) = A(Y)[\beta]$ contains $\mathfrak{m}_y$. Thus, we can identify the fiber of y with the maximal ideals of

$$\frac{A(Y)[\beta]}{(\mathfrak{m}_y)A(Y)[\beta]}.$$

Let us take a different viewpoint of this ring: Since $A(Y)[\beta] \cong \frac{A(Y)[t]}{(f)}$, via $t \mapsto \beta$. We get an isomorphism

$$\frac{A(Y)[\beta]}{(\mathfrak{m}_y)A(Y)[\beta]} \cong \frac{A(Y)[t]/(f)}{(\mathfrak{m}_y)(A(Y)[t]/(f))} \cong \frac{k[t]}{(f_y)}$$

where the last isomorphism is induced by $A(Y) \ni x \mapsto x(y) \in k$. Thus, we have shown that the fiber of y corresponds to the maximal ideals of $\frac{k[t]}{(f_y)}$. These, in turn, correspond to the maximal ideals of $k[t]$ which contain f_y . Since k is algebraically closed, this equals the number of distinct roots of f_y .

- (iii) $f_y \in k[t]$ has distinct roots if and only if the polynomial $\Delta(f_y) \in k[t]$ is not the zero polynomial. Furthermore, the morphism $A(Y)[t] \rightarrow k[t]$ induced by evaluating coefficients at y , maps $\Delta(f)$ to $\Delta(f_y)$ since the discriminant is given by a polynomial on the coefficients of f , and these coefficients depend polynomially on y . Hence we must show that $\Delta(f) \in K(y)[t]$ is not the zero polynomial. But since $\text{char}(k) = 0$, the extension $k(X)/k(Y)$ is separable and thus f has distinct roots in the algebraic closure of $k(X)$ since it is irreducible. □

Exercise 12.4. Let $X \subset \mathbb{P}^n$ be a projective variety of dimension m ; assume that

$$n > m + 1.$$

Consider the *incidence variety*

$$I := \{(P, Q, R) \in X \times X \times \mathbb{P}^n \mid P \neq Q, R \in L_{PQ}\},$$

where L_{PQ} is the line through P and Q .

- (i) Show that $\dim I \leq 2m + 1$.
(ii) Consider the projection

$$\text{pr} : I \rightarrow \mathbb{P}^n$$

to the third factor. Show that for a general point S of $\mathbb{P}^n$ the fibre

$$\text{pr}^{-1}(S)$$

has dimension at most $m - 1$.

(iii) Show that for $S \in \mathbb{P}^n$, projection from S induces a map

$$\pi_S : X \rightarrow \mathbb{P}^{n-1}$$

which is injective over an open subset of X .

Hint: what is the image of $\text{pr}^{-1}(S)$ via the first projection

$$I \rightarrow X?$$

Remark 12.5. I believe the statement as was written in (iii) is not true, for example if X is simply a line through S then the projection from S is a constant map, and thus certainly not injective. More generally the statement fails if X contains a line through S . However, the statement is true if we only require it for a general S rather than all S .

(iv) Assuming $\text{char}(k) = 0$, deduce via Exercise 2 that any m -dimensional variety is birational to a hypersurface in $\mathbb{P}^{m+1}$; this in fact holds in arbitrary characteristic.

Solution. (i) We should first clarify that I is a variety. Note that the condition $R \in L_{PQ}$ can be phrased in terms of the rank of a certain matrix being 2:

$$\text{rank} \begin{pmatrix} P & Q & R \end{pmatrix} = 2,$$

Since we assume $P \neq R$, this can be phrased by requiring that all (2×2) minors of this matrix disappear. Thus we have :

$$I = ((X \times X) \setminus \Delta_X) \cap \{(P, Q, R) \in X \times X \times \mathbb{P}^n \mid \text{rank} \begin{pmatrix} P & Q & R \end{pmatrix} \leq 2\},$$

which is a open subset of the closed subset

$$S = \{(P, Q, R) \in X \times X \times \mathbb{P}^n \mid \text{rank} \begin{pmatrix} P & Q & R \end{pmatrix} \leq 2\}$$

We claim that S is also irreducible. Let $C(X) \subseteq \mathbb{A}^{n+1}$ be the cone over X and $\pi := C(X) \setminus \{0\} \rightarrow X$ denote the projection. We Then have a morphism

$$\Phi : C(X) \times C(X) \times \mathbb{A}^2 \rightarrow \mathbb{A}^{n+1} \times \mathbb{A}^{n+1} \times \mathbb{A}^{n+1}, (P, Q, s, t) \mapsto (P, Q, sP + tQ).$$

Furthermore, the preimage of $(\mathbb{A}^{n+1})^3 \setminus \{0, 0, 0\}$ is an open subset of $C(X) \times C(X) \times \mathbb{A}^2$ and hence a quasi projective variety. Furthermore, this variety projects surjectively to S which implies that S is irreducible.

Now we can clarify the dimension. Since products are unique up to unique isomorphism we can identify $X \times X \times \mathbb{P}^n \cong (X \times X) \times \mathbb{P}^n$ and hence we may consider the restriction

$$\pi_{X \times X} : I \rightarrow (X \times X) \setminus \Delta_{X \times X}$$

Interpreting this map as a dominating map to $X \times X$, and noticing that the fibers are one dimensional, The theorem of the dimension of fibers tells us that:

$$\dim I - \dim X \times X \leq 1 \Rightarrow \dim I \leq 2m + 1,$$

In fact if we use the second part of the theorem We see that this is actually an equality unless I is empty which is equivalent to X being a single point.

- (ii) if $pr : I \rightarrow \mathbb{P}^n$ is not dominating then the claim is true, since the generic fiber is empty in this case. Otherwise we can use the second part of the theorem of the dimension of fibers to conclude that a generic point in $S \in \mathbb{P}^n$ satisfies:

$$\dim pr^{-1}(S) = \dim I - n.$$

Combining this with part (i) and the assumption $n > m + 1$, we receive:

$$\dim pr^{-1}(S) = \dim I - n \leq 2m + 1 - n < 2m + 1 - m - 1 = 2m.$$

Since this inequality is strict, and both sides are integers, we have $\dim pr^{-1}(S) \leq m - 1$.

- (iii) We will assume S is a point with the properties from (ii), as a general point satisfies these properties. Furthermore, we assume $S \notin X$ so that the projection is actually a morphism rather than a rational map. Then

$$pr^{-1}(S) = \{(P, Q, S) \in X \times X \times \mathbb{P}^n \mid S \in L_{P,Q}\},$$

Has dimension at most $m - 1$. Consider the projection

$$\pi : X \times X \times \mathbb{P}^n \rightarrow X, (P, Q, R) \mapsto P,$$

and its restriction to I :

$$\pi : I \rightarrow X.$$

Since $\dim pr^{-1}(S) \leq m - 1$, we have that the closure: $\overline{\pi(pr^{-1}(S))} \subseteq X$ has codimension atleast one, and its complement $U = X \setminus \overline{\pi(pr^{-1}(S))}$ is a nonempty open subset of X . On the otherhand:

$$\pi(pr^{-1}(S)) = \{P \in X \mid \exists Q \in X : S \in L_{P,Q}\}.$$

Thus, for all points in $P \in U$, we have that the line $L_{P,S}$ contains no points in X other than P . Therefore, $pr|_U$ is an injective map.

- (iv) Let $X \subseteq \mathbb{P}^n$ be a m dimensional Variety. Then X is birational to its closure so let us assume wlog that X is closed in $\mathbb{P}^n$. We can prove the by induction over $n - m$. If $n = m$, Then $X \cong \mathbb{P}^n \cong H_0 \subseteq \mathbb{P}^{n+1}$ where H_0 is the linear subspace given by $\{X_0 = 0\}$. Similarly, if $n = m + 1$, then we are done so let us assume $n > m + 1$ and so the hypothesis from the previous tasks are true. It suffices to prove that the projection π_S constructed in (iii) induces a birational from X to its image, since then we reduce the quantity $n - m$ and the claim follows from induction. Let Y denote the closure of $\pi_S(X)$, since the fibers are generically zero dimensional, we have $\dim Y = \dim X$. Thus the degree $d = [k(X) : k(Y)]$ is finite and π_S is birational if and only if this degree is one. To prove this we will show that there exists an open affine variety $Y' \subseteq Y$, such that $X' := \pi_S^{-1}(Y') \subseteq X$ is also affine and such that $\pi_S|_{X'}$ is a finite morphism. Then, the claim follows from exercise (2) since d will equal the number of points in a generic fiber which we have shown to be one. Via some change of coordinates let us assume $S = [1 : 0 : \dots : 0]$ and the projection is given by restricting the map

$$\pi_S : \mathbb{P}^n \setminus \{S\} \rightarrow \mathbb{P}^{n-1}, [x_0 : \dots : x_n] \mapsto [x_1 : \dots : x_n],$$

Let $X_0, \dots, X_{n-1}$ be homogeneous coordinates for $\mathbb{P}^n$ and $Y_1, \dots, Y_n$ be homogeneous coordinates for $\mathbb{P}^{n-1}$. Then we have

$$\pi_S^{-1}(\{Y_i = 0\}) = \{X_i = 0\} \setminus \{S\}.$$

Choose an i such that $Y' = Y \cap \{Y_i = 0\} \neq \emptyset$, and observe:

$$X' := \pi_S^{-1}|_X(Y') = (X \cap \{X_i \neq 0\}) \setminus \{S\} = X \cap \{X_i \neq 0\},$$

is also affine. In particular we have a module extension $A(X')/A(Y')$. Since $A(X')$ is a finitely generated k -Algebra we have $\alpha_1, \dots, \alpha_r \in A(X)$ such that :

$$A(X') = k[\alpha_1, \dots, \alpha_r] = A(Y')[\alpha_1, \dots, \alpha_r].$$

Since the extension $K(X')/K(Y')$ is finite, the α_i all satisfy equations

$$f_i = \sum_j T^j a_{i,j} \in k(Y')[T].$$

By passing to the open affine subset U where all the a_i are in $A(U)$ and the leading coefficients are all units, we may assume that $f_i \in A(Y)[t]$ and monic, such that the α_i are actually integral. As we have seen in Exercise 2, the preimage of this U , is an affine subset W of X' , and $A(W)/A(U)$ is a module finite extension. Hence we have all the assumptions from (2) and the claim follows. $\square$

Exercise 12.6. Let $Q \subset \mathbb{P}^{n+1}$ be an irreducible quadric hypersurface, i.e.,

$$Q = \mathcal{Z}(f)$$

for an irreducible homogeneous polynomial of degree 2.

- (1) Let $P \in Q$ be a smooth point. Identify the space of lines in $\mathbb{P}^{n+1}$ through P with $\mathbb{P}^n$. Show that for lines L in a non-empty open

$$U \subset \mathbb{P}^n,$$

we have

$$L \cap Q = \{P, P'\}, \quad P \neq P'.$$

- (2) Assuming that k has characteristic 0, show that Q is rational, i.e., that it is birational to $\mathbb{P}^n$.

Solution. The identification of lines through P with $\mathbb{P}^{n-1}$ works via projection from P is illustrated in Figure (2)

- (1) Via some change of coordinates, assume $S \notin \{X_0 = 0\}$ and let the identification of lines through P be given via projection through P to $\{X_0 = 0\}$. For $R \in \mathbb{P}^n \cong \{X_0 = 0\} \subseteq \mathbb{P}^n$ be any point. The line identified with P is given by $L_{P,R}$ and it is given by

$$L : \{\lambda P + \mu R \mid [\lambda, \mu] \in \mathbb{P}^1\}$$

where here this sum is interpreted as follows: Choose any representative in $\mathbb{A}^{n+1}$ of the two points, form their linear combination, and projectivise. Thus, we can find:

$$L \cap Q = V_L(f|_L) = f(\lambda P + \mu R) \in k[\lambda, \mu]_2,$$

As a homogeneous quadratic in two variables, f factors into a product of two homogeneous linear forms and thus f intersects L twice when counted with multiplicity. Furthermore, this intersection occurs with multiplicity two, if and only if $\Delta(f(\lambda P + \mu R)) = 0$, which is a polynomial in the coordinates of R . It is not however, the zero polynomial: P is smooth point of Q and so not every line through P is also tangent to Q . In particular, we may choose a line through P not tangent to Q and see that P is not a double intersection point of $L \cap Q$, since if it were, then $L \in T_P Q$.

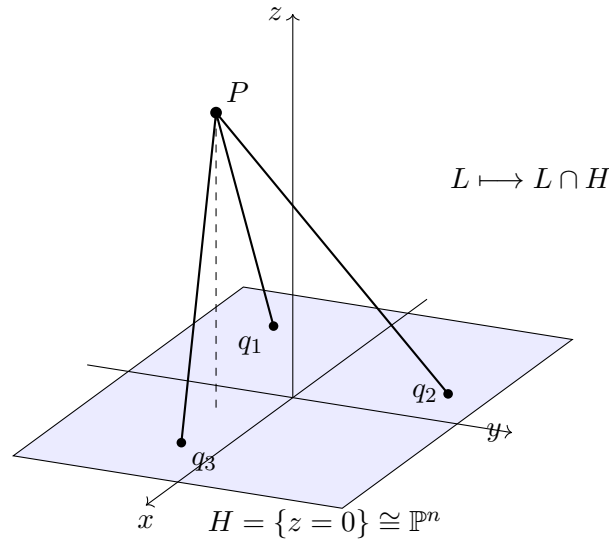


Figure 2: Lines through P are identified with their intersection points with $H = \{z = 0\}$.

- 2 This follows the same as the last part of problem (3), noting that projection from P induces a birational map to $\mathbb{P}^{n-1}$. Again this comes down to localizing in the image such that the projection becomes a finite map of affine varieties, and then applying Problem (2).

□

§13 Lecture 13

In the last lecture we introduced what it meant for an affine variety to be smooth at a certain point. Intuitively this should be a local property at p , and hence for any variety, one should be able to check smoothness by passing to open covers. However, as it stands now, our definition depends on the embedding of our variety and it is determined in terms of global equations. In this lecture, we will show that the definition we introduced last time is actually equivalent to a certain invariant of the local ring $\mathcal{O}_p$. Hence, for general varieties, we will use this invariant to define smoothness.

Definition 13.1. Let X be a quasi projective variety, and $p \in X$. Furthermore, let $\mathfrak{m}_p \subseteq \mathcal{O}_{X,p}$ be the maximal ideal of the local ring of X at p . The **Zariski cotangent space** is the k -vector space

$$\mathfrak{m}_p / \mathfrak{m}_p^2.$$

The **Zariski tangent space** is the dual of this space:

$$(\mathfrak{m}_p / \mathfrak{m}_p^2)^\vee = \text{Hom}_{k\text{-Vec}}((\mathfrak{m}_p / \mathfrak{m}_p^2), k)$$

Let us now show the connection of the tangent space and the zariski tangent space. Recall that $T_p X$ is the linear subspace $\mathbb{A}^n$ cut out by the equations $d_p f_1, \dots, d_p f_r$. Suppose now that $f \in I_X(p)$. Then f defines an element of $(T_p X)^\vee$, via

$$\Phi: I_X(p) \rightarrow (T_p X)^\vee, f \mapsto d_p f|_{T_p X}$$

Here $(T_p X)^\vee$ consist of maps $x \mapsto A(x - p)$, where A is a linear map to k .

Lemma 13.2

The map Φ is well defined and induces an isomorphism

$$\Phi: I_X(p)/I_X(p)^2 \xrightarrow{\sim} (T_p X)^\vee$$

Proof. We have a well defined morphism

$$\Phi: I_X(p) \rightarrow (T_p X)^\vee, f \mapsto d_p f|_{T_p X}$$

Suppose $I(X) = (f_1, \dots, f_r)$ and $f \in I(X)$. Then we have

$$f = \sum_{i=1}^r h_i f_i, \quad h_i \in k[x_1, \dots, x_n].$$

Taking differentials at p , we have

$$d_p f = \sum_{i=1}^r f_i(p) d_p h_i + \sum_{i=1}^r h_i(p) d_p f_i = \sum_{i=1}^r h_i(p) d_p f_i.$$

Hence $d_p f$ is a linear combination of $d_p f_i$, which by definition all vanish on $T_p X$. Thus Φ sends the ideal of X to zero. which means that the map

$$I_X(p) \rightarrow (T_p X)^\vee, f \mapsto d_p f$$

is well defined. This map is surjective as the elements $d_p(x_j - p_j) = x_j - p_j$ form a basis of $(T_p X)^\vee$ and they are in the image.

Assume for a moment that p is the origin and $f \in \ker \Phi$. Let us write $f = l + g$, where l is linear and $g \in I_{\mathbb{A}^n}(p)^2$. Then we find $d_p f = d_p l$. Since $\Phi(f) = d_p f = 0$, we must have that $d_p l = 0$ and thus $d_p l = \sum_{i=1}^r \lambda_i d_p f_i$ with $\lambda_i \in k$. Consider now $l' = l - \sum_{i=1}^r \lambda_i f_i$. then l' has no linear part at p which means $l' \in I_{\mathbb{A}^n}^2(p)$. Therefore $f = l' + g + \sum_{i=1}^r \lambda_i f_i$ belongs to $I_{\mathbb{A}^n}(p)^2 + I(X) = I_X(p)^2$. This means $\ker \Phi = I_X(p)^2$. For p not equal to the origin the statement follows in the same way, but by developing in terms of the monomials $x_i - p_i$ instead of the canonical coordinates x_i . $\square$

Lemma 13.3

there is a natural isomorphism of k vector spaces.

$$I_X(p)/I_X(p)^2 \xrightarrow{\sim} \mathfrak{m}_p/\mathfrak{m}_p^2,$$

where $\mathfrak{m}_p$ denotes the maximal ideal of $\mathcal{O}_{X,p}$.

Proof. Let S denote the multiplicative set $A(X) \setminus I_X(p)$ of regular functions on X which do not vanish at p . Consider now the exact sequence of $A(X)$ modules

$$0 \longrightarrow I_X(p)^2 \longrightarrow I_X(p) \longrightarrow I_X(p)/I_X(p)^2 \longrightarrow 0.$$

Since localization is an exact functor, it produces an exact sequence

$$0 \longrightarrow \mathfrak{m}_p^2 \longrightarrow \mathfrak{m}_p \longrightarrow S^{-1}(I_X(p)/I_X(p)^2) \longrightarrow 0.$$

Thus we obtain an isomorphism

$$\mathfrak{m}_p/\mathfrak{m}_p^2 \cong S^{-1}(I_X(p)/I_X(p)^2)$$

Now notice that $(I_X(p)/I_X(p)^2) \cong I_X(p) \otimes_{A(X)} A(X)/I_X(p)$ and S already acts on the second one via invertible elements, hence localizing doesn't actually change this module. Hence we get an isomorphism of k vector spaces as desired $\square$

Corollary 13.4

Let X be an affine variety and $p \in X$. Then there is a natural isomorphism

$$(T_p X)^\vee \cong \mathfrak{m}_p/\mathfrak{m}_p^2.$$

In particular, $T_p X$ does not depend on the specific embedding of X , and it is a local invariant. Applying the dual, we have a natural isomorphism $T_p X \cong (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$. If X is any quasi projective variety, we can define its tangent space at a point p to be the tangent space in any affine cover. Any of these choices are naturally isomorphic to dual space to $\mathfrak{m}_p/\mathfrak{m}_p^2$. We then say that X is smooth at a point p if and only if we have an equality of dimensions: $\dim_k T_p X = \dim X$.

Corollary 13.5

Let X be a any quasi projective variety. Then

$$\dim X \leq \dim T_p X.$$

Proof. Set

$$n := \dim_k T_p X = \dim_k (\mathfrak{m}_p/\mathfrak{m}_p^2).$$

Choose $f_1, \dots, f_n \in \mathfrak{m}_p \subseteq \mathcal{O}_{X,p}$ whose images form a basis of $\mathfrak{m}_p/\mathfrak{m}_p^2$. Nakayama's lemma ensures that they generate $\mathfrak{m}_p$ as an $\mathcal{O}_{X,p}$ module. The statement then follows from the following:

Let $(R, \mathfrak{m})$ be a local ring and $(f_1, \dots, f_n) = \mathfrak{m}$. Then $\dim R \leq n$.

Let us prove this by induction over n . If $n = 0$, then R is a field and it has dimension 0. Suppose now $n > 0$. and consider the ring $R/(f_n)$. It is still a local ring, with maximal ideal generated by $(f_1, \dots, f_{n-1})$. By induction, we have

$$\dim R/(f_n) \leq n - 1.$$

On the other hand, By Krull's PID theorem, we have

$$\dim R/(f_n) \geq \dim R - 1.$$

From this, the result follows. $\square$

This leads to the more general definition:

Definition 13.6. A noetherian local ring $(R, \mathfrak{m})$ is called **regular**, if its dimension is equal to the minimum number of generators of the maximal ideal.

Corollary 13.7

Let X be quasi projective variety and p be a point in X . Then X is smooth at p if and only if $\mathcal{O}_{X,p}$ is a regular local ring.

Proof. Since smoothness is local we are free to assume that X is an affine variety. Suppose first that X is smooth at p , then we have

$$\dim \mathcal{O}_{X,p} = \dim X = \dim T_p X = \dim(\mathfrak{m}_p / \mathfrak{m}_p^2).$$

On the other hand $\dim(\mathfrak{m}_p / \mathfrak{m}_p^2)$ is equal to the minimum number of generators of $\mathfrak{m}_p$ by Nakayama's lemma. Hence $\dim \mathcal{O}_{X,p}$ is a regular local ring.

Suppose now that $\dim \mathcal{O}_{X,p}$ is a regular local ring. Setting $n = \dim X = \dim \mathcal{O}_{X,p}$, we have that

$$\mathfrak{m}_p = (f_1, \dots, f_n).$$

This then implies

$$\dim_k \mathfrak{m}_p / \mathfrak{m}_p^2 \leq n.$$

However, by the previous proposition, the reverse inequality holds as well. Hence, we find

$$\dim T_p X = \dim X.$$

□

Although we have now developed a nice intrinsic theory for tangent spaces and smoothness, the theory we have developed so far is not very practical for actually determining smoothness in practice. Indeed, determining Krull dimensions, and minimal systems of generators in ideals can be hard. Luckily, inspired from analysis, there is a rather easy way to determine this in theory. Suppose we have $I(X) = (f_1, \dots, f_r)$. Then with respect to this system of generators, we may form the **Jacobian** matrix

$$J_p(X) := J_p := \left(\frac{\partial f_i}{\partial x_j(p)} \right)_{i,j}$$

This matrix clearly depends on choice of generators. However, as it turns out, it contains an invariant which is independent of generators.

Proposition 13.8

The variety X is smooth at a point p if and only if

$$\text{rank } J_p(X) = n - \dim X$$

Proof. The matrix J_p determines a linear map

$$J_p: k^n \mapsto k^r, \quad a = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \mapsto J_p \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n \frac{\partial f_1}{\partial x_i}(p) a_i \\ \vdots \\ \sum_{i=1}^n \frac{\partial f_r}{\partial x_i}(p) a_i \end{pmatrix} = \begin{pmatrix} d_p f_1(a+p) \\ \vdots \\ d_p f_r(a+p) \end{pmatrix}$$

From the last expression, we see that the kernel of this map is naturally identifiable, to $\ker(d_p f_1, \dots, d_p f_r)$, via a translation by p . In particular,

$$n - \text{rank } J_p = \dim \ker J_p = \dim T_p X$$

Thus we see that X is smooth at p if and only if

$$n - \text{rank } J_p = \dim X$$

This is equivalent to the claim. $\square$

In particular, since we have $\dim X \leq \dim T_p X$, the proof of the above shows

$$\text{rank } J_p \leq n - \dim X.$$

To check smoothness of a quasi projective variety, one can simply pass to an open cover of affine charts and apply the Jacobian criterion there.

Example 13.9

Using the Jacobian criterion, we obtain a rather surprising result, namely a method of checking whether a certain set of elements generate an ideal. Suppose we have X to be an affine variety and

$$X = Z(f_1, \dots, f_r),$$

But we don't know if $I(X) = (f_1, \dots, f_r)$. Hence we are asking whether the ring $k[x_1, \dots, x_n]/(f_1, \dots, f_r)$ is a reduced ring. We already know from Remark 12.11, that

$$Z(d_p f_1, \dots, d_p f_r) \subseteq T_p X,$$

Consider now the matrix

$$J'_p = \left(\frac{\partial f_i}{\partial x_j}(p) \right)_{i,j}$$

It has rank $n - m_p$ where $m_p := \dim_k Z(d_p f_1, \dots, d_p f_r)$. Suppose at every point we have that this matrix has rank $n - \dim X$. Then we must have that the rank of the true jacobian of X also has this rank, since it has at most this rank by the comment above, and also at least this rank since it is obtained by adding columns to J'_p . But then all the local rings $\mathcal{O}_{p,X}$ are regular local rings. These are just the reductions of the localisations:

$$R := (k[x_1, \dots, x_n]/(f_1, \dots, f_r))_{\mathfrak{m}_p}$$

These two rings clearly have the same Krull dimension since the nilradical is contained in the intersection of all prime ideals and hence prime chains are in one to one correspondence. Any set of generators for the maximal ideal of R generate the maximal ideal of $\mathcal{O}_{X,p}$. Conversely, suppose that $\{\bar{g}_1, \dots, \bar{g}_t\}$ generate the maximal ideal of $\mathcal{O}_{X,p}$. Then for the maximal ideal $\mathfrak{m}_R$ of R , we have

$$\mathfrak{m}_R = \langle g_1, \dots, g_t \rangle_R + \mathcal{N}(R),$$

where $\mathcal{N}(R)$ is the nilradical of R . But then by Nakayama's lemma, we have $\mathfrak{m}_R = \langle g_1, \dots, g_t \rangle_R$. Thus, the minimum number of generators of the maximal ideal agrees for both of these rings. Thus, since $\mathcal{O}_{X,p}$ is a regular local ring, R is as well. Since local rings are integral domains by the Auslander-Buchsbaum theorem, R is integral and hence reduced.

§14 Lecture 14

In this lecture we started by proving an application of the theory developed thus far. Namely we prove Bertini's theorem

Theorem 14.1 (Bertini's theorem)

Let $X \subseteq \mathbb{P}^n$ be a smooth projective variety of dimension ≥ 1 . Then, for a general hyperplane $H \subseteq \mathbb{P}^n$, The intersection $H \cap X$ is smooth. Furthermore, the ideal $I(X \cap H)$ is generated by the ideal of X and the linear equation of H .

To prove this theorem, we will need some preliminaries. Firstly, we should clarify what we mean by general hyperplane. Note that a hyperplane $H \subseteq \mathbb{P}^n$ is given by an equation

$$H = Z_{\mathbb{P}}^n(a_0X_1 + \dots a_nX_n),$$

where not all a_i are zero. Furthermore, two such equations define the same hyperplane if and only if they differ in a nonzero scalar $\lambda \in k^\times$. Thus the set of hyperplanes in $\mathbb{P}^n$, denoted by $(\mathbb{P}^n)^\vee$, is naturally identified with $\mathbb{P}^n$ and under this identification carries a Zariski topology. In this context, it becomes clear what we mean with general hyperplane: There exists a nonempty open subset $U \subseteq (\mathbb{P}^n)^\vee$, such that for all $H \in U$, the intersection $H \cap X$ is smooth.

To prove this we adapt the Jacobion criterion to the projective form;

Lemma 14.2

Let $X \subseteq \mathbb{P}^n$ be a projective variety, and let $I(X) = (F_1, \dots, F_r)$, for homogeneous polynomials F_i . Furthermore denote by $d = \dim X$ the dimension of X . For a point $p = [p_0 : \dots : p_n] \in X$, consider the jacobian matrix

$$\tilde{J}_P := \left(\frac{\partial F_i}{\partial x_j}(\hat{p}) \right)_{i=1, \dots, r}^{j=0, \dots, n},$$

where $\hat{p}$ is any representative in $\mathbb{A}^{n+1}$ of p . Then the rank of this matrix depends only on $p \in \mathbb{P}^n$, and X is smooth at p is and only if

$$\text{rank } \tilde{J}_p = n - d.$$

Proof. For two different representatives of $\hat{p}$, the correspoining matrices differ from the original one in a nonzero scalar in every row. Hence, the rank is invariant under change of representative. Suppose $p = [p_0, \dots p_n] \in X$ and for sake of simplicity let us suppose that $p_0 \neq 0$, hence we may choose a representative $p = [1 : t_1, \dots : t_n]$. Since smoothness is a local property, we can pass to the open subset $U = X \cap U_0$, which is an affine variety, with

$$I(U) = (F_1(T_0, T_1, \dots, T_n), \dots F_r(1, T_1, \dots, T_n)),$$

where $T_i = \frac{x_i}{x_0}$ are affine coordinates for $\mathbb{A}^n$ (in particular $T_0 = 1$). For each of the F_i , we may apply Eulers formula to obtain

$$\deg(F_i) \cdot F_i = \sum_{j=0}^n x_j \frac{\partial F_i}{\partial x_j}.$$

Let us denote $f_i(T_1, \dots T_n) = F_i(T_0, \dots T_n)$. The affine jacobian matrix at p is then

$$J_p = \left(\frac{\partial f_i}{\partial T_j}(p) \right)_{i=1, \dots, r}^{j=1, \dots, n},$$

However, we have

$$\frac{\partial f_i}{\partial T_j}(t) = \frac{\partial F_i}{\partial x_j}(1, T_1, \dots, T_n)$$

Now we claim

$$\text{rank } J_p = \text{rank } \tilde{J}_p,$$

Indeed, these matrices differ only in the fact that $\tilde{J}_p$ has an extra column, corresponding to taking derivatives with respect to x_0 . On the other hand, from eulers formula, we have

$$\frac{\partial F_i}{\partial x_0} = \deg(F_i)F_i - \sum_{j=1}^n x_j \frac{\partial F_i}{\partial x_j}.$$

Evaluating at p , we see that the coloumn corresponding to x_0 , is a linear combination of the other columns, and hence removing it does not change the rank of the matrix. $\square$

With this projective version of the Jacobion criterion, we can prove Bertini's theorem.

Proof of Bertini's Theorem. Let $I(X) = (F_1, \dots, F_r)$ for homogeneous polynomials F_i . Furthermore, let H be a hyperplane with coordinates in $(\mathbb{P}^n)^\vee$, given by $H = [h_0 : \dots : h_n]$. If X is not contained in H , then $X \cap H$ is a union of irreducible projective varieties all of dimension $d - 1$, where $d = \dim X$. For $P \in H \cap X$, if we find that

$$\text{rank} \begin{pmatrix} \frac{\partial F_i}{\partial x_j}(p) \\ h_0 \dots h_n \end{pmatrix} = n - (d - 1).$$

then the result would follow from the lemma 14.2 and Example 13.9. moreover if this condition holds for any $p \in H \cap X$, then $I \cap X$ is a disjoint union of smooth varieties.

Since X itself was smooth, the above condition is equivalent to

$$(h_0, \dots, h_n) \notin \langle (\frac{\partial F_i}{\partial x_j})_j | i = 1, \dots, r \rangle_k.$$

Consider now the incidence variety:

$$Z := \{(p, H) \in \mathbb{P}^n \times (\mathbb{P}^n)^\vee \mid p \in H\}$$

If $x_0, \dots, x_n$ are homogeneous coordinates for $\mathbb{P}^n$ and similarly $a_0, \dots, a_n$ for $(\mathbb{P}^n)^\vee$, then Z is cut out by the bihomogeneous equation

$$Z = Z(\sum_{i=0}^n a_i t_i).$$

This allows us to define:

$$Z_X = \{(p, H) \in X \times (\mathbb{P}^n)^\vee \mid p \in X\}$$

, which is also an algebraic set. It comes with two projections.

$$\begin{array}{ccc} & Z_X & \\ \swarrow \pi_X & & \searrow \pi_{(\mathbb{P}^n)^\vee} \\ X & & (\mathbb{P}^n)^\vee \end{array}$$

For a point $p \in X$, we have $\pi_X^{-1}(p) \cong \{H \mid p \in H\} \cong \mathbb{P}^{n-1}$, while for a hyperplane, we have

$$\pi_{(\mathbb{P}^n)^\vee}^{-1}(H) = H \cap X.$$

Consider now the set Consider now the set

$$W = \{(p, H) \in Z_x \mid (h_0, \dots, h_n) \notin \langle (\frac{\partial F_i}{\partial x_j}(p))_{j|i=1, \dots, r} \rangle_k\}.$$

For a given hyperplane $H \in (\mathbb{P}^n)^\vee$, the condition of $H \cap X$ being smooth is the equivalent to H not being in $\pi_{(\mathbb{P}^n)^\vee}(W)$. For a fixed $p \in X$ let $W_p := W \cap \pi^{-1}(p)$. From the condition above, we see that it is isomorphic to $\mathbb{P}^{n-d-1}$ inside the $\mathbb{P}^{n-1}$ of hyperplanes through p . Applying the theorem of the dimension of fibers, we obtain

$$\dim W \leq \dim X + (n - d - 1) = n - 1.$$

Thus, $\dim \pi_{(\mathbb{P}^n)^\vee}(W) \leq n - 1$ and it is therefore a proper closed subset of $(\mathbb{P}^n)^\vee$. Thus the complement of this set is a nonempty open with the desired properties. $\square$

This concludes our discussion of Bertin's theorem. In the following we introduce the notion of normal varieties, which will allow us to begin our discussion of divisors.

Definition 14.3. Let R be an integral domain. We say that R is **normal** if R is integrally closed in its field of fractions.

Example 14.4

Any UFD is integrally closed.

Translating this to the world of affine varieties, we define:

Definition 14.5. An affine variety is called normal its coordinate ring is closed.

Being normal is actually a local property:

Lemma 14.6

An integral domain R is normal if and only if all of its localizations at maximal ideals $R_{\mathfrak{m}}$ are normal.

Proof. Suppose first that R is a normal ring and $\mathfrak{m}$ is a maximal ideal of R . Note that $R_{\mathfrak{m}}$ and R share the same field of fractions F . Suppose now that $x \in F$ satisfies an equation

$$x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 = 0, \quad a_i \in R_{\mathfrak{m}}.$$

Let $d \in R - \mathfrak{m}$ be a common denominator of the a_i . Multiplying by d^n , we see that xd is in F and integral over R . Thus, by assumption, it is in R . But since d is a unit in $R_{\mathfrak{m}}$, this means that x is in $R_{\mathfrak{m}}$.

Suppose now R is an integral domain such that all of its localisations at maximal ideals are normal. Suppose furthermore, that $x \in F$, satisfies, fractions F . Suppose now that $x \in F$ satisfies an equation

$$x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 = 0, \quad a_i \in R.$$

Then the assumption states that $x \in \bigcap_{\mathfrak{m}} R_{\mathfrak{m}}$. Let us write $x = \frac{A}{B}$, for $A, B \in F$. If B is not a unit in R , then it is contained in some maximal $\mathfrak{m}$. For this maximal ideal, we claim $x \notin R_{\mathfrak{m}}$. Otherwise

$$\frac{1}{B} = \frac{C}{D}, D \notin \mathfrak{m} \Leftrightarrow D = BC.$$

which contradicts the choice D . $\square$

Remark 14.7. In fact, the above proof didn't really require that $\mathfrak{m}$ was maximal, the same proof goes through assuming only that it is prime.

Corollary 14.8

An affine variety is normal if and only if all its localizations are normal rings

Since we have a local condition, we may use this to extend to quasi projective varieties,

Definition 14.9. A quasi projective variety is called normal if all its local rings $\mathcal{O}_{X,p}$ are normal rings.

Note that these are, in fact, local rings. For local rings, we have

Lemma 14.10

Let $(R, \mathfrak{m})$ be a local ring. Then the following implications hold

$$(R, \mathfrak{m}) \text{ is regular} \Rightarrow (R, \mathfrak{m}) \text{ is UFD} \Rightarrow (R, \mathfrak{m}) \text{ is normal}$$

The first implication is called the Auslander Buchsbaum theorem, which is not easy to prove. Let us specialize this to rings of dimension one. For such rings we have:

Proposition 14.11

Let $(R, \mathfrak{m})$ be a local ring of dimension one, then the following are equivalent:

- 1 R is regular
- 2 $\mathfrak{m}$ is principal
- 3 R is a UFD
- 4 R is normal

Proof. $1 \Leftrightarrow 2$ is clear. Suppose now that $\mathfrak{m} = (t)$ is principal and $x \in R$ is not a unit. Then $x = tx_1$. If $\tilde{x}$ is a unit, we have found a factorisation of x . Otherwise continue this process to obtain an increasing chain

$$(x) \subseteq (x_1) \subseteq (x_2) \subseteq \dots,$$

Since R is noetherian, this chain terminates and we arrive at the factorization needed. In particular it is of the form $x = t^n \varepsilon$, for a unit ε ,

If $\mathfrak{m}$ is principal, R is a UFD, and R is normal as discussed above. The final implication can be found in Atiyah Macdonald. $\square$

We obtain from this

Corollary 14.12

Let X be a normal variety and $Z \subseteq X$ a subvariety of codimension one. Then $\mathcal{O}_{X,Z}$ is a regular local ring of dimension one.

Proof. The ring $\mathcal{O}_{X,Z}$ depends only on open subsets intersecting Z , hence we may assume wlog that X is affine. In this case, we know that $\mathcal{O}_{X,Z} = A(X)_{I(Z)}$, and since $A(X)$ is normal, this is a normal local ring. Furthermore, it has dimension $\text{codim} Z = 1$, so the proposition follows from the above theorem. $\square$

Applying this to the case when X is a curve and Z is a point in X we obtain:

Corollary 14.13

Let X be a normal quasi projective curve, Then X is smooth.

More generally, we have:

Theorem 14.14

Let X be a normal quasiprojective variety, then every irreducible component of $\text{Sing}(X)$ has at least codimension two.

Proof. Let $Z \subseteq X$ be any subvariety of codimension 1, we want to show that Z is not contained in $\text{Sing}(X)$. Passing to an open subset of X , we may assume that X is an affine variety, and thus $Z \subseteq X$, is a closed affine subvariety. Since X is a normal variety the ring $\mathcal{O}_{X,Z}$ is a regular local ring of dimension one, and thus its maximal ideal is generated by a single element π_Z . While π_Z may not be regular on X , up to shrinking of X , we may assume $\pi \in A(X)$: Therefore, $Z = Z_X(\pi_Z)$, Since Z is irreducible and (π_Z) is a prime ideal. Since Z is an affine variety, it contains a smooth point p , and as such, the local ring $\mathcal{O}_{Z,p}$ is a regular local ring of dimension $n - 1$, where $n = \dim X$. Therefore, its maximal ideal has generators $\bar{g}_1, \dots, \bar{g}_{n-1}$. Note now, that we have a closed immersion $Z \hookrightarrow X$, which induces a surjection

$$\mathcal{O}_{X,p} \twoheadrightarrow \mathcal{O}_{Z,p}.$$

The kernel of this map is precisely (π_Z) . Since it is surjective, we may choose lifts $g_1, \dots, g_{n-1}$ of the elements $\bar{g}_1, \dots, \bar{g}_{n-1}$. With this in mind, we obtain:

$$\mathcal{O}_{X,p}/(\pi_Z, g_1, \dots, g_n) \cong \mathcal{O}_{Z,p}/(\bar{g}_1, \dots, \bar{g}_{n-1}) \cong k.$$

Thus the maximal ideal of $\mathcal{O}_{X,p}$ is generated by n elements and as such $\mathcal{O}_{X,p}$ is a regular local ring, and p is a smooth point of X . $\square$

§15 Lecture 15

We continue with a few more results about normal varieties. An important result for normal varieties is that rational maps are defined in codimension one:

Lemma 15.1

Let X be a normal variety, Y a projective variety, and $f: X \dashrightarrow Y$ a rational map. Then every irreducible component of $X \setminus \text{dom}(f)$ is of codimension at least two. In particular, for any codimension one subvariety of $Z \subseteq X$, $\text{dom}(f)$ meets Z .

Proof. Without loss of generality, we may assume that $Y = \mathbb{P}^n$. and as such, f is given by

$$f = [f_0 : \cdots : f_n],$$

for rational functions $f_i \in K(X)$. Let $Z \subseteq X$ be any irreducible subset of codimension one. Since X is normal, $\mathcal{O}_{X,Z}$ is a DVR of $k(X)$ with a principle maximal ideal generated by some π . and as such each f_i can be written as $f_i = u_i \pi^{q_i}$ for $u_i \in \mathcal{O}_{X,Z}^\times$ and $q_i \in \mathbb{Z}$ unique (except of course for those entries which correspond to the zero function but these are not all entries). Up to reordering, let us assume $q_0 \leq q_i$ for all $i = 1, \dots, m$. And as such, f is represented by

$$f = [u_0 : u_1 \pi^{q_1 - q_0} : \cdots : u_n \pi^{q_n - q_0}]$$

since $u_0 \in \mathcal{O}_{X,Z}^\times$, it does not vanish on all of Z , and thus this expression shows that f is defined on a nonempty open subset of Z . $\square$

Again, applying this to curves we get:

Corollary 15.2

Let X be a smooth curve, and Y a projective variety. Then any rational map $X \dashrightarrow Y$ is a morphism.

Corollary 15.3

Two smooth projective curves C_1, C_2 are birational if and only if they are isomorphic.

The above shows that normal varieties have very nice properties; Rational maps are defined always on large sets and we have a well defined notion of the order of a function along codimension 1 varieties. In particular, for curves the properties of normal varieties transfer to smooth curves and curves. As we will see later, The notion of normalness is the right framework in which to do birational geometry.

Before we delve into the theory of divisors and birational geometry, we will discuss normalization. Recall commutative algebra:

Theorem 15.4

Let R be a finitely generated integral k -algebra, K its fraction field, and L/K a finite extension. Let S denote the integral closure of R in L , i.e. the ring obtained adding all elements of the integral L over R . Then S is a normal domain with a fraction field L , which is a finitely generated k -algebra and a finite R -module.

Definition 15.5. Let X be a variety, a **normalization** of X is a finite birational morphism $\tilde{X} \dashrightarrow X$, such that $\tilde{X}$ is a normal variety

Theorem 15.6 (Universal property of normalization)

Let X be an affine variety, and let

$$\nu: X^\nu \rightarrow X$$

be its normalization. If Y is a normal variety and

$$\phi: Y \rightarrow X$$

is a dominant morphism, then there exists a unique morphism

$$\psi: Y \rightarrow X^\nu$$

such that

$$\phi = \nu \circ \psi.$$

Proof. One checks that the integral closure of $A(X)$ in $k(X)$ satisfies these properties. $\square$

Weil Divisors

With the theory of normal varieties introduced, we are able to have a substantial discussion of Divisors. While these are defined for quasi projective varieties, most interesting results will hold only for normal varieties.

Definition 15.7. Let X be a quasi projective variety of dimension $n > 0$. A **Prime Divisor** of X is an irreducible closed subset of X that has codimension one. We denote by $Div(X)$ the free abelian group generated by these prime divisors. Furthermore an element $D \in Div(X)$ is called **effective** if all of its coefficients are nonnegative.

Recall that if k is a field, a discrete valuation on K is a surjective map $\nu: K^\times \rightarrow \mathbb{Z}$, such that:

- $\nu(xy) = \nu(x) + \nu(y)$
- $\nu(x + y) \geq \min\{\nu(x), \nu(y)\}$

If $k \subseteq K$, we say that ν is a valuation over k , if $\nu(k^\times) = \{0\}$.

Lemma 15.8

Let K be a field and ν a discrete valuation on K . Then the ring

$$\mathcal{O}_\nu := \{x \in K \mid \nu(x) \geq 0\},$$

is a regular local ring of dimension one. such a ring is called a discrete valuation ring.

Conversely, if $R \subseteq K$ is a regular local ring of dimension one with fraction field K , then there exists a discrete valuation ν , with $R = \mathcal{O}_\nu$.

Proof. It is easy to see that $\mathcal{O}_\nu$ is a local ring, with maximal ideal

$$\mathfrak{m}_\nu := \{x \in K \mid \nu(x) > 0\} \cup \{0\}.$$

Indeed, every element not in this set has an inverse in K of order 0 and thus the inverse is again in $\mathcal{O}_\nu$. Furthermore, since ν is surjective, there is an element $x \in k$ with $\nu(x) = 1$. From this, it follows easily that $\mathfrak{m}_\nu = (x)$ and so $\mathfrak{m}$ is also principal, which forces $\mathcal{O}_\nu$ to be of dimension one and thus also regular.

Suppose now that R is a regular local ring of dimension one with fraction field K . Let t be a generator of the maximal ideal of R . Then every element in $x \in K^\times$ can be written as

$$x = t^n u, \quad n \in \mathbb{Z} \quad u \in R^\times.$$

For a unique n and u . This allows us to define a valuation

$$\nu: K \rightarrow \mathbb{Z}, \quad t^n u \mapsto n.$$

□

Let us draw the connection to divisors. Assume for the rest of this lecture that X is a normal variety. Let $D \subseteq X$ be a prime divisor. Then we know that $\mathcal{O}_{X,D}$ is a regular local ring of dimension one. Thus, by the above, it is a discrete valuation ring of $k(X)$. In particular, it defines a valuation

$$\text{ord } K(X) \rightarrow \mathbb{Z}, \quad f \mapsto \text{ord}_D(f) := \nu_{\mathcal{O}_{X,D}}(f)$$

In more down to earth language, we have the following construction: Choose a generator π_D of the maximal ideal of $\mathcal{O}_{X,D}$. Then, any $f \in k(X)^\times$, has a unique representation $f = \pi_D^n u$, where $u \in \mathcal{O}_{X,D}^\times$ and $n \in \mathbb{Z}$. The above order then maps $f \mapsto n$.

Since we can now evaluate the order of a rational function on prime divisors, we can associate to any rational function divisor:

Definition 15.9. Let $f \in k(X)^\times$ be a rational function. The divisor of f , denoted $\text{div}(f)$ is defined as

$$\text{div}(f) = \sum_{\substack{D \subseteq X \\ D \text{ prime divisor}}} \text{ord}_D(f)$$

A priori, this could be an infinite sum; therefore, we need to prove that it is well defined.

Lemma 15.10

Let $f \in K(X)$, Then

$$\nu_D(f) = 0$$

for all but finitely many prime divisors.

Proof. Note that for a prime divisor D , $\nu_D(f)$ depends only on an open subset of X that intersects D (by the definition of $\mathcal{O}_{X,D}$). Hence, If U is an open affine subset of X , then $\nu_D(f)$ can be recovered in U for all prime divisors except for those which do not intersect U , i.e those contained in $X \setminus U$. Now $X \setminus U$ has finitely many irreducible components, and thus it suffices to prove the claim for U . Thus, we assume wlog, that X is affine and $f = \frac{A}{B}$ for $A, B \in A(X)$.

Since ν_D is a valuation, we have

$$\nu_D(f) = \nu_D(A) - \nu_D(B).$$

it therefore suffices to show that $\nu_D(A) = 0 = \nu_D(B)$ holds for all but finitely many prime divisors of X . However, $\nu_D(A) \neq 0$ occurs if and only if A vanishes on D and so D is a component of $V(A)$. Since $V(A)$ has finitely many components, there are finitely many such D . Applying the same argument for B , we obtain the claim. $\square$

Using the above lemma, we have group homomorphism

$$\Phi : k(X)^\times \rightarrow \text{Div}(X), f \mapsto \text{div}(f).$$

Definition 15.11. A divisor D is called **principle** if it is of the form $D = \text{div}(f)$ for some $f \in k(X)^\times$. Two divisors D, D' are called linearly equivalent if $D - D'$ is principle.

The reason we introduce divisors is that they allow us to capture the nature of rational functions in very simple notation; for example, a rational function vanishes along a divisor if and only if its coefficient along that divisor is positive. Later, we will prescribe to any divisor a certain space of rational functions encoded by this divisor. As it turns out, linearly equivalent divisors will induce isomorphic function spaces. This motivates the definition of the class group of X .

Definition 15.12. The class group of X , denoted $\text{Cl}(X)$, is defined as the divisor group module linear equivalence. I.e:

$$\text{Cl}(X) = \text{coker}(\Phi : k(X)^\times \rightarrow \text{Div}(X))$$

Remark 15.13. Note that if $X \cong X'$, then the class groups are isomorphic. Hence we have defined an invariant in the category of normal varieties.

Example 15.14

For any $f \in k(X)$ let us view it as a map $f: X \dashrightarrow \mathbb{P}^1$. Suppose D, D' are two effective divisors, then $D \sim D'$ if and only if there exists a $f \in k(X)$ such that D are the zeros of f and D' are the poles of f .

Example 15.15

Consider $X = \mathbb{A}^n$. Since $k[x_1, \dots, x_n]$ is a UFD, X is a normal variety. Furthermore, The prime divisors D are all of the form $V(f)$ for irreducible f . In particular, all prime divisors are principle and $\text{Cl}(X) = 0$.

More generally if X is affine and $A(X)$ is a UFD, then $\text{Cl}(X) = 0$.

Remark 15.16. In fact, for an affine normal variety, we have $\text{Cl}(X) = 0$ if and only if $A(X)$ is a UFD.

To end this lecture, recall from commutative algebra:

Theorem 15.17

Let A be a normal integral domain, then

$$A = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \operatorname{ht}(\mathfrak{p})=1}} A_{\mathfrak{p}} \subseteq \operatorname{Frac}(A).$$

Corollary 15.18

Let X be a quasi projective normal variety and $f \in k(X)$, then f is regular on X if and only if $\operatorname{div}(f) \geq 0$.

§16 Lecture 16

In the Last lecture we have introduced the class group of a normal variety. Although we have seen that, for example, if X is affine with $A(X)$ being a UFD, then $\operatorname{Cl}(X)$ vanishes; in general, we do not yet have tools to calculate this class group. We start this lecture by calculating some examples and introducing some general tools for computing the class group

Example 16.1

For projective space we have $\operatorname{Cl}(\mathbb{P}^n) \cong \mathbb{Z}$ and any Linear hyperspace can be chosen as a generator.

Proof. Let $H = Z(h) \subseteq \mathbb{P}^n$ be any hyperplane and $D \subseteq \mathbb{P}^n$ and prime divisor. Then D is the zero set of a homogeneous polynomial F of degree d for some d . Consider now the function

$$f := \frac{F}{h^d} \in k(\mathbb{P}^n).$$

By construction, we then have

$$\operatorname{div}(f) = D - d,$$

thus every divisor is equivalent to a multiple of D . Hence the class of H generates $\operatorname{Cl}(X)$. We need to show that the group is torsion free, i.e $Hm \sim 0$ can only occur if $m = 0$. Suppose that $\operatorname{div}(f) = mH$. If $m \geq 0$, then f is regular on all of $\mathbb{P}^n$ and thus a nonzero constant in particular $m = 0$. if $m < 0$, then $\frac{1}{f}$ is constant and the same argument applies. $\square$

Suppose now, that Y is a prime divisor of $\mathbb{P}^n$. Then we know that $Y = V(F)$ for a irreducible homogeneous polynomial F , which is unique up to units. Thus we can define $\deg Y = \deg F$. Hence, in the case of $\mathbb{P}^n$ we can define a map

$$\Phi: \operatorname{Div}(\mathbb{P}^n) \rightarrow \mathbb{Z}, \sum n_i Y_i \mapsto \sum n_i \deg(Y_i)$$

The kernel of this map are precisely the principal divisors.

Now let us introduce some tools for computing the class group.

Lemma 16.2

Let $U \subseteq X$ be a open subset and $Z := X \setminus U$. If $Z_1, \dots, Z_r$ are the irreducible components of Z which have codimension one in X , then there is an exact sequence of abelian groups:

$$\bigoplus_{i=1}^r \mathbb{Z}Z_i \longrightarrow \text{Cl}(X) \longrightarrow \text{Cl}(U) \longrightarrow 0$$

Remark 16.3. Note that since X is normal, U is also normal. Also it is important to remember that the first map is not injective in general, there is no guarantee that these components don't share some relations in $\text{Cl}(X)$.

Proof. Let us define the maps and then show exactness. For the first map we simply map Z_i to its class in $\text{Cl}(X)$. Now let us consider the second map: Note that the irreducible subsets of codimension one in U are all obtained by taking a prime divisor in X and intersecting it with U . Furthermore, any prime divisor which is not contained in Z , produces a prime divisor in U in this way. For the remaining prime divisors Z_i , we map them to zero. A priori this is only defined for the divisor group:

$$\Phi: \text{Div}(X) \rightarrow \text{Div}(U), D \mapsto \begin{cases} 0 & \text{if } D \subseteq Z \\ D \cap U & \text{otherwise} \end{cases}$$

the kernel of this map is then clearly the subgroup generated by the Z_i . This map clearly factors over the class group, since $k(U) = k(X)$ and hence factoring we obtain the map. Thus, we actually have a well defined map

$$\Phi: \text{Cl}(X) \rightarrow \text{Cl}(U)$$

and its kernel is that generated by the Z_i . □

Proposition 16.4

For any normal quasi projective variety, $\text{Cl}(X \times \mathbb{A}^1) \cong \text{Cl}(X)$.

Proof. First, we should prove that $X \times \mathbb{A}^1$ is normal. Since being normal is a local property, we may assume that X is affine, and thus it has a normal coordinate ring $A(X)$. Then, the coordinate ring of $X \times \mathbb{A}^1$ is given by $A(X)[t]$. Hence the result will follow from

Lemma 16.5

If A is a normal ring the, $A[t]$ is as well.

Proof. Let $F = \text{Frac}(A)$. Then we have $\text{Frac}(A[t]) = F(t)$. Suppose $f \in F(t)$ satisfies a monic equation with coefficients in $A[t]$. Write f as

$$f = a_n t^n + \cdots + a_1 t + a_0,$$

and the equation as

$$G(X, t) = X^m + b_{n-1} X^{n-1} + \cdots + b_1 X + b_0, \quad b_i \in A[t]$$

Plugging $t = 0$, we arrive at a monic polynomial with coefficients in A , which annihilates a_0 . Thus, $a_0 \in A$. hence it suffices to show that

$$f - a_0 = t(a_n t^{n-1} + \cdots + a_1) \in A[t].$$

For this, we can show simply that

$$a_n t^{n-1} + \cdots + a_1 \in A[t],$$

which follows from induction. □

The main idea is to construct a map $\text{Cl}(X) \rightarrow \text{Cl}(X \times \mathbb{A}^1)$. To do this we need to understand the prime divisors in $\mathbb{A}^1 \times X$. For the moment let us assume that X is affine and as such we want to understand the height one primes in $A(X \times \mathbb{A}^1) = A(X)[t]$. Let D be a prime divisor of $X \times \mathbb{A}^1$ and let D' denote the closure of $\pi_X(D)$ in X . Consider the restriction of π to D and apply the theorem of the dimension of fibres to find that

$$\dim D - \dim D' \geq \dim \pi_X^{-1}(x) = 1.$$

and as such $\dim D' \geq \dim D - 1 = \dim X - 1$. Thus, prime divisors of $X \times \mathbb{A}^1$ are of two types

Type one divisors : $\pi(D)$ is dense in X

Type two divisors : $\pi(D)$ is dense in a prime divisor of X .

Note that if $Y \subseteq X$ is a prime divisor, then $\pi^{-1}(Y) = Y \times \mathbb{A}^1$ is prime divisor in $X \times \mathbb{A}^1$ of type two. Thus, we can define a map

$$\Phi : \text{Div}(X) \rightarrow \text{Div}(X \times \mathbb{A}^1), D \mapsto \pi_X(D).$$

Now every rational function on X may be pulled back to a rational function on $X \times \mathbb{A}^1$, and as such, this map factors over the Class group, so we have a well defined map .

$$\Phi : \text{Cl}(X) \rightarrow \text{Cl}(X \times \mathbb{A}^1), D \mapsto \pi_X(D).$$

We claim that it is surjective. Suppose first that $\mathfrak{p} \subseteq A(X)[t]$ is any height one prime corresponding to a prime divisor Y . If $\mathfrak{p} \cap A(X) \neq (0)$, then $\mathfrak{p}$ corresponds to a divisor of type two. The prime ideal $\mathfrak{p} := \mathfrak{p} \cap A(X)$ is a height one prime of $A(X)$ and $\mathfrak{p} = (\mathfrak{q})^e$ as such Y is in the image of Φ .

If, instead, $\mathfrak{p} \cap A(X) = 0$ consider the multiplicative set $S := A(X) \setminus \{0\} \subseteq A(X)[t]$. Localizing $A(X)[t]$, at S , we obtain

$$S^{-1}(A(X)[t]) \cong k(X)[t].$$

Since $\mathfrak{p}$ does not meet S it corresponds to a prime $\mathfrak{q} := S^{-1}\mathfrak{p}$ of this ring. Since $k(X)[t]$ is a euclidean ring and hence a PID, $\mathfrak{q}$ is generated by a single prime element $f \in k(X)[t]$. As such, $\mathfrak{q}$ is height one, as any nontrivial ideal containing $\mathfrak{q}$ would correspond to a factor of f . We may regard this f as a rational function on $X \times \mathbb{A}^1$. Let us compute its divisor. We may write f as First, we claim:

$$\nu_{\mathfrak{p}}(f) := 1.$$

Indeed, suppose π is a generator of $\mathcal{O}_{X,Y} \subseteq k(X)(t)$ and write $f = \pi^n u$, for $n \in \mathbb{Z}$, and u a unit in $\mathcal{O}_{X,Z}$. We may assume that $\pi \in A(X)[t]$ since the nonzero elements of $A(X)$ do not vanish on all of Y and thus are units. Furthermore, we may write f as a quotient

$$f = \frac{A}{B}, A \in A(X)[t], B \in A(X) \setminus \{0\}$$

In particular, since B does not vanish on all of Y , this expression shows that f is regular on an open subset of Y and the integer n in $f = \pi^n u$ is nonnegative. Thus, $f = \pi \cdot \pi^{n-1}u$. The division algorithm in $k(X)[t]$ then shows that the quotient: $f/\pi = \pi^{n-1}u$ remains in $A(X)[t]$. But since f is prime this forces $n = 1$.

Now we may write

$$\text{div}(f) = Y + \sum_{D \neq Y} n_D D$$

for $n_D \in \mathbb{Z}$. We claim that the only prime divisors D for which $n_D \neq 0$, are of type two. Otherwise, f would be irreducible in the polynomial ring $k(X)[t]$. Thus Y is linearly equivalent to a divisor of type two which shows that Φ is surjective.

Now Let us consider the general case where X is not affine. In this case we can still construct the map Φ_X as above, but we do not know if it is surjective. Let $U \subseteq X$ be an affine subset of X and let $\bar{Z}_1, \dots, \bar{Z}_r$ be the irreducible components of codimension one of $X \setminus U$. Consider the open subset $\pi_X^{-1}(U)$, it has codimension one components $Z_1, \dots, Z_r$, where $Z_i = \pi_X^{-1}(\bar{Z}_i)$. Thus, by lemma 16.2, we get a commutative diagram with exact rows:

$$\begin{array}{ccccccc} \bigoplus_{i=1}^r \mathbb{Z} & \longrightarrow & \text{Cl}(X) & \longrightarrow & \text{Cl}(U) & \longrightarrow & 0 \\ \downarrow = & & \downarrow \Phi_X & & \downarrow \Phi_U & & \\ \bigoplus_{i=1}^r \mathbb{Z} & \longrightarrow & \text{Cl}(X \times \mathbb{A}^1) & \longrightarrow & \text{Cl}(U) & \longrightarrow & 0 \end{array}$$

Since Φ_U is surjective, a simple diagram chase shows that Φ_X is surjective as well.

To finish the proof, we need to show, that Φ_X is injective. Let

$$D = \sum_{i=1}^n n_i \bar{Y}_1 \in \text{Div}(X)$$

be a divisor such that

$$\sum_{i=1}^n n_i Y_1 \sim 0 \in \text{Div}(X \times \mathbb{A}^1).$$

By assumption, we have

$$D = \text{div}(f), f \in k(X)(t).$$

Since the Y_i are all of type two, f may be chosen as an element in $k(X)[t]$. Let U be the open subset of X for which all coefficients of f are invertible, and by shrinking, assume U is affine. As such, $f \in A(U)[t]$, $f = \sum f_i t^i$. Then the set

$$(Z_{U \times \mathbb{A}^1}(f)) \cap \{u\} \times \mathbb{A}^1$$

corresponds to the zeros of $\sum f_i(u)t^i$. Consider now the set of $u \in U$, such that not all $f_i(u)$ are zero for $i > 0$. This set is open in U , and thus either empty or dense. Assuming it is not empty, choosing such a u , we have that

$$(Z_{U \times \mathbb{A}^1}(f)) \cap \{u\} \times \mathbb{A}^1$$

is a finite set. Thus, there exists an irreducible component Y_U of $(Z_{U \times \mathbb{A}^1}(f))$ such that $Y_U \cap \{u\} \times \mathbb{A}^1$ is finite. Then the projection to U , restricted to Y_U has zero dimensional fiber at u and therefore Y_U has the same dimension as U and thus the same dimension as X by the theorem on fibers. Thus, the closure of Y_U is a prime divisor in $X \times \mathbb{A}^1$ of type two along which f is zero. But $\text{div}(f) = D$ is a combination of type two divisors which is a contradiction.

Thus f is actually an element of $k(X) \subseteq k(X)[t]$ and the claim follows. $\square$

§17 Lecture 17

We began by briefly showing how the theorem we proved last lecture could be used in practice to compute a class group

Example 17.1

Consider the set $Q = (zy - xw) \subseteq \mathbb{P}^3$. We know that Q is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$ via the Segre-embedding. Note that $\mathbb{P}^1 \times \mathbb{P}^1$ is locally near each point, isomorphic to $\mathbb{A}^1 \times \mathbb{A}^1 \cong \mathbb{A}^2$, and thus, it is a normal variety. Hence it makes sense to talk about its Class group. Choose a points p_1, p_2 in each copy of $\mathbb{P}^2$ and set

$$Z_1 = \{p_1\} \times \mathbb{P}^1, \quad Z_2 = \mathbb{P}^1 \times \{p_2\},$$

and let U_1, U_2 denote the complements of Z_1, Z_2 respectively. Notice that $U_i \cong \mathbb{P}^1 \times \mathbb{A}^1$, and thus by the previous lemma

$$\text{Cl}(U_i) \cong \text{Cl}(\mathbb{P}^1) \cong \mathbb{Z}.$$

Furthermore, the class groups are generated by $\mathbb{A}^1 \times \{p\}$ and $\{p\} \times \mathbb{A}^1$.

Applying the localization exact sequence, we get a commutative diagram

$$\begin{array}{ccccccc}
 & & \mathbb{Z} \cdot Z_1 & & & & \\
 & & \downarrow & \searrow \alpha_1 & & & \\
 \mathbb{Z} \cdot Z_2 & \longrightarrow & \text{Cl}(Q) & \longrightarrow & \text{Cl}(U_2) \cong \mathbb{Z} & \longrightarrow & 0 \\
 & \searrow \alpha_2 & \downarrow & & & & \\
 & & \text{Cl}(U_1) \cong \mathbb{Z} & & & &
 \end{array}$$

Thus, we find that $\text{Cl}(Q) \cong \mathbb{Z} \times \mathbb{Z}$

The reason that divisors and the class group are important tools is that they allow us to elegantly talk about maps into projective space. The easiest example of this are the spaces $L(D)$ for a divisor D . For the following let X be a smooth quasi projective variety and thus, in particular, normal.

Definition 17.2. The space $L(D)$ is defined as

$$L(D) = \{f \in k(X) \mid f = 0 \text{ or } \operatorname{div}(f) + D \geq 0\}.$$

Remark 17.3. It is easy to see that $L(D)$ is a k vector space. Furthermore, if $D \leq D'$ then $L(D) \subseteq L(D')$.

Remark 17.4. The spaces $L(D)$ are finite dimensional. The proof of this was skipped in the lecture but it is not hard for the case when X is a curve so we briefly sketch it here. By the previous remark it suffices to prove this for when D is effective. Then one proceeds by induction and noticed that $L(0) = k$, and that when increasing D by P , that the dimension can increase by at most one. We denote by $l(D)$ the dimension of $L(D)$.

Definition 17.5. Let D be a divisor on X . The **complete linear system** $|D|$ is defined as the set

$$|D| := \{D' \sim D \mid D' \geq 0\}.$$

Proposition 17.6

There is a natural bijection between $|D|$ and the projectivisation $\mathbb{P}(L(D))$, of $L(D)$.

Proof. First note that if $L(D) = 0$, then $|D| = \emptyset$ and $\mathbb{P}(L(D)) = \mathbb{P}(0) = \emptyset$ and the claim is true. Suppose now that $L(D) \neq 0$, and define the map

$$\Phi : L(D) \setminus \{0\} \rightarrow |D|, f \mapsto \operatorname{div}(f) + D.$$

First, this map is surjective: If $D' \sim D$ then $D' = D + \operatorname{div}(f)$ for some $f \in k(X)^\times$. But, in fact, $f \in L(D)$, as $\operatorname{div}(f) + D = D' \geq 0$.

Suppose now that $\operatorname{div}(f) + D = \operatorname{div}(g) + D$, then $\operatorname{div}(f/g) = 0$. and so $f/g \in L(0)$, That is to say, that f/g defines a regular function on the projective variety X which we already know only has constant global sections. Thus $f/g \in k^\times$. and $f = \lambda g$ for some $\lambda \in k^\times$. $\square$

Example 17.7

Let $X = \mathbb{P}^1$. Then a divisor on $\mathbb{P}^1$ is nothing but a $\mathbb{Z}$ -linear combination of points on X . let $P = [1 : 0] \in X$. Let us compute $L(dP)$, $d > 0$. For this, first note, that

$$k(X) \cong k(U_0) = k(t),$$

where $t = \frac{X}{Y}$ is an affine coordinate for U_0 . It is not hard to see, via a change of variables, that t has order -1 at the point P (compute this in the affine chart U_1). consider now a function $f \in L(dP)$. By assumption it is regular on U_0 , and thus it is written as a polynomial in t :

$$\begin{aligned} f &= a_n t^n + \cdots + a_1 t + a_0 \\ &= a_n \left(\frac{X}{Y} \right)^n + \cdots + a_1 \left(\frac{X}{Y} \right) + a_0 \\ &= \left(\frac{Y}{X} \right)^{-n} \left(a_n + \cdots + a_1 \left(\frac{Y}{X} \right)^{n-1} + a_0 \left(\frac{Y}{X} \right)^n \right) \end{aligned}$$

The right factor is a unit in $\mathcal{O}_{X,P}$ and thus, since $\frac{Y}{X}$ is a uniformizer at P , we have shown that the order of f at P is $-n$. Hence

$$L(dP) = k \oplus k \cdot t \oplus \cdots \oplus k \cdot t^d = k[t]_{\leq d}$$

The projectivization of this is d dimensional and so $|dP| \cong \mathbb{P}^d$.

Example 17.8

Let us consider the case again $X = \mathbb{P}^n$. For this variety, we already know that $\text{Cl}(X) \cong \mathbb{Z}$, and two divisors are equivalent if and only if they map to the same integer under the map Φ constructed in example 16.1. If $D \in \text{Div}(X)$ is any divisor of negative degree, then $D + \text{div}(f)$ will also have negative degree and hence can never be effective. As such, $|D| = \emptyset$. Suppose now that $\deg D > 0$. Then $|D|$ is nonempty, and its elements are all contained in

$$\mathcal{O}_{\mathbb{P}^n}(d) := \{D \mid D \geq 0, \Phi(D) = d\}.$$

We claim This space may be identified with degree d hypersurfaces in $\mathbb{P}^n$ (not necessarily irreducible). That is to say, there is a bijection:

$$\mathcal{O}_{\mathbb{P}^n}(d) \cong \mathbb{P}(k[x_0, x_1, \dots, x_n]_d).$$

Consider the map

$$\Psi: (k[x_0, x_1, \dots, x_n]_d) \setminus \{0\} \rightarrow \mathcal{O}_{\mathbb{P}^n}(d), \quad F = \prod_i F_i^{\nu_i} \mapsto \sum_i \nu_i V(F_i)$$

This map is clearly surjective. Suppose now that $\Psi(F) = \Psi(G)$ for two degree d homogeneous polynomials. Then $\frac{F}{G}$ defines a rational function on $\mathbb{P}^n$. It is not hard to see that it has divisor $\Psi(F) - \Psi(G) = 0$. Thus it is a global section and therefore a constant. Thus, F and G differ in a nonzero scalar. Thus, we have induced a bijection

$$\mathcal{O}_{\mathbb{P}^n}(d) \cong \mathbb{P}(k[x_0, x_1, \dots, x_n]_d).$$

$k[x_0, x_1, \dots, x_n]_d$ has basis equal to monomials of degree d in $n + 1$ variables. Using some combinatorics, one may conclude that

$$\mathcal{O}_{\mathbb{P}^n}(d) \cong \mathbb{P}^{\binom{n+d}{d}-1}$$

Thus, in particular, for all divisors of degree d , we have

$$L(D) \cong k[x_0, \dots, x_n]_d$$

We now introduce a slight generalization of complete linear systems.

Definition 17.9. Let X be a smooth projective variety. A **linear system** δ on X is a projective linear subspace of a complete linear system $|D|$. And as such, δ is isomorphic to $\mathbb{P}^r$ for some $r \leq l(D) - 1$. This r is called the dimension of δ .

Definition 17.10. Let δ be a linear system on X and $D = \sum_i n_i Y_i \in \delta$, where $n_i > 0$. We say that a point $p \in X$ is contained in D , if it is contained in one the Y_i .

Remark 17.11. If $p \in D \geq 0$, then the functions in $L(D)$ can have poles at p . Hence the points in D track the poles of the functions in $L(D)$.

Definition 17.12. Let δ be a linear system on X . We say that δ has a **base point**, if there exists a $p \in X$, such that $p \in D$ for all $D \in \delta$. δ is called base point free if it does not have base points.

Lemma 17.13

Suppose δ is a linear system and $P \in X$ is not a base point of δ . Then

$$\delta_P := \{D \in \delta \mid P \in D\}$$

is a linear system of X and P is a basepoint of δ_P . Furthermore, if δ has dimension $r > 0$, then δ has dimension $r - 1$.

Proof. By definition δ is a projective subvector space of a linear system $|D|$. Since we have assumed P not to be a basepoint of δ , there exists a $D' \in \delta \subseteq |D|$ such that $P \notin D'$. Let us assume $D' = D$, since the linear systems are the same. Then, δ is identified with a subvector space V of $L(D)$ of dimension $r + 1$. Choosing a basis

$$f_0, \dots, f_r$$

be a k basis of V . Then, every function f in V is uniquely represented as a k linear combination

$$f = \sum_{i=0}^r \lambda_i f_i.$$

Such a function then defines a unique divisor

$$D_\lambda := \text{div}(f) + D,$$

and every divisor in δ is obtained in this way. Then, the condition $P \in D_\lambda$ is equivalent to $p \in \text{div}(f)$ since P does not appear in D . Also, since $P \notin D$, all functions in $L(D)$ are regular at P .

$$\sum_{i=0}^r \lambda_i f_i(P) = 0,$$

This is a linear constrain on the λ_i and it is not trivial since atleast one of the f_i does not vanish at P since P was assumed not to be a basepoint. $\square$

Remark 17.14. In the following we will consider smooth quasi projective varieties rather than just normal ones. For such varieties many calculations become easier. Crucially, we have the following lemma

Lemma 17.15

Let X be a smooth projective variety and $Z \subseteq X$ a prime divisor. Then every point $P \in X$, has an affine neighborhood U_P such that $I_{U_P}(U_P \cap Z) = (F_{U_P, Z})$ is a principle ideal of $A(U_P)$.

Proof. If $P \notin Z$, then the statement is essentially obvious: take any open affine neighborhood of P that is contained in $X \setminus Z$. As such, $U_P \cap Z = \emptyset$ and $I_{U_P}(U_P \cap Z) = (1)$. Assuming now that $P \in Z$, consider the regular local ring $\mathcal{O}_{X, P}$, of dimension $n := \dim X$. By the Auslander-Buchsbaum theorem from commutative algebra, $\mathcal{O}_{X, P}$ is a UFD. Since p is contained in Z , Z corresponds to a height one prime of this ring (if one prefers, pass to an open cover to assume X is affine and Z as well). In a UFD, a height one prime is principal, and as such, Z corresponds to a principal ideal $(\pi_Z) \subseteq \mathcal{O}_{X, P}$. Passing to an open affine subset U of X , we may assume that π_Z is a regular function on U . It still generates the maximal ideal of $Z \cap U$ in U . $\square$

The crucial part in the above proof was that $\mathcal{O}_{X,p}$ was regular and hence a UFD. for normal varieties, this cannot be ensured in general, although the only points where this could go wrong must necessarily be singular points.

The previous lemma allows us the following construction called the pullback of a divisors

Let $\varphi: X \rightarrow Y$ be a regular map of smooth quasi projective varieties. If $Z \subseteq Y$ is a prime divisor, take an open cover $Y = \bigcup_i U_i$ of affine open subsets such that $Z \cap U_i$ is determined by a single regular $\psi_i \in A(U_i)$, as guaranteed by the lemma above. Define the open sets

$$V_i := \varphi^{-1}(U_i)$$

. the ψ_i pullback to regular functions $\varphi^*\psi_i \in \mathcal{O}_{V_i}$. Using this we define

$$\varphi^*(Z) := \sum_{\substack{W \subseteq X \\ W \text{ prime divisor}}} n_W W,$$

where the coefficient n_W is defined as follows: choose a V_i such that $W \cap V \neq \emptyset$. Then define n_W as the valuation of $\varphi^*\psi_i$ defined by the local ring $\mathcal{O}_{V_i, V_i \cap W}$. n_W is nonnegative, as $\varphi^*\psi_i$ is regular on an open neighborhood intersecting W nontrivially. In the lecture it was claimed that it is easy to see that this is independent of choices, but I am not sure why. I'm guessing one can equally define this as a ramification of a mapping of local rings.

Extending linearly, we get a map of abelian groups

$$\varphi^* \operatorname{Div}(Y) \rightarrow \operatorname{Div}(X), D \mapsto \varphi^*(D).$$

Example 17.16

let X be a smooth quasi projective variety and let $\varphi : X \rightarrow \mathbb{P}^n$ be a regular map. φ is called nondegenerate if it is not contained in a hyperplane. For such a φ , Consider the set of divisors:

$$\delta := \{\varphi^*(H) \mid H \in \mathbb{P}^n \text{ } H \text{ is hyperplane}\}.$$

(Here, by hyperplane, we mean one-dimensional subspaces). Then δ is a base point free linear system of dimension n on X

Proof. By definition of the pullback these divisors are all effective. Furthermore, if H, H' are two hyperplanes, then they have linear equations L, L' which, since φ is nondegenerate, do not vanish on the entire image of φ . That is to say that the rational function $\frac{L}{L'}$ pulls back to an element $f \in k(X)^\times$. Then

$$\text{div}(f) = \varphi^*(H) - \varphi^*(H').$$

Thus, the divisors appearing in δ are all linearly equivalent to one another. Suppose now that $\varphi^*(H) = \varphi^*(H')$, then $L/L' \in \mathcal{O}_X(X) = k$, and therefore L and L' differ by a nonzero constant scalar. the linear system δ is a subset of $|D|$ where we take $D = \varphi^*H$ for some hyperplane H . Using the identification, $|D| \cong \mathbb{P}(L(D))$, let us try to see what δ corresponds to. Consider again the map

$$L(D) \setminus \{0\}, \rightarrow |D|, f \mapsto \text{div}(f) + D.$$

we have

$$\begin{aligned} \text{div}(f) + D \in \delta &\Leftrightarrow \text{div}(f) = \varphi^*(H') - \varphi^*(H) \text{ for some } H' \\ &\Leftrightarrow \text{div}(f) = \text{div}\left(\varphi^*\left(\frac{H}{H'}\right)\right) \end{aligned}$$

Thus, the set of f which correspond to a members of δ is identified with the elements of $L(D)$ which are the pullbacks of functions of the form $\frac{H'}{H}$. The set of such pullbacks is obviously a vector space; hence, δ is a linear system. Lastly, we may identify δ with the set of numerators of the form H' . The set of all such H' has projective dimension n . Lastly, δ is basepoint free since if $P \in X$, let $Q := \varphi(P)$ and choose a hyperplane H which does not contain Q . Its pullback does not contain P . $\square$

Using the construction above, we have seen that any regular nondegenerate map $\varphi : X \rightarrow \mathbb{P}^n$ defines , n base point free linear system on X of degree n . In fact, we claim that this is a bijection. Suppose for this, that δ linear system of dimension n and let $P \in X$. We have seen before that P is not a base point of δ if and only if

$$\delta_P := \{D \in \delta \mid P \in D\}$$

is a subvector space of δ of codimension one. Choosing a basis of δ , this defines a bijection

$$\varphi : X \setminus \{\text{base points of } \delta\} \rightarrow \{H \subseteq \delta, \text{codim } H = 1\}$$

Identify δ with a dimension $n + 1$ subspace of $L(D)$ for some divisor D and choose a basis $f_0, \dots, f_n$ of V . Then, we have seen that $f \in V$ corresponds to a function in δ_P if and only if its coefficients λ_i in the expansion,

$$f = \sum_{i=0}^n \lambda_i f_i$$

, satisfy

$$\sum_{i=0}^n \lambda_i f_i(P) = 0.$$

Thus, P defines a nontrivial element of the dual $V^\vee$ given by

$$\Phi_P: V \rightarrow k, \sum_{i=0}^n \lambda_i f_i \mapsto \sum_{i=0}^n \lambda_i f_i(P).$$

I.e the map above is given by

$$X \setminus \{\text{basepoint of } \delta\} \rightarrow V^\vee, P \mapsto \Phi_P$$

Since we have chosen a basis of V , this gives rise to a canonical basis of $V^\vee$, constructing an isomorphism $V \cong k^{n+1}$. And in this basis the map is given by

$$P \mapsto \Phi_P = (f_0(P), \dots, f_n(P))$$

By projectivizing, we obtain a map.

$$\varphi_\delta: X \setminus \{\text{base points of } \delta\} \rightarrow \mathbb{P}^n, P \mapsto [f_0(P) : \dots : f_n(P)].$$

From this presentation, it is evident that this map is well defined and regular. We claim it is nondegenerate. Suppose not, i.e., for all non δ -base points, we have $\sum_{i=0}^n \lambda_i f_i(P) = 0$ for fixed λ_i . Consider the function $f = \sum_i \lambda_i f_i$. It is zero as a member of V , and thus, since the f_i are linearly independent, $f = 0$ and all $\lambda_i = 0$.

One may easily verify that the base point free degree system of dimension n induced by the pullback φ_δ is again δ . This long and intricate discussion can be summarized in the following theorem:

Theorem 17.17

Let X be a smooth quasi projective variety, then there is a bijection

- Base point free dim n linear systems δ and a choice of basis of the corresponding $V_\delta \subseteq L(D)$.
- Regular nondegenerate morphism $\varphi: X \rightarrow \mathbb{P}^n$.

Example 17.18

Consider $X = \mathbb{P}^n$ and consider the complete linear system

$$\mathcal{O}_{\mathbb{P}}^n(d)$$

We know that $\mathcal{O}_{\mathbb{P}}^n(d)$ is a basepoint free linear system of degree $\binom{n+d}{d}$. Identifying it with the projectivization of degree d monomials with the canonical basis, this is just the veronese embedding of degree d

§18 Lecture 18

For the rest of this course we will only study curves. In fact we will mainly focus on smooth projective curves as these have very nice properties. To list a few of them:

- An algebraic curve is normal if and only if it is smooth
- smooth projective curves are birational if and only if they are isomorphic
- For smooth curve X , and $p \in X$ the local ring $\mathcal{O}_{X,p}$ is a discrete valuation ring of $k(X)$ with $k \subseteq \mathcal{O}_{X,p}$. A DVR with the second property is said to be over k . We will see that any such discrete valuation ring corresponds to a point in X .
- A nonconstant morphism $\varphi : X \rightarrow Y$ between projective curves is either constant or surjective. In the latter case it is quasi finite.

The second property means that studying smooth projective curves is the same as studying their function fields, which are fields of transcendence degree one over k . For the third point let us prove:

Lemma 18.1

Let X be a smooth projective curve and $V \subseteq k(X)$ be a DVR with $k^\times \subseteq \mathcal{V}$. Then there exists a $p \in X$, such that $V = \mathcal{O}_{X,p}$. Thus, there is a bijection:

$$\{P \in X\} \leftrightarrow \{\text{DVR's of } k(X) \text{ over } k\}$$

Proof. Consider the map

$$\Phi : X \rightarrow \{\text{DVR's of } k(X) \text{ over } k\} \quad P \mapsto \mathcal{O}_{X,p}.$$

First we claim that Φ is injective. Let $P \neq Q$ be two points on X . Since X is projective it is a closed subset of some $\mathbb{P}^n$. Thus, we can choose a hyperplane of $\mathbb{P}^n$ which contains neither P nor Q , and dehomogenize $\mathbb{P}^n$ at this hyperplane to assume that X is affine. In the affine case $\mathcal{O}_{X,P} = A(X)_{\mathfrak{m}_P}$ and $\mathcal{O}_{X,Q} = A(X)_{\mathfrak{m}_Q}$. For two different points, the maximal ideals are different and hence the localizations are different. Thus, we need to show that this map is surjective. Note that it is crucial that we assume X to be projective and smooth, otherwise X may be missing some points which are seen by $k(X)$. Let $V \subseteq k(X)$ be any discrete valuation ring over k and ν the corresponding valuation. Since X is projective let us choose a closed embedding

$$\Psi : X \hookrightarrow \mathbb{P}^n, P \mapsto [f_0(P) : \cdots : f_n(P)] \quad f_i \in k(X).$$

In fact let us assume that this embedding is nondegenerate and so $f_i \in k(X)^\times$. Notice that the f_i then satisfy the equations of $I_{\mathbb{P}^n}(X)$. Up to reordering, we can assume $\nu(f_0) \leq \nu(f_i)$ for all $i = 1, \dots, n$. Thus, $\nu(f_i/f_0) \geq 0$. Then restricting to the open subset $\{f_0 \neq 0\}$, Ψ can be written as

$$P \mapsto [1 : \frac{f_1}{f_0} : \cdots : \frac{f_n}{f_0}],$$

which is precisely the embedding of this subset into $U_0 \cong \mathbb{A}^n$. Note now that f_i/f_0 are congruent to values $\alpha_i \bmod \mathfrak{m}_V$ since $V/\mathfrak{m}_V \cong k$. Which allows us to consider the points

$$P = [1 : \alpha_1 : \cdots : \alpha_n] \in X_0 := X \cap U_0.$$

Denoting by $g_i = f_i/f_0$, we have a surjective morphism

$$k[g_1 : \cdots : g_n] \rightarrow \mathcal{O}(X_0)$$

since X_0 is a closed subset of $\mathbb{A}^n$. Finally, localizing the polynomial ring $k[y_1, \dots, y_n]$ of $\mathbb{A}^n$ at the point P , and then quotienting out by the ideal of X_0 , we obtain a surjective morphism

$$\mathcal{O}_{\mathbb{A}^n, P} \rightarrow \mathcal{O}_{X, P}.$$

But by construction the kernel of this is just $\mathfrak{m}_V$ and so $\mathcal{O}_{X, P} = V$. $\square$

Corollary 18.2

Let X be a smooth projective curve and U a smooth affine curve such that $k(U) \cong k(X)$. Then there exists an embedding $U \hookrightarrow X$.

Proof. Since U is smooth, all of its points determine DVR'S over $k(U) \cong k(X)$. For $k(X)$ all DVR'S correspond to points. Thus we obtain a set theoretic map

$$\Phi: U \rightarrow X, P \mapsto Q \in X \text{ such that } \mathcal{O}_{U, P} = \mathcal{O}_{X, Q}.$$

As we have seen this map is injective as different points induce different maximal ideals and thus different local rings. Since it is injective, it is automatically continuous since the preimage of finite sets is finite. To show that it is a morphism. we need to show that it pulls back regular functions to regular functions. Suppose $V \subseteq X$ is any open set, then we have

$$\Phi^*(\mathcal{O}(V)) = \Phi^*\left(\bigcap_{Q \in V} \mathcal{O}_{X, Q}\right) = \bigcap_{Q \in \Phi^{-1}(V)} \mathcal{O}_{U, Q} = \mathcal{O}(\Phi^{-1}(V)).$$

We claim the image of Φ is open. Indeed, consider the ring

$$A(U) = \bigcap_{Q \in \Phi(U)} \mathcal{O}_{X, Q}.$$

we know that $A(U)$ is a finitely generated k -algebra and thus let us choose $x_1, \dots, x_n$ to be generators. Each of these x_i have finitely many poles in X and thus, the set $S = X \setminus \bigcap_{i=1}^n \text{dom}(x_i)$ is finite. We claim

$$\Phi(U) = \bigcap_{i=1}^n \text{dom}(x_i)$$

If $Q \in \Phi(U)$, then there exists P with $\Phi(P) = Q$ and thus $\mathcal{O}_{U, P} = \mathcal{O}_{X, Q}$. Since $x_i \in \mathcal{O}_{U, P}$, the x_i are regular at Q . Conversely, suppose $P \in \bigcap_{i=1}^n \text{dom}(x_i)$, then $x_i \in \mathcal{O}_{X, P}$. Thus $A(U) \subseteq \mathcal{O}_{X, P}$. Define $\mathfrak{m} = A(U) \cap \mathfrak{m}_P$, which is a maximal ideal of $A(U)$. Then $A(U)_{\mathfrak{m}} = \mathcal{O}_{X, P}$ and $Q = V(\mathfrak{m})$ maps to P . We will see in the exercises, that this means that $V = \Phi(U)$ is affine and $A(V) \cong A(U)$. $\square$

We have the following theorem as well, although to prove it we need the existence of normalization of projective curves.

Corollary 18.3

Any smooth affine curve U admits smooth projective compactification: There exists a smooth projective curve X and an open embedding $U \hookrightarrow X$ with $k(U) \cong k(X)$. Moreover, this embedding is unique up to unique isomorphism.

To discuss morphisms of curves better, we extend the notion of finiteness to quasi projective varieties.

Definition 18.4. Let $\varphi: X \rightarrow Y$ be a dominant morphism of quasi projective varieties. We say that φ is finite if for all open affine subsets U of Y , the set $V := \varphi^{-1}(U)$ is also affine and

$$\varphi|_V: V \rightarrow U,$$

is a finite map of affine varieties.

Remark 18.5. We should check at this point that this extends the definition we gave before for affine varieties...

We already know that if $\varphi: X \rightarrow Y$ is a nonconstant morphism between curves, then φ is quasi finite which follows from the dimension of fibers theorem. As it turns out, in the smooth projective case we can do better:

Proposition 18.6

Let $\varphi: X \rightarrow Y$ be a non constant morphism between smooth projective curves. Then φ is finite.

To prove the theorem we will need the following lemma, which we prove in the exercises.

Lemma 18.7

Let X be a smooth projective curve and $U \subseteq X$ be a proper open subset. Then U is affine.

Proof of Proposition 18.6. Since φ is non constant we already know that it is surjective and in particular dominating. Suppose now that $U \subseteq Y$ is any affine open subset. Then U is not all of Y since Y is projective. Since the Zariski topology on curves is the cofinite topology, U is obtained by removing finitely many points from Y . Applying the theorem of the dimension of fibers to these points, they all have finite, nonempty fibers; therefore, $V = \varphi^{-1}(U)$ is obtained by removing finitely many points from X and is therefore affine by the lemma above. The map φ induces a finite algebraic extension

$$k(Y) \subseteq k(X).$$

Let A denote the integral closure of $A(V)$ in $k(X)$. From commutative algebra we then have that $A(V)/A$ is a finite module extension. Furthermore, $A(V) \subseteq A(U)$ and X is normal hence $A(U)$ is integrally closed. In particular, $A \subseteq A(U)$ holds as well. We claim $A = A(U)$. Note that for any maximal ideal $\mathfrak{m} \in A$, we have that $A_{\mathfrak{m}}$ is a DVR of $k(X)$ containing $A(V)$. Since $A(V)$ is the intersection of all DVR's containing it, we find $A(V) \subseteq A$ as well. Thus $A(V) = A$ and the claim follows. $\square$

Definition 18.8. Let $\varphi: X \rightarrow Y$ be a nonconstant morphism of smooth projective curves. Furthermore let $P \in X$ and $Q = \varphi(P)$. Then φ induces an inclusion, $\mathcal{O}_{Y,Q} \subseteq \mathcal{O}_{X,Q}$ which maps the maximal ideal to a subset of the other maximal ideal. Let π_Q be a uniformizing parameter at Q , we define the ramification index of φ at P to be

$$\dim_k \left(\frac{\mathcal{O}_{X,P}}{(\pi_Q)} \right) = \nu_P(\pi_Q).$$

Theorem 18.9

Let $\varphi: X \rightarrow Y$ be a non constant morphism between smooth projective curves of degree $d = [k(X) : k(Y)]$. Then for every $Q \in Y$ we have

$$\sum_{P \in \varphi^{-1}(Q)} e_P = d.$$

Proof. TO DO!!!!

□

Lemma 18.10

Let R be A DVR and M finitely generated torsion free R module. Then M is *free*

Proof. Let $\mathfrak{m}$ be the maximal ideal of R and choose a basis $\bar{f}_1, \dots, \bar{f}_n$ of the k vector space

$$M \otimes_R \frac{R}{\mathfrak{m}}$$

For each of the $\bar{f}_i$ choose lifts $f_i \in M$. By Nakayama's lemma they generated M as a R -module. We claim the generation is free. Suppose we had some nontrivial relation

$$\sum_{i=1}^n \lambda_i f_i = 0.$$

reducing mod $\mathfrak{m}M$, we obtain a relation that $\lambda_i \in \mathfrak{m}$ for all i . choosing a uniformizer t of $\mathfrak{m}$ we can write $\lambda_i = rt^{n_i}$ for $n_i > 0$ and a unit r . let d be the lowest exponent appearing, then we have:

$$t^d \sum_{i=1}^n \frac{\lambda_i}{t^d} f_i = 0 = \sum_{i=1}^n \mu_i f_i = 0$$

Since M is torsion free, we have that $\sum_{i=1}^n \mu_i f_i = 0$ and not all the $\mu_i \in \mathfrak{m}$. Reducing mod $\mathfrak{m}M$ we obtain a contradiction.

□

§19 Lecture 19

We will now discuss divisors on curve. Since prime divisors of curves are simply points, the divisor group of a curve consists of finite sums of points. To each such divisor we can assign a degree:

Definition 19.1. Let X be a smooth projective curve. The degree map of X is given by

$$\deg : \text{Div}(X) \rightarrow \mathbb{Z}, \sum_{P \in X} n_P P \mapsto \sum_{P \in X} n_P.$$

We have already introduced the pullback of divisors. Let us see what this looks like for Curves.

Definition 19.2. Let $\varphi : X \rightarrow Y$ be a non constant morphism of smooth projective curves. The pullback on divisor is the map given by:

$$\varphi^* : \text{Div}(Y) \rightarrow \text{Div}(X), P \mapsto \sum_{Q \in \varphi^{-1}(P)} e_Q Q.$$

This map maps divisors of degree n to divisors of degree dn where d is the degree of φ .

Lemma 19.3

Let $f \in k(Y)$, then $\varphi^*(\text{div}(f)) = \text{div}(\varphi^*(f))$.

Proof. We have:

$$\text{div}(\varphi^*(f)) = \sum_{Q \in X} \nu_Q(\varphi^* f) Q = \sum_{P \in Y} \sum_{Q \mapsto P} \nu_Q(\varphi^* f) Q = \sum_{P \in Y} \sum_{Q \mapsto P} e_Q \nu_P(f) Q = \varphi^*(\text{div}(f)).$$

□

In particular we see that the pullback of a principle divisor is again principle and so the pull back actually induces a map of class groups.

Corollary 19.4

Any principal divisor on a smooth projective curve has degree zero. Thus the degree map factors over the class group.

Proof. Let $f \in k(X)^\times$ be non constant. Then f actually represents a morphism $f : X \rightarrow \mathbb{P}^1$. consider the rational function $t \in k(\mathbb{P}^1) \cong k(\mathbb{A}^1) = k(t)$. Then f is the pullback of t . By the previous proposition we have

$$\text{div}(f) = \text{div}(f^*(t)) = f^*(\text{div}(t)) = f^*([0] - [\infty]).$$

the right hand side has degree $d \cdot 0 = 0$.

□

To fully understand the spaces $L(D)$, we need to talk about differentials. Recall that for a smooth manifold the De rahm cohomology is isomorphic to the singular cohomology. Thus, the space of differnetials captures topological holes. For the algebraic setting the differentials will turn out to capure similar invariants.

Definition 19.5. Let K be a field containing k . The Kähler differentials of K over k is the free K - vector space generated by symbols dx for $x \in K$, subject to the relations:

$$d(x + y) = dx + dy, d(x \cdot y) = ydx + xdy, d\lambda$$

for all $x, y \in K$ and $\lambda \in k$. This space is denoted by $\Omega_{K/k}$.

Remark 19.6. The space of Kähler differentials comes with a derivation $d : K \rightarrow \Omega_{K/k}$ and it is initial in this sense. I.e it satisfies the universal property for initial objects.

Lemma 19.7

Let $K = k(t_1, \dots, t_n)$ i.e K is the fraction field of a polynomial ring over k , then

$$\Omega_{K/k} \cong K dt_1 \oplus \dots \oplus K dt_n.$$

Proof. First, we show that the dt_i generate K . For this it suffices to show that all monomials

$$d(t_1^{n_1} \dots t_n^{n_n})$$

are in the span. We can do this via induction over $\sum n_i$. If this sum is zero or 1 the claim is clear. Otherwise assume wlog that $n_1 > 0$. Then by the Leibnitz rule we have

$$d(t_1^{n_1} \dots t_n^{n_n}) = t_1 d(t_1^{n_1-1} \dots t_n^{n_n}) + (t_1^{n_1-1} \dots t_n^{n_n}) dt_1.$$

By induction the dt_1 generates both of the summands. Now we need to show that the dt_i are linearly independent. Consider the derivation on K given by

$$d : K \rightarrow K^n, f \mapsto \begin{bmatrix} \frac{\partial f}{\partial t_1} \\ \vdots \\ \frac{\partial f}{\partial t_n} \end{bmatrix}$$

This map is clearly surjective and it is a derivation. Thus, it factors of $\Omega_{K/k}$ and therefore $\dim_K \Omega_{K/k} \leq n$. By linear algebra this shows that the dt_i are a basis. $\square$

The above shows that for $\mathbb{P}^1$, its space of differentials is a one dimensional $k(t)$ vector space. Geometric intuition from differential geometry tells us that this should always be the case. This is true as we shall see in the next proposition.

Proposition 19.8

Let K/k be a finitely generated field extension of transcendence degree 1. Then $\Omega_{K/k}$ is one dimensional.

Proposition 19.9

Since k is algebraically closed it is possible to choose $t \in K$ such that $K/k(t)$ is a separable finite extension. Thus, we may apply the theorem of the primitive element to find that $k(t)(\alpha)$. Let $f \in k(t)[X]$ be the minimal polynomial of α . By multiplying out denominators we may assume $f \in k[t][X]$. Furthermore since the extension $K/k(t)$ was separable, f has no double roots. Consider now the affine curve with coordinate ring $k[t, \alpha] = \frac{k[t, X]}{(f)}$. We know that

$$\frac{\partial f}{\partial X}(\alpha) \neq 0.$$

However we have

$$0 = d(f(t, \alpha)) = \frac{\partial f}{\partial t}(t, \alpha) dt + \frac{\partial f}{\partial X}(t, \alpha) d\alpha$$

Since the second coefficient is nonzero, we can solve this for $d\alpha$ and see that dt generates $\Omega_{K/k}$.

Let K/k be a finitely generated field extension of transcendence degree 1. Then $\Omega_{K/k}$ is a one dimensional K vector space.

Proof. First we claim that there exists $t \in K$ such that $K/k(t)$ is a separable extension. If $\text{char}(k) = 0$ this is obvious so assume $\text{char}(k) = p > 0$. Since K is a finitely generated k algebra of transcendence degree 1, we may write

$$K = k(x, y_1, \dots, y_n),$$

where x is transcendental over k and the y_i are algebraic over $k(x)$. We prove the induction over n ; if $n = 0$, there is nothing to prove, so suppose $n > 0$. Without loss of generality let the y_i all be transcendental over k , otherwise they are in k since k is algebraically closed. Also assume that none of the y_i are already in $k(x)$ since otherwise the claim follows by induction. Thus, there exists a polynomial

$$f \in k(x)[Y_1, \dots, Y_n], \quad f(x, y_1, \dots, y_n) = 0.$$

By multiplying out a denominator, assume $f \in k[x][Y_1, \dots, Y_n]$. Furthermore, assume that f is irreducible, since otherwise one of its factors vanishes on the y_i and we may replace f by this factor. The crucial point now is that not all the variables $X, Y_1, \dots, Y_n$ can occur only in powers of p , since otherwise the polynomial f is contained in

$$k[X^p, Y_1^p, \dots, Y_n^p],$$

which, since k is algebraically closed, consists only of reducible polynomials. Let us assume Y_n doesn't appear in a p 'th power (note that x could also be the variable which does not appear only in p 'th powers, but we can replace x by any of the Y_i since we assume them all to be transcendental over k). Then The polynomial

$$g \in k(x_1, y_1, \dots, y_{n-1})[Y], \quad g = f(x, y_1, \dots, y_{n-1}, Y)$$

is irreducible, nonzero, and vanishes on y . Furthermore, since its derivative with respect to Y is nonzero, it has distinct roots and y_n is therefore separable over $k(x, y_1, \dots, y_{n-1})$. The claim then follows by induction and the fact that a tower of separable extensions is separable.

Now we can prove the actual statement. By the above discussion, we can write $K = k(t, \alpha)$ where α is separable over $k(t)$. Let $f \in k[t, Y]$ be the minimal polynomial of α , and so in particular its derivative with respect to Y does not vanish at α . By the universal property of Kähler differentials, there exists a K linear map

$$\Phi : \Omega_{k(t, Y)/k} \rightarrow \Omega_{K/k}, \quad \begin{cases} dt \mapsto dt \\ dY \mapsto d\alpha. \end{cases}$$

Note that $f(t, \alpha) = 0$ and so $d(f(t, Y))$ maps to zero under this map. so:

$$\frac{\partial f}{\partial t}(t, \alpha)dt + \frac{\partial f}{\partial Y}(t, \alpha)d\alpha = 0.$$

On the other hand, by the choice of α , we know that $\frac{\partial f}{\partial Y}(t, \alpha) \neq 0$. Thus we can rearrange the above to get

$$d\alpha = -\frac{\frac{\partial f}{\partial t}(t, \alpha)}{\frac{\partial f}{\partial Y}(t, \alpha)}dt.$$

since Φ is surjective, this proves that $\dim_K \Omega_{K/k} \leq 1$. To finish the proof we need to show that $dt \neq 0$. By the universal property of Kähler differentials it suffices to find a derivation

$$\delta : K \rightarrow M,$$

which does not vanish on t . To do this, notice $K = \frac{k(t)[Y]}{(f(t,Y))}$. So we simply construct the derivation on $k(t,Y)$ and show that it vanishes on f to ensure it is well defined on K . The following derivation does the trick:

$$\delta_f : k(t,Y) \rightarrow k(t,Y), G \mapsto \frac{\partial G}{\partial t} - \frac{\frac{\partial f}{\partial t}}{\frac{\partial f}{\partial Y}} \cdot \frac{\partial G}{\partial Y}.$$

□

This leads us to the space of rational differentials:

Definition 19.10. Let C be a smooth projective curve. The one dimensional $K(C)$ module $\Omega(C) := \Omega_{K(C)/k}$ is called the space of **rational differentials** on C .

Definition 19.11. A rational differential $\omega \in \Omega(C)$ is called **regular** at a point $P \in C$ if it can be written as

$$\omega = \sum_i f_i dg_i, \quad f_i, g_i \in \mathcal{O}_{C,P}.$$

The set of regular differentials at P is denoted by $\Omega_{C,P}$ and it is naturally a $\mathcal{O}_{C,P}$ module.

Proposition 19.12

The $\mathcal{O}_{C,P}$ module $\Omega_{C,P}$ is free and of rank one,

Proof. let $t \in \mathfrak{m} \subseteq \mathcal{O}_{C,P}$ be a uniformizing parameter. We claim that dt generates $\Omega_{C,P}$. By definition it is clear that the symbols $dg, g \in \mathcal{O}_{C,P}$ generate $\Omega_{C,P}$ so we only need to show that these lie in the $\mathcal{O}_{C,P}$ span of dt . Furthermore, if $u \in \mathcal{O}_{C,P}^\times$, then $du = d(u - u(p))$ and so it suffices to show that dt generates the dg for $g \in \mathfrak{m}$. Such g can be written as $t^n u$ where u is a unit in $\mathcal{O}_{C,P}$. Differentiating this we have

$$dg = t^n du + u dt(t^n) = t^n du + u t^{n-1} dt.$$

Modding this relation by $\mathfrak{m}\Omega_{C,P}$ we see that dt generates $\frac{\Omega_{C,P}}{\mathfrak{m}\Omega_{C,P}}$. By Nakayama's lemma it follows that dt generates $\Omega_{C,P}$. Furthermore, $\Omega_{C,P}$ is clearly torsion free since it is a subset of a $K = \text{Frac}\mathcal{O}_{C,P}$ vector space. Thus, by the results from the last section, it is free and of rank at most one. To show that the rank is equal to one, we need to prove that $\Omega_{C,P}$ is nonzero. By the universal property of differentials, it suffices to prove a nonzero derivation $\mathcal{O}_{C,P} \rightarrow M$. Here is such a derivation:

$$\mathcal{O}_{C,P} \rightarrow k, u \mapsto \left(\frac{u - u(P)}{t}\right)(P)$$

one easily checks that this is a derivation and maps t to 1. □

The above shows in particular, that dt is nonzero and thus a $k(C)$ basis of $\Omega(C)$. In particular every rational differential $\omega \in \Omega(C)$ is uniquely written as $f_\omega dt$ and a rational differential is regular at P if and only if $f_\omega \in \mathcal{O}_{C,P}$. With this in mind, we can define the order of a rational differential.

Definition 19.13. Let ω be a rational differential of C and $t \in k(C)$ a uniformizer of C at a P . Then the order of ω at P . Write $\omega = f_\omega dt$ for a unique f . Then we define

$$\text{ord}_P(\omega) := \text{ord}_P(f_\omega)$$

Remark 19.14. This is independent of choice of uniformizer: if t' is another uniformizer, then $t' = tu$ for a unit in $\mathcal{O}_{C,P}$. Taking differential we have

$$dt' = udt + tdu.$$

write $du = gdt$ for $g \in \mathcal{O}_{C,P}$. to find

$$f dt' = f(u + tg)dt,$$

and then see that $\text{ord}_P(f) = \text{ord}_P(f(u + tg))$.

Definition 19.15. Let ω be a nonzero rational differential on C , we define its divisor as:

$$\text{div}(\omega) := \sum_{P \in C} \text{ord}_P(\omega) P.$$

Divisors of this form are called **canonical divisors**.

Remark 19.16. We should check that the above sum is actually a divisor, i.e it is a finite sum. This can be shown as follows: note that $dt = d(t - t(Q))$ for any Q where t is regular (which is all but finitely many points) hence it suffices to show that $t - (Q)$ is a uniformizer at Q for all but finitely many t . Consider now the morphism

$$C \rightarrow \mathbb{P}^1, P \mapsto t(P).$$

Then, then points where $t - t(Q)$ is not a uniformizer are the ramification points of this mapping of which there are finitely many. This can be seen by the fact that degree d finite morphisms admit d distinct pre images for a nonempty open subset.

Notice that any two canonical divisors are equivalent and thus have the same degree. Furthermore if $\text{div}(\omega) = D + \text{div}(f)$ then $D = \text{div}(\frac{1}{f}\omega)$ is also canonical. Thus the canonical divisors form a class in $\text{Cl}(X)$. We will denote this class by K_C or when we want to choose any representative we will also sometimes call this divisor K_C .

The canonical divisor allows us to define an invariant of a smooth projective curve called the genus.

Definition 19.17. Let C be a smooth projective curve. The **genus** of C is the nonnegative integer defined as

$$g(C) := g := \dim_k L(K_C) = l(K_C)$$

where K_C is any canonical divisor of C .

Remark 19.18. Alternatively, we may identify $L(K_C)$ with the space of regular differentials on C since if $\text{div}(f) + K_C \geq 0$, then fk_C is a regular differential on C .

Let us compute a first example of a canonical divisor.

Example 19.19

Let $C = \mathbb{P}^1$ and identify $K(C)$ with the function field $k(t)$ where t is an affine coordinate for $U_0 \cong \mathbb{A}^1$. Then for any $p \in \mathbb{A}^1$, we have $t - p$ is a uniformizing parameter at p . and as such

$$dt = 1 \cdot d(t - p) \Rightarrow \text{ord}_P(dt) = 0$$

On the otherhand, at ∞ , we have that $u = \frac{1}{t}$ is a local uniformizer. Thus,

$$dt = -\frac{1}{u^2} du \Rightarrow \text{ord}_\infty(dt) = -2.$$

In total,

$$\text{div}(dt) = -2[\infty],$$

which shows that $L(K_C) = 0$ and therefore, $g(C) = 0$.

§20 Lecture 20

We arrive at the last lecture of this semester. In this lecture, we discuss the Riemann-Roch theorem and its applications. We will postpone a proof of the Riemann Roch theorem till next semester when we introduce sheaf cohomology. It should however be remarked, that it does have an elementary proof which can be found for example in fulton's Algebraic Curves. Let us state the theorem now.

Theorem 20.1 (Riemann-Roch Theorem)

Let C be a smooth projective curve and D a divisor on C . Then

$$l(D) - l(K_C - D) = \deg D + 1 - g$$

Remark 20.2. Notice that $l(D) = 0$ if $\deg D < 0$. Thus for $\deg D > \deg K_C$ the formula simplifies to

$$l(D) = \deg D + 1 - g \text{ for large } D.$$

Furthermore, the theorem tells us that the dimension grows roughly linearly in D except for a few small cases. Notice also, that the genus contributes negatively to $l(D)$. So having a larger genus creates more obstructions to having functions with certain local datum.

Let us explore some further consequences.

Corollary 20.3

The canonical divisor K_C has degree $2g - 2$.

Proof. The Riemann Roch theorem states

$$l(D) - l(K_C - D) = \deg D + 1 - g,$$

if we let $K_C = D$, we have

$$l(K_C) - l(0) = \deg K_C + 1 - g.$$

Recalling that $l(K_C) = g$ and $L(0) = k$, we have

$$g - 1 = \deg K_C + 1 - g \Leftrightarrow \deg K_C = 2g - 2.$$

□

Corollary 20.4

Let D be a divisor of degree $\deg D > 2g - 2$, then

$$l(D) = \deg D + 1 - g.$$

Proof. If $\deg D > 2g - 2$, then we have $\deg(K_C - D) < 0$ and the claim follows. □

Corollary 20.5

Let $P_1, \dots, P_n$ be points on C . Then there exists a function $f \in k(C)^\times$ which is regular on $C \setminus \{P_1, \dots, P_n\}$ and has poles at the P_i . Furthermore, $C \setminus \{P_1, \dots, P_n\}$ is affine.

Proof. Let $n = 2g$, then by the above corollary, and the Riemann roch

$$l(nP_1) = n + 1 - g > 1$$

Thus there is a nonconstant function f_1 in this space, and it must have a pole at P_1 and be regular outside of P_1 . Do the same for $P_2, \dots, P_n$ to obtain functions $f_1, \dots, f_n$. The function

$$f := f_1 + \dots + f_n,$$

is regular outside of $\{P_1, \dots, P_n\}$. Furthermore, for any of the P_1 choose a local paramter t_i and use this to write:

$$f_i = t_i^{-n_i} u_{i,i}, \quad f_j = t^{n_j} u_{i,j} \quad \text{for } j \neq i$$

where $n_i > 0$, $n_j \geq 0$ and the $u_{i,j}$ are units. Thus

$$f = f_1 + \dots + f_n = t^{-n_i} \left(u_{i,i} + \sum_{j \neq i} u_{i,j} t^{n_i + n_j} \right) \Rightarrow \text{ord}_{P_i}(f) = \text{ord}_{P_i}(f_i) < 0.$$

Thus, f has the desired properties. Now consider the divisor $D = mP_1 + \dots + mP_n$. and the linear system $|D|$. We know that $|D|$ sperates points and tangents (see exercise sheet) and therefore $|D|$ corresponds to an embedding

$$\Phi: X \hookrightarrow \mathbb{P}^r.$$

since $f_0 = 1 \in L(D)$, extend this to a basis $f_0, f_1, \dots, f_r$ of $L(D)$ to obtain, in these coordinates, the embedding locally represeneted by

$$\Phi: X \hookrightarrow \mathbb{P}^r, P \mapsto [1 : f_0(P) : \dots : f_r(P)]$$

This representation is only not valid on $X \setminus \{P_1, \dots, P_r\}$ and in fact

$$X \setminus \{P_1, \dots, P_r\} = \Phi^{-1}(\mathbb{P}^r \setminus \{X_0 = 0\}) \cong \Phi(X) \cap \mathbb{A}^r,$$

which is affine. □

Example 20.6

Consider the curve $C \subseteq \mathbb{P}^2$ given by $C = Z_{\mathbb{P}^2}(X_0X_2^2 - X_1^3 + X_0^2X_1)$. On the affine chart $X_0 \neq 0$, C the points in C all satisfy

$$f(x, y) = y^2 - x^3 - x = 0$$

where $x = X_1/X_0$ and $y = X_2/X_0$. Thus, as a rational function on C , the function f is zero. Differentiating this relation, we obtain

$$0 = df = 2ydy - (3x^2 - 1)dx.$$

Let us assume $\text{char}(k) \neq 2$ to simplify the discussion. we will calculate the divisor of dx . First we will calculate the order at points in the affine $C' = C \cap \{X_0 \neq 0\}$. Suppose $p = (p_1, p_2)$ is any point in C' then we have:

$$\mathcal{O}_{C,p} = \left(\frac{k[x, y]}{(y^2 - x^3 - x)} \right)_{(x-p_1, y-p_2)}$$

And so the maximal ideal of this ring is $\mathfrak{m}_p = (x - p_1, y - p_2)\mathcal{O}_{C,p}$. Then $x - p_1$ is a uniformizer of this ring if and only if it does not correspond to the tangent line of C at p . which is given by

$$T_p C = Z(df_p) = Z(f_x(p)(x - p_1) + f_y(p)(y - p_2))$$

This line is not equal to $Z(x - p_1)$ if and only if $f_y(p) = 2p_2 \neq 0$. Hence unless $p_2 = 0$ $x - p_1$ is a uniformizer and

$$dx = d(x - p_1) = 1 \cdot d(x - p_1) \Rightarrow \text{ord}_P(dx) = 0.$$

If $p_2 = 0$, then P is one of the three points: $P_1 = (0, 0)$, $P_2 = (1, 0)$ and $P_3 = (-1, 0)$. For there $y - p_2$ is a uniformizer and we have

$$dx = \frac{2y}{3x^2 - 1} dy \Rightarrow \text{ord}_{P_i}(dx) = \text{ord}_{P_i} \left(\frac{2y}{3x^2 - 1} \right) = 1,$$

where we use $\text{char}(k) \neq 2$ to see that the denominator does not vanish at all of these three points.

Now let us calculate the order of dx at points of $C \cap \{X_0 = 0\}$. For such points we have

$$0 = X_0X_2^2 - X_1^3 + X_0^2X_1 = X_1^3 \Rightarrow P = [0 : 0 : 1].$$

Thus, there is only one intersection point and it is contained in the affine $X_2 \neq 0$. Let $s = X_0/X_2$ and $t = X_1/X_2$. These elements satisfy:

$$s - t^3 + s^2t = 0 \Rightarrow ds - 3t^2dt + 2tsds + s^2dt = 0$$

Furthermore, $x = \frac{s}{t}$. we see that at $(0, 0) = [0 : 0 : 1]$, that t is a uniformizing parameter. Solving the above relation for ds , we have

$$ds = \frac{3t^2 - s^2}{1 + 2ts} dt.$$

Then we conclude for x :

$$dx = d\left(\frac{1}{t}s\right) = sd\left(\frac{1}{t}\right) + \frac{1}{t}ds = \frac{-s}{t^2}dt + \frac{1}{t} \frac{3t^2 - s^2}{1 + 2ts} dt =$$

As we have seen, linear systems can be used to study maps to projective space. With the Riemann-Roch theorem, at hand, this view point becomes even more fruitful.

Definition 20.7. A divisor D on C is called **very ample** if it is base point free and it induces a closed embedding $X \hookrightarrow |D|^\vee$. D is called **ample** if there exists a positive integer m , such that mD is very ample.

Lemma 20.8

Let D be a divisor on C , then the complete linear system $|D|$ is basepoint free if and only if $|D - P|$ has dimension one less than that of $|D|$.

Proof. This is simply a weaker version of the lemma we have already proven for linear systems, applied to $\delta := D$, and noting that δ_P is either δ or has dimension one less. $\square$

Theorem 20.9

Let D be a divisor on C , then D is very ample if and only if for all $P, Q \in X$, we have

$$\dim |D - P - Q| = \dim |D| - 2.$$

Proof. This was not properly proven but the idea is that for $P \neq Q$ the above implies injectivity and for $P = Q$ it implies injectivity of tangent spaces. $\square$

Corollary 20.10

Let D be a divisor on C . If $\deg D \geq 2g + 1$, then $|D|$ is very ample.

Proof. Notice that $\deg D - P - Q \geq 2g - 1 < 2g - 2$. Thus the correction term in the Riemann-Roch vanishes for and we have

$$\dim |D - P - Q| = l(D - P - Q) - 1 = \deg(D) - 2 + 1 - g - 1 = l(D) - 1 - 2 = \dim |D| - 2.$$

$\square$

An immediate corollary is :

Theorem 20.11

A divisor on a projective curve is ample if and only if it has positive degree.

Example 20.12

let C be a projective curve of with $g = 0$, then we claim $C \cong \mathbb{P}^1$. Indeed, consider the divisor $D = P$ It has degree $1 \geq 2g + 1$, and thus it is very ample by the theorem above. Therefore, it defines an embedding into $\mathbb{P}^r$, where $r = \dim |D|$. On the otherhand, using the Riemann-Roch theorem we have

$$|D| = l(D) - 1 = \deg D + 1 - l(K_C - D) - 1 = 1.$$

Where we used the fact that the canonical divisors have degree $2g - 2 = -2$ and so $\deg(K_C - D) < 0$. Thus this defines an embedding to $\mathbb{P}^1$. But it is also surjective as a nonconstant map between projective curves. Hence it is an isomorphism. In fact the arguments above show that the following are equivalent:

- $C \cong \mathbb{P}^1$
- The genus of C is zero
- there exists a point P , with $l(P) > 1$
- for all points P , $l(P) > 1$.
- there exists a point P , with $l(P) = 2$
- for all points P , $l(P) = 2$

The only part of this equivalence which we have not argued is that if there exists a point with $l(P) > 1$, then $X \cong \mathbb{P}^1$. Indeed let $f \in L(P)$ be nonconstant, then we may view f as a morphism to $\mathbb{P}^1$. Then f is a finite map, and the degree of f is the number of points in a generic fiber. However, outside of P , f is a map to $\mathbb{A}^1$, with no poles. and one zero. then for $p \in \mathbb{A}^1$, the function $f - f(P)$ still has only one pole and thus has only one zero. Thus, f is injective, and every fiber has one point in the preimage. In particular, f induces an isomorphism of function fields and thus an isomorphism to $\mathbb{P}^1$.

Example 20.13

Let C be a curve of genus 1 now. Then Every divisor of degree $\geq 2g + 1 = 3$ is very ample. Suppose that $P \in C$ is any point. Applying the Riemann-Roch theorem , we have

$$l(3p) - l(K_C - 3p) = 3 + 1 - 1 = 3.$$

On the otherhand, $\deg K_C - 3p = 2g - 2 - 3 = -3 < 0$ and as such $l(3P) = 3$. Thus, the very ample linear system $|3p|$ defines an embedding φ to a curve $X \subseteq \mathbb{P}^2$. Suppose now that L is any line in $\mathbb{P}^2$, we know that the degree of the X is given by the number of intersection points of X and L when counted with multiplicity. However for such a line we have $\varphi^*(L) \in |3P|$ and thus has degree 3. Thus L intersects X three times.

Example 20.14

Let C be a curve of genus $g \geq 2$